SOUTH AFRICAN
INSTRUMENTATION AND CONTROL

SAIMC
Official publication of the SAIMC

VOLUME 28
NO 3 MARCH 2012



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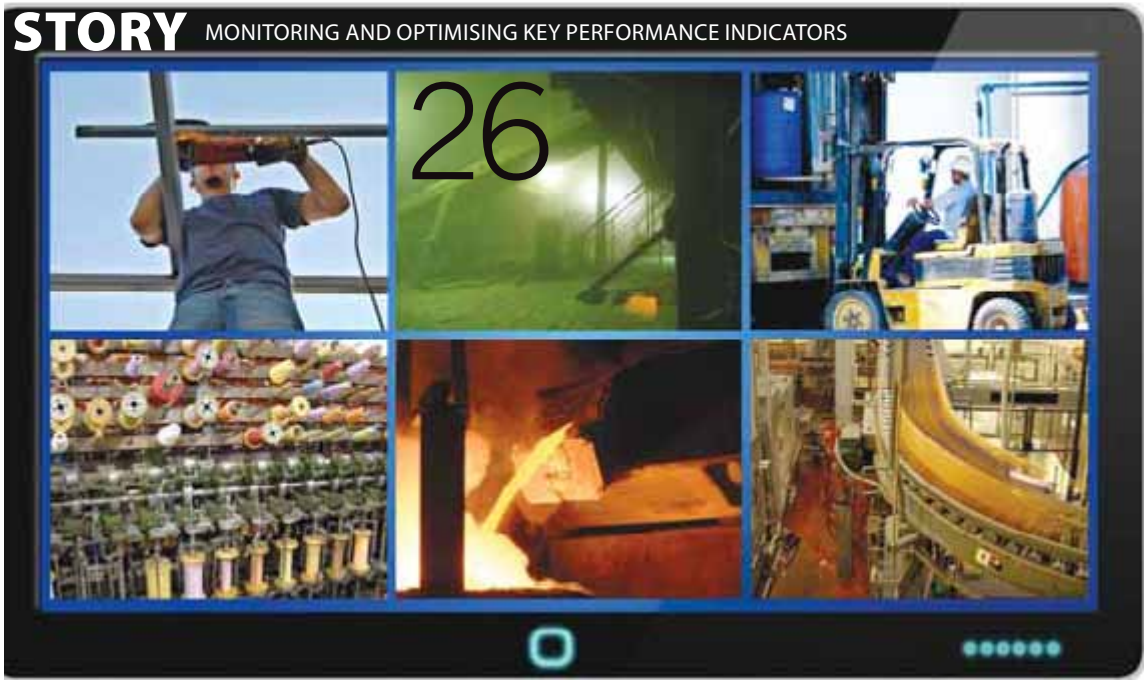
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contents

VOLUME 28 NO 3 MARCH 2012

SOUTH AFRICAN INSTRUMENTATION AND CONTROL

COVER STORY MONITORING AND OPTIMISING KEY PERFORMANCE INDICATORS









28 POWER GENERATION & DISTRIBUTION

41 FACILITIES MANAGEMENT

46 FIELDBUS & INDUSTRIAL NETWORKING

58 IT IN MANUFACTURING

61 DATA-LOGGING & RECORDING

62 PRESENCE SENSORS & VISION SYSTEMS

72 SAFETY SYSTEMS

86 MASS MEASUREMENT

90 CONTROL LOOP

04 News digest

06 News & events

20 SAIMC news

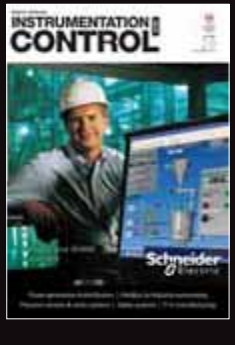
23 Jim Pinto

24 System integrators

92 Product news

OUR COVER

In today's challenging economic climate, the ability to capture real-time and historical operational data, and analyse and visualise that data in the right context, plays an increasingly important role in a company's success. With its ability to provide manufacturers with a plant's complete digital history, Vijeo Historian V4.30 offers a powerful plant historian that supports the monitoring and optimisation of critical key performance indicators. More on page 26.



Not another same-old, same-old year



The latest *Control* magazine list of top automation companies is out. Siemens, ABB and Emerson retain their positions in the top 3 by global ranking, while in North America; it is Emerson, Rockwell and ABB that fill the podium spots. Jim Pinto reviews the results in this issue and also gives us his prognostications for 2012.

The forecast is that this will not be another same-old, same-old year for industrial automation. When one gazes into the Pinto 'crystal ball' it looks like 2012 could be the year of the industrial network. industrial wireless and industrial Ethernet are both expected to gain further acceptance, while traditional centralised control could start to give way to more autonomous peer-to-peer style architectures. The popularity of mobile devices as the operator interface of choice is also likely to expand quickly.



On the corporate front, Pinto believes that a number of the automation majors are hoarding cash because they are worried about uncertainty in global economic conditions. This is preventing an acquisition binge that will happen sooner rather than later in the futurists' view of things.

IT'S BEEN A BUSY MONTH

Closer to home, it was announced at a press conference on 15 February that Adroit Technologies has been appointed as an authorised distributor for sub-Saharan Africa of the Mitsubishi Electric range of automation products. This follows the announcement in March last year of the acquisition by Mitsubishi – ranked 6 globally in the Top 50 list – of a 14,9 % shareholding in Adroit. The *SA Instrumentation and Control* team had the opportunity to talk to Adroit MD Dave Wibberley and Misubishi's Hartmut Puetz about the nature of the agreement, the plans for setting up the distribution infrastructure and what it will take to kick on and achieve the targets. Due to a clash of deadlines for this issue, we will be publishing the article on this in-depth discussion in next month's magazine.

The SAIMC has also been busy recently in the run-up to the Society's AGM later this month. The regional branches have held their local elections and we have all the details of the 2012 committees. President, Johan Maartens, writes of 2012 as a year of great new beginnings for the SAIMC. The organisation has signed a Memorandum of Understanding with the international Automation Federation, giving it access to the worldwide automation group's insight, infrastructure and method. This can only have a positive influence on one of the major challenges to be addressed locally this year – the issue of training.

We look forward to keeping you updated on this vital element for economic growth in our region. In light of the infrastructure expansion plans being proposed by government and by the private sector, the importance of training of C&I professionals is about to become an item of national importance and the SAIMC will have a pivotal role to play. This has all the makings of not being another same-old, same-old year in the automation industry.

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PUBLISHERS
Technews Publishing (Pty) Ltd,
Box 385, Pinegowrie 2123

PRINTED BY:
Intrepid Printers (Pty) Ltd
9 Grix Road, Willowton, Pietermaritzburg

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FIELDBUS FOUNDATION 2012 GENERAL ASSEMBLY

The Fieldbus Foundation is to hold its 2012 General Assembly from 6-8 March in São Paulo, Brazil. This key automation industry event will attract the world's leading suppliers and end users of Foundation fieldbus.

The 2012 General Assembly, themed 'In a World of Choices, Foundation Brings it all Together,' the event will include a comprehensive, end user-oriented agenda consisting of fieldbus project case studies and tabletop exhibitions from leading automation vendors across the globe, as well as a demonstration of the new Foundation for Remote Operations Management technology.

WIKA ACQUIRES SCANDURA



The acquisition of Scandura has taken WIKA a step further towards strengthening its leading role in the field of calibration technology. Scandura, founded in Italy in the 1950s, established itself internationally as a manufacturer of special calibration applications. The portable calibrators, in particular, meet the highest demands and safety standards, and they offer reliable measuring equipment (for example with ATEX certificates and for critical applications in the process industry). The complete calibration systems enjoy a special reputation as well. Calibration software rounds out the portfolio.

Scandura will be integrated into the calibration technology division of the WIKA Group.

METSO TO SUPPLY LARGEST RECOVERED FUEL FIRED BOILER

Metso is to supply Mälarenergi's combined heat and power plant (CHP) in Sweden with the world's largest recovered fuel fired boiler. The new boiler with a fuel input of 167 MW will utilise circulating fluidised bed (CFB) technology and it will serve as a base unit to meet the district heating power needs of the surrounding municipalities. Metso's delivery will account for about 30 per cent of the €300 million modernisation project of the CHP plant.

The principal fuel for the new boiler will be recovered fuel prepared from municipal waste, the new boiler also having possibility to burn biofuel. Compared with current boilers, the new system will give the plant a more flexible fuel mix. The modernised CHP plant will be commissioned around the middle of 2014.

SKF A SUPPLIER TO PARISIAN ELECTRIC CAR PROJECT



SKF has announced that it has supplied advanced technological products and solutions to the Bolloré Group for use in their innovative electric cars, part of the Autolib car sharing programme to be launched in Paris and the surrounding area.

The programme calls for 3000 cars on the road by mid-2012 and is anticipated to reduce carbon dioxide emissions by 22 000 tons per year. The SKF solution is based on ceramic bearings and advanced SKF sensor technology developed to fit the electric power-train giving high standards of efficiency and reliability in the traction system.

VICTORY FOR RS SPONSORED SOLAR CAR



RS Components recently supported the solar car team from Japan's Tokai University, who participated in the World Solar Challenge – a car race held in Australia last year. The race car used HIT solar cells supported with high-capacity lithium-ion batteries provided by Panasonic. The body was formed from super lightweight Toray carbon fibre material, which enabled the weight of the vehicle to be reduced to 140 kg.

After a gruelling race over 3000 km and four days, Tokai University's solar car team came in first. This marks their second consecutive victory at the race, made even more challenging this time by bushfires in the Northern Territory.

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Adroit Technologies now authorised distributor of Mitsubishi Electric



Dave Wibberley (front left) with Hartmut Puetz at the press conference. Adroit team members in background.

Adroit Technologies has been appointed as an authorised distributor for sub-Saharan Africa of the Mitsubishi Electric range of automation products. The product range includes programmable logic controllers (PLCs), process automation controllers (PACs), remote I/O, variable speed drives, operator panels and power monitoring devices.

In March of 2011, Mitsubishi Electric Europe acquired 14.9% of the shareholding in Adroit Technologies. Since this acquisition it has successfully been promoting the Mitsubishi Adroit Process Suite (MAPS), as well as Adroit's world-class Scada, Alarm Management and Scada Intelligence software, throughout Europe. In the 1990s Mitsubishi held a leading position in southern Africa and it is part of the EMEA strategy to have a direct automation presence in the region once again. It believes through its share in Adroit Technologies, with its leading position and knowledge of the sub-Saharan African market, it will once again be in a strong position to offer the level of service that customers expect from a global leader in the automation market. The joint goal is to become the leading platform that powers industry in southern Africa.

Various key personnel, who have many years of experience with the Mitsubishi products, have joined Adroit Technologies to offer a full range of services including:

- Pre-sale and post-sales assistance.
- Product training.
- Standard and customised service level maintenance agreements.
- Turn-key custom solutions.

ABOUT MITSUBISHI

Established in 1921 and with current revenues of around €40 billion, Mitsubishi Electric is one of the leading global electrical and electronics manufacturers. Established in 120 countries, employing more than 100 000 people, the company prides itself on the highest quality automation products that drive production and processes around the world.

ABOUT ADROIT

Established in 1983, Adroit Technologies is the developer of the Adroit suite of automation software. Products include the Adroit Scada/HMI software, Adroit Alarm Management tools, Adroit Scada Intelligence, Adroit Information Portal and the MAPS Integrated Process Suite. With over 5000 solutions globally, Adroit software is found around the world running any operation that requires real-time visibility. The products allow customers to understand and optimise operations, offering high returns on automation system investments.

For more information contact Jan Hendrik Rust, Adroit Technologies, +27 (0)11 658 8100, jan-hendrikr@adroit.co.za, www.adroit.co.za





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ANA-DIGI TO BULK SHIP VSD DRIVES

Ana-Digi is now bulk shipping its variable speed drives products in order to keep up with the demand from distribution and integration partners. "This is principally for LS Industrial Systems VSDs, but it also impacts on Power Electronics' large VSDs over 100 kW, as these will be warehoused in the same locations across the country," says Ana-Digi Systems' Peter Gross.

"Together with our partner network, we provide a unique systems solution to the mining, water, HVAC and manufacturing industries that offers energy savings and facility optimisation. This move is an extension to add even more value for our end users through the shorter delivery times from the increased stockholding at our strategically located distributors."

For more information contact Ana-Digi Systems, 086 132 3334, info@anadigi.co.za, www.anadigi.co.za

CRAIGCOR DISTRIBUTION HAS RELOCATED

CraigCor Distribution has relocated to 234 Albert Amon Road, Meadowdale, Gauteng. The new facilities, situated along the R24 between the N3 and R21, include a warehouse, dedicated product display area, equipment testing facility and an in-house training room.

CraigCor's John Lingenfelder notes that the move had much to do with necessity: "Behind all strategic decisions at CraigCor lies our 4 Point Promise, perhaps the most important of which is stockholding. While our old building served us well, it had just become too small to hold the adequate inventory required to serve our clients.

"The first order of business in the new building will be to bolster our stock levels. With the additional space we are looking to increase inventory by 40% in the first quarter, once again reconfirming our commitment to our clients' operations."

For more information contact John Lingenfelder, CraigCor Distribution, +27 (0)11 574 5300, jm.lingenfelder@craigcor.co.za, www.craigcor.co.za



ROCKWELL AUTOMATION IN CONTACT ACROSS BORDERS



Rockwell Automation hosted successful two-day technology updates in neighbouring Botswana and Namibia focused primarily on the water, wastewater and mining industries.

Following on the success of the road show presented in Botswana in 2010, 45 local delegates joined Rockwell Automation, its channel partners, Systems Automation & Investment (SAI), Kalahari Control & Instrumentation, and JRK Investments, as well as Rockwell Encompass Partners, Endress + Hauser and ProSoft, in Gaborone. Rockwell Automation, in conjunction with its local channel partners EcoServe and Electro Tech, then continued to host its first event of this kind in Namibia to 30 delegates in Windhoek.

Both events placed particular emphasis on process automation and software integration, products such as Allen Bradley's ControlLogix and MicroLogix 800, as well as variable speed drives (VSDs) and medium voltage (MV) drives were showcased.

"These events provide us with the ideal opportunity to share valuable industry skills and knowledge with our end-user customers," says Colin Morris, Rockwell Automation export sales manager, sub-Saharan Africa. "This helps them to stay abreast of new developments without having to travel across international borders. We have clearly seen the positive interest expressed by customers in this region to attend these events."

In addition to presentations by representatives of Rockwell Automation and industry, hands-on demonstration units provided delegates with first-hand experience of the Allen-Bradley equipment.

"We have already seen positive feedback from both these events, with promising leads being received. Based on this success, we are planning technology updates for other sub-Saharan regions," concludes Morris.

For more information contact Colin Morris, Rockwell Automation, +27 (0)11 654 9700, cmorris@ra.rockwell.com, www.rockwellautomation.co.za

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INVENSYS EXPANDS VIRTUALISATION TECHNOLOGY FOR HIGH AVAILABILITY AND DISASTER RECOVERY

Invensys Operations Management has expanded its certification for virtualisation technology to include high availability, disaster recovery and fault tolerance in supervisory control applications, leveraging both the VMware and Microsoft Hyper-V virtualisation platforms. The company's ArchestrA System Platform 2012 and Wonderware InTouch 2012 software are now certified for the latest VMware solutions, including VMware vSphere version 5.0 and ESXi version 5.0 for mission-critical applications.

"Historically, high availability and disaster recovery solutions in supervisory control systems were expensive to implement, not only because of hardware and software costs, but also because of additional administrative burdens," says Maryanne Steidinger, director of product marketing, Invensys Operations Management. "Now, after a rigorous validation period, our ArchestrA System Platform 2012 and Wonderware InTouch 2012 software are also certified for disaster recovery and high availability using VMware virtualisation. All this is possible on commercial operating systems using COTS hardware, further reducing cost and easing implementation of mission-critical applications."

Virtualisation software, like that offered by Microsoft and VMware, transforms a computer's hardware, i.e. CPU, hard drive and network controller, to create a virtual computer that can run its own operating system and applications. By sharing hardware resources with each other, multiple operating systems can run simultaneously on a single physical computer. By eliminating dependencies between the physical hardware and the software, customers have more choices to manage their applications, servers and equipment. One of the benefits is the ability to move virtual machines between host computers, which enable a variety of different fail-safe scenarios to be implemented, each providing options for different levels of redundancy that make systems more resilient, less prone to equipment failure, less prone to site failure and simpler to upgrade.

"At OpsManage'11 last year, we set up and demonstrated a scenario where an entire primary system in Florida failed over to a backup disaster recovery system in California," says Steidinger. "Customers were impressed with the ease and speed with which the backup system took control, using only commercial hardware. That's the flexibility they are looking for to modernise and optimise their business. As the first industrial automation provider to be certified for high availability, disaster recovery and fault tolerance on the two major virtualisation platforms, we look forward to continuing to offer the products our customers need to achieve the highest levels of compliance, collaboration and operational excellence."

For more information contact Jaco Markwat, Invensys Operations Management, +27 (0)11 607 8100, tech@invensys.co.za, www.iom.invensys.co.za

AC/DC DYNAMICS STARTS APPRENTICESHIP PROGRAMME



January 2012 sees the introduction of an apprenticeship programme at AC/DC Dynamics in Edenvale. The picture shows the 22 apprentices in the class of 2012, along with their manager Brian Pillay, at the issuing of the tool kits ceremony. Eight ladies are on the team, which is encouraging to see in the usually male dominated electrical engineering sector. The apprentice courses include electrician, electronics mechanician and armature winder training.

For more information contact Roger Burrows, AC/DC Dynamics, +27 (0)10 202 3300, roger@acdc.co.za, www.acdc.co.za

REGISTRATION OPEN FOR EMERSON'S GLOBAL USERS EXCHANGE IN EUROPE

Online registration is now open for Emerson's first Global Users Exchange in Europe (www.emersonexchange.org/emea). The three-day event, aimed at existing and potential users of Emerson Process Management products and services, is being held from 29-31 May at the Hotel Maritim, Düsseldorf, Germany. Delegates can either book for the full conference programme or choose to attend specific days based on the agenda of events.

Delegates booking the full conference will have access to all conference presentations, courses and technical exhibition, meals, exhibition reception and networking events. In addition, delegates that register before 30 March will benefit from a reduced 'Early Bird' registration fee.

"Interest in the European Global Users Exchange has been high and the response to the call for presentations has exceeded our expectations with over 300 abstracts being received," said Bob Sharp, president, Emerson Process Management Europe. "We know that delegates will be looking for ways to improve energy efficiency with enhanced automation, meet new regulatory requirements, increase yields, or simply get more out of their automation investments. Many of the presentations will follow these themes."

Tailored to meet the needs of users in Europe, the Middle East and Africa, the three-day conference will start at midday on Tuesday, 29 May and finish on Thursday, 31May. Delegates will be able to choose from around 100 themed presentations with some being translated into German and Russian. The programme also includes industry forums, technology exhibits on both evenings and networking events where delegates can meet fellow automation professionals and process industry experts.

Shaped by users, for users, the Emerson Global Users Exchange provides a platform for customers to share their learning by delivering presentations to their colleagues. The event is managed by a 'Board of Directors', comprising users from a range of industries who shape the conference content and guide the direction of the users group at large.

For more information contact Mark Tapson, Emerson Process Management, +27 (0)11 451 3700, mark.tapson@emerson.com, www.emersonprocess.com

THE PATH TO PERMANENT ENTERPRISE EFFICIENCY

As the global community unites to fix the emission mistakes of the last 50, Schneider Electric South Africa has a clear view of the energy challenge. "We need new solutions, new thinking and new companies to lead us into an era in which we can truly do more while consuming less," says Gys Snyman, vice president energy efficiency at Schneider Electric South Africa.

The solutions to the energy challenge are multiple and complex. In most cases, they rely on a stronger and more efficient collaboration between key stakeholders, including facility and security managers, IT managers and machine or production managers. "Requisite levels of business efficiency involve system dynamics across platforms and providers like never before," adds Snyman.

Doing more with less is a familiar mantra at companies in South Africa and around the world, especially when it concerns energy use. Managers at all levels are looking for meaningful ways to reduce energy consumption and boost performance for the sake of the environment as well as the bottom line.

According to the World Energy Outlook 2009, as energy demands escalate from all aspects of commerce and culture, the most acute pain point is that due to intrinsic inefficiencies, 33 units of energy consumed at the point of use require 100 units of primary energy – the coal used to supply that energy unit. "It follows that the war on waste will deliver the best returns for business," says Snyman. "But before you can drive business efficiencies and save energy, you have to be able to see it, that's where solutions from Schneider Electric add value.

"Our EcoStruxure Active Energy Management architecture is Schneider Electric's response to the energy challenge as it turns a business into an efficient enterprise. That is, the solution is our method to create intelligent energy management systems that are simple, save money, and most importantly, reduce waste by enabling guaranteed compatibility between five critical domains: power, process and machine, IT room, building and security management.

"It is a proven methodology that empowers organisations to identify potential savings through increased visibility of their energy consumption, electrical equipment benchmarking and estimated ROI reporting. As a comprehensive, integrated approach designed for the digital economy, it can deliver capital and operational savings of up to 30 percent."

For more information contact Jacqui Gradwell, Schneider Electric SA, +27 (0)11 254 6400, jacqui.gradwell@schneider-electric.com, www.schneider-electric.co.za



INSTROTECH TO REPRESENT BEANAIR

French wireless sensor networks company, BeanAir, is now represented locally by Instrotech. BeanAir's technology is based on IEEE 802.15.4 & GSM/GPRS wireless protocol and is present on advanced WSN applications auch as embedded measurement, engineering structure and technical building management, industrial process as well as isolated sites and harsh environments.

The company manufactures wireless sensors for both dynamic (accelerometer, strain gauge) as well as static measurement (temperature, force, tilt). BeanAir's non-intrusive, low power wireless measurement offering comprises:

- Humidity Wireless Sensor : Measures humidity, relative humidity, moisture within porous elements in order to ascertain the air quality inside the building
- Wireless temperature sensor: measurement of ambient air temperature, surface temperature, find heat losses and reduce internal heat loads.
- Wireless electrical measurement device: current loop, voltage measurement of peak power consumption and power leak detection.
- Wireless flow sensor: detects water leaks in pipes.
- Wireless pressure sensor: measures atmospheric pressure, liquid pressure, detect and measure airflow and locate water leaks in pipes.
- Computer system: display and configure wireless sensors network from a Web server, real time wireless sensor monitoring.
- The wireless sensors coordinator: builds and supervises the wireless sensor networks.

"Instrotech's representation of BeanAir is a great addition to our service offering. There is synergy between our corporate philosophies of service delivery and technical excellence that makes us proud to represent BeanAir in southern Africa," says Paul Mason, Instrotech's managing director.

For more information contact Pieter Deysel, Instrotech, +27 (0)11 462 1920, info@instrotech.co.za, www.instrotech.co.za



VEGA HOSTS TECHNOLOGY MORNING



John Groom (l) and Gunter Kech.

Representative from industries across the manufacturing and production spectrum were able to experience Vega's technology first hand at a recent product training morning held at the Roodepoort Country Club. The day was hosted by Vega's South African subsidiary Vega Instruments SA. On hand to introduce the company and the technology were Vega executive managing director Günter Kech and product manager Florian Burgert, Vega Instruments SA was represented by MD John Groom and national sales manager Clint Viviers.

Burgert introduced delegates to the different housings available in the Vega range, including suspension pressure transmitters, differential pressure sensors and sensors using ceramic-capacitive measuring cells. Depending on the applications, housings are available in plastic, aluminium and stainless steel, all double chambered for extra protection. Live demonstrations illustrated the qualities of the ceramic Certec cell and remote configuration using the VegaConnect technology.

Viviers introduced delegates to Vega's wireless communication technology. PlicsMobile is based on the GSM/GPRS standard and reduces on-site service visits through remote diagnosis and teleservice. Its application is ideal for mobile silos, deep wells, high containers, river and sea levels, waste oil collection sites and material recycling facilities.

NEW CAPABILITIES

Groom expanded on Vega Instruments SA's new capabilities and renewed vigour now that the January 2012 move to the Honeydew premises is complete. The expanded service department is able to carry out 90% of instrument repairs within 7 working days and new machinery installed will allow for the local manufacture of flanges, process components and sockets. "We are in the business not only of selling instruments, but also selling ourselves as relationship builders and trustworthy and reliable partners. Our goal is to enable our customers to get on with the business of running their plants to the fullest capability."

For more information contact John Groom, Vega Instruments SA, +27 (0)11 795 3249, john.groom@vega.com, www.vega.com

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By devising the SINUMERIK 828D, we have managed to develop a panel-based CNC that is tailor-made for use in milling and turning machines. The product combines CNC, PLC, operating and axis control functions for six CNC measuring circuits into one compact unit, which is designed to fit any operator panel housing. In addition, the SINUMERIK 828D can understand other CNC programs from other manufacturers due to the online ISO dialect interpreter.

Maintenance friendly - no battery (durable data buffering by NV-RAM technology!), no fan, no hard disk

User-friendly - full QWERTY keyboard, mechanical keys with protection foil, IP65

Communicative - front interfaces IP65, RJ45 Ethernet, USB 2.0, CF-Card

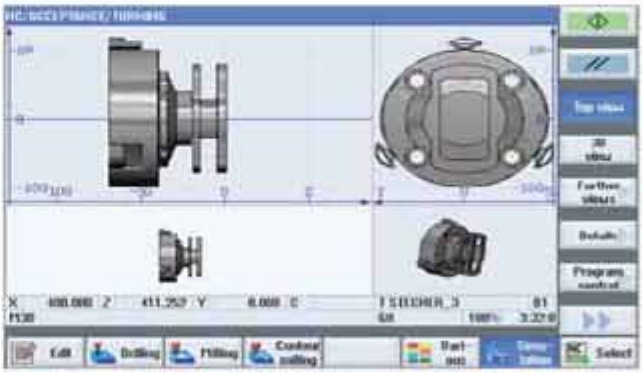
Clever - 3/8" threads for optional devices

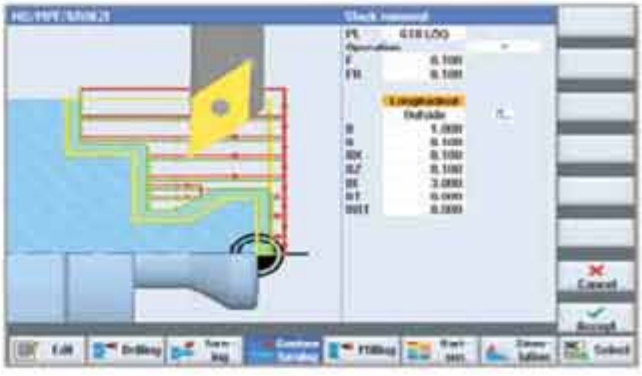
Robust - panel front made of magnesium die-cast.

For more information please call 011 652 3640









Totally Integrated Automation

CHEVRON REFINERY UPGRADE BY SERA DOSETECH

sera DoseTech South Africa was recently selected as the chemical dosing pump supplier for Cape Town's Chevron Refinery demineralisation boiler feed water plant upgrade.

The demineralisation process requires raw water to flow over resin in a cation and anion column. The minerals are attracted to the resin stripping them from the water ensuring that the water entering the boiler tubes is mineral free. This is important to prevent scaling and corrosion of the boiler tubes which will result in poor heat transfer and tube failure.

In the process the demineralisation resin needs to be regenerated (the minerals need to be removed from the resin). This process is carried out by dosing 98% sulphuric acid into a water stream, the diluted acid is then pumped through the resin, stripping the minerals and cleaning the resin. Once the resin is clean, caustic soda is dosed into the water stream, the diluted caustic is then pumped through the resin to rinse and neutralise the acid water, before returning the demineralisation plant back into service.

The boiler feed water reliability is critical to the refinery's production reliability. With reliability as the key focus, sera dosing pumps were chosen as the preferred dosing pump supplier. The model R410.2-940 ML pump was installed for both chemical applications and the multilayer diaphragm design is perfect for the extremely corrosive sulphuric acid, with the multi layer diaphragm the pump gearbox will be protected should the diaphragm fail.

For more information contact Wally Feldon, sera DoseTech SA, +27 (0)11 397 5120, wally@sera-web.co.za, www.sera-web.co.za



CONFIGURED DRIVE PACKAGES FROM ROCKWELL AUTOMATION

To help smooth out the process of specifying and ordering AC drives and related ancillaries, Rockwell Automation has introduced a configured drives service designed to fulfil the needs of a wide variety of industrial applications, saving customers time and effort. The options are designed to add functionality or features, in line with the demands of the end application.

"Although there is still room for an off-the-shelf product offering, we believe that we are able to match the needs of our customers more closely with the configured drives service. This service provides customers with peace of mind, as they can rest assured that all chosen components and options will work together in harmony," says Adrian van Wyk, business manager components, Rockwell Automation.

This configured drive approach allows OEMs or end user engineers to choose a base drive package and then enhance it with numerous options in line with the needs of the application. The beauty of this approach is that many of the combinations offered in the configuration process are already installed and working in unison in industrial applications across the world, thus removing the headaches associated with a normal mix-and-match approach.

The 'starter package' includes a basic configured drive, input fusing, RFI filter and enclosure. Options are designed to add functionality or features, in line with the demands of the end application. These options can include power features such as circuit breakers, isolators, input, output and bypass contactors and line reactors.

There are various control and communications choices available, which include control interfaces (analogue and digital) and serial communications (such as DeviceNet and ControlNet). The operator interfaces that can be utilised include HMIs, door-mounted devices and meters, while motor interface options include encoders, resolvers, thermistors, RTDs, cooling blowers and space heaters.

All packages are supplied in cabinets, with either IP21 or IP54 ratings, and are supplied complete with control circuit transformer and cooling fans appropriate for the frame size. All configured drives are assembled and tested in accordance with ISO9001 and comply with the relevant EEC directives.

For more information contact Adrian van Wyk, Rockwell Automation, +27 (0)11 654 9700, avanwyk@ra.rockwell.com, www.rockwellautomation.co.za



HONEYWELL INTRODUCES COMPREHENSIVE CONTROL SOLUTION



Tightening capital budgets and the need to maximise revenue are driving the process industries towards unified controller platforms for automating plant operations. Not only does a unified platform ensure Common Industrial Protocol (CIP) support for increased network efficiency and reduced network engineering time, it also confines costs to one common, scalable hardware suite and software package. An added advantage is that it gives the control designer a consistent look and feel across the automation disciplines. The alternate option of using separate best-in-class controllers increases costs and makes coordination between processes a challenge.

The MasterLogic PLC from Honeywell is a unified PLC platform backed by a powerful and versatile CPU with complimentary software – the SoftMaster. The PLC is modular and scalable with online maintenance and troubleshooting. SoftMaster provides the engineer with an integrated PLC engineering environment – an all-in-one window for ladder programming, configuration/ setup of CPU or other special/communication modules, debugging, monitoring, troubleshooting, documentation and maintenance. A vast library of standard function blocks allows the engineer to create new user-defined function blocks.

The MasterLogic PLC is an excellent choice for automation requirements where the primary mode of control will be through high-speed logic. Facilitated by a powerful processor, the MasterLogic CPU provides high-speed execution of program instructions at the rate of 42 nanosec/step. Backplane communication data transfer is facilitated by the 7 MB program memory, 2 MB data memory, and 16 MB built-in flash memory. In addition, dedicated intelligent communication modules (Ethernet, Serial, Profibus) offer co-processing assistance to the main controller CPU.

This advanced controller can be used anywhere efficient and comprehensive solutions are required. In southern Africa, the MasterLogic has been successfully applied in the mining, metals and minerals industry; food and beverage as well as in the sugar sector with stand alone boilers.

For more information contact Mohammed Sali-Ameen, Honeywell Southern Africa, +27 (0)11 695 8000, mohammed.sali-ameen@honeywell.com, www.honeywell.co.za

INDUSTRIAL AUTOMATION CONFERENCE – JOHANNESBURG 22–24 MAY

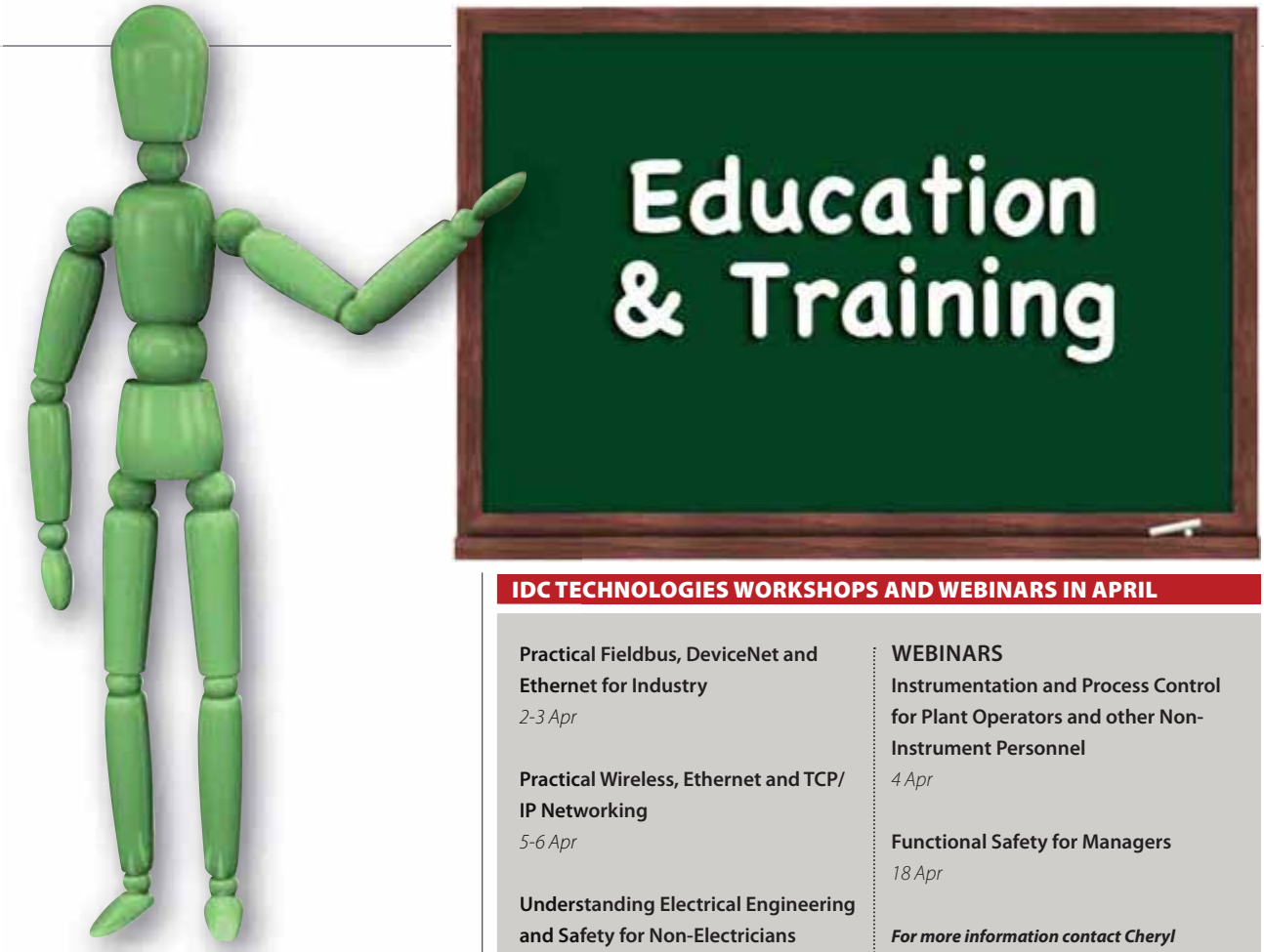
IDC Technologies has selected an outstanding line-up of speakers for the Industrial Automation Conference 2012 to be held in Johannesburg on 22, 23 and 24 May. All of the speakers will be using practical case studies to illustrate their real life experiences with industrial automation technology. All of the speakers will be using practical case studies to illustrate their real life experiences with industrial automation technology.

Need to be armed with practical and competent advice in order to make the correct decisions? This conference will be of interest to any organisation that relies on industrial automation for growth and business survival. Tactical not theoretical solutions are the key to success. Topics include:

- **Distributed network protocols – which one to use? (DNP3, IEC60870 and IEC61850).**
- **Network audit case study.**
- **Redundant load balancing for industrial networks.**
- **Taking industrial Ethernet networks to the next level – design and installation principles.**
- **New innovations in scada & PLC.**
- **Industrial wireless technology.**
- **Artificial intelligence – leveraging relationships.**
- **Secure OPC process data transfer and centralisation with minimal bandwidth usage.**
- **Ethernet network redundancy in a VLAN segregated environment.**
- **Increasing demand for reliable wireless radio connectivity between RTUs and scada.**
- **Leading-edge technologies enable a new generation of intelligent plant floor operations.**
- **Cyber security in a modern process network.**
- **Choosing the right scada system.**
- **Remote monitoring of critical plant assets.**

For more information contact IDC Technologies, +27 (0)11 024 5529, sarah.montgomery@idc-online.com, www.idc-online.com





FESTO DIDACTIC COURSES IN APRIL

Pneumatics (1) Basic PN111:
3-5 Apr **Durban**
18-20 Apr **East London**
24-26 Apr **Jhb**
28-30 Apr **PE**

Pneumatics (2) Maintenance PN121:
2-4 Apr **PE**
3-5 Apr **Jhb**
24-26 Apr **CT**

Electro – Pneumatics PN211:
11-13 Apr **Jhb**

Hydraulics (1) Basic HY511:
11-13 Apr **Jhb**
18-20 Apr **Dbn**

Hydraulics (2) Advanced HY521:
24-26 Apr **Jhb**

Process Instrumentation PA211:
18-20 Apr **Jhb**

Maintenance Strategies TPM LP141:
16-17 Apr **Jhb**

For more information contact Helen Pretorius, Festo, +27 (0)11 971 5586, didactic@festo.co.za, www.festo.co.za

IDC TECHNOLOGIES WORKSHOPS AND WEBINARS IN APRIL

Practical Fieldbus, DeviceNet and Ethernet for Industry
2-3 Apr

Practical Wireless, Ethernet and TCP/IP Networking
5-6 Apr

Understanding Electrical Engineering and Safety for Non-Electricians
9-10 Apr

Practical Distributed Control Systems (DCS) for Engineers & Technicians
12-13 Apr

WEBINARS

Instrumentation and Process Control for Plant Operators and other Non-Instrument Personnel
4 Apr

Functional Safety for Managers
18 Apr

For more information contact Cheryl Reyneke, IDC Technologies, +27 (0)11 024 5520, cheryl@idc-online.co.za, www.idc-online.com

MICHAEL BROWN'S PRACTICAL CONTROL COURSES

Michael Brown is holding another series of his well known courses on practical process control. The courses are unique and invaluable to novices, as well as to experienced practitioners in the field of regulatory control optimisation.

The series consists of two courses:

- Introduction and appreciation of practical process control: 1 day for non-instrument personnel who work in process plants, but have no previous experience or practical training in control, such as: process engineers; operators; production managers; metallurgists and chemical engineers. It is also recommended that people who are attending the Parts 1+2 the following week should attend the Introductory.
- Practical Process Control Parts 1 & 2: 5 days for instrument and control personnel like mechanicians, technicians and engineers. Even people with years of experience in this field have found the course an eye-opener. Subjects covered include: loop problems and troubleshooting; tuning and control of processes with difficult dynamics and techniques like feed-forward. (It should be noted that the two parts have been combined into one session, and may no longer be taken separately.)

The first course series will take place in Johannesburg on Friday 13 April at a cost of R2290-00 and the second will run from Monday 16 to Friday 20 April at a cost of R13 117-00 (prices excl. VAT). Booking should be made early as the courses are often oversubscribed.

For more information contact Michael Brown Control Engineering, 082 440 7790, michael.brown@mweb.co.za, www.controlloop.co.za



MAGNET OPENS A TRAINING DIVISION

Suppliers of electrical equipment and industrial instrumentation, Magnet, recently opened a specialist training division. "This is an exciting development for the electrical sector where there is growing demand for practical training in product selection, energy efficiency, installation, repairs and safety inspection," says managing director Brian Howarth. "These training programmes will cover all areas in the industry, including electrical installations in hazardous locations."

The company has appointed Brian Bilton to head up this new department. Brian is a former Durban Electricity Department installation inspector. Magnet will offer 'Certificate of Compliance' courses, encompassing the requirements for verification and certification contained in SANS 10142, and SANS 10108.

A full calendar is scheduled for 2012, with courses to be run at Magnet branches in Durban, Richards Bay and Johannesburg.

For more information contact Brian Bilton, Magnet, +27 (0)31 274 1096, brianb@magnetgroup.co.za, www.magnetgroup.co.za

OMRON TRAINING IN APRIL

DURBAN

Introduction to Drives 1 (ID1)
2 Apr

JOHANNESBURG

PLC introduction training (PL1)
16-17 Apr

Enhanced PLC training (PL2)
18-20 Apr

Advanced PLC training (PL3)
11-12 Apr

Human Machine Interface (HM1)
23-24 Apr

For more information contact Dave Gould, Omron Electronics, 086 066 7661, dave.gould@eu.omron.com, www.industrial.omron.co.za

APPOINTMENTS

AC/DC Dynamics has appointed Quintin Tesselaar as Gewiss product specialist.


Quintin Tesselaar.

APC by Schneider Electric has appointed Lucian Reynard as industry account manager for South Africa.


Lucian Reynard.

Raegan Ramsunder has been appointed field sales engineer at ifm electronic's head office in Centurion.


Raegan Ramsunder.

Christopher Kidson has been appointed field sales engineer at ifm electronic's head office in Centurion.


Christopher Kidson.

Ralph White has been appointed field sales engineer at ifm electronic's head office in Centurion.


Ralph White.

Cube Technologies has appointed Tanis-Leigh Williams as national sales and training executive sub-Saharan Africa.


Tanis-Leigh Williams.

Skills shortage: a multifaceted approach is required

Honeywell Process Solutions recently hosted a panel debate on the shortage of engineering skills. Held during its 2011 Users' Group Conference (HUG) for Europe, Middle East and Africa (EMEA), the panellists and audience debated numerous causes and solutions for the problem, including: gender disparity, age, government intervention, corporate training programmes, quality of education, and motivation of young people into the field of engineering.

By Graeme Bell

The full debate can be viewed online (video or audio podcast) at http://instrumentation.co.za/+c16219c.

Some highlights from the debate include:

Alberto Leni (Saipiem)

Reacting to the relatively low percentage of women present in the audience (with no women represented on the panel), Leni said, "In southern Europe, things are improving. When I began in the mid '90s I estimate that only 1-2% of engineers were female. Now things are a little bit better … we have 7-8% female engineers.

Neil Stander

Is our tertiary education of engineers correct? Is there enough practical during the initial stages of their training to get them up to speed quicker?

Ahmed Abdullah (GPIC)

On the subject of age, "In the Arab Gulf countries approximately 40% of our engineers are expatriates from Europe, mainly from Asia, India and so on. Of the expatriates, 80% of them are aged above 48. With the local engineers, the majority are aged between 30 and 40.

Helen Tunnicliffe (TCE)

On the subject of whose responsibility it is to attract more students to the subject of engineering?: "Why are the engineering firms so reluctant to take on graduates to train them up? Other professions don't have that problem. If you take the law profession, the big firms will take people on and train them. Accountancy is another. These professions readily accept that that is going to happen… they accept that the movement of graduates and younger employees is quite fluid. Why do engineering firms not accept this?"

Mohammed Saleh (ZADCO)

"I was attracted to engineering first of all because I liked engineering. Then the oil companies in Abu Dhabi, sponsored high school students (like myself) to attract them to the engineering discipline. They pay for our tuition



Graeme Bell: Editorial Director, Technews Publishing.

fees. Of course in return for that we have to work for a couple of years. That approach will help solve the problem.

Ursula Kok (BP Belgium)

"As part of my job as an engineer, I am also a schooling co-ordinator. I go to primary and secondary schools to promote to them, what it means to be an engineer.

"I usually start the session with the kids by asking them, "Do you know what a doctor does?" They can always answer correctly. "Do you know what a journalist does?" Also, no problem answering. Do you know what somebody in a bank does? They can answer that too.

"But as soon as I ask them "What does an engineer do? I usually get answers such as: 'They fix engines' or 'They change light bulbs.' The children and young-adults just don't have a clue what engineering is about. Many of these 'kids' are just a few years away from applying for a place in a university.

"Young people are not attracted to engineering simply because they don't know what engineering is. I think the industry is not making enough effort to advertise to younger kids what it's like to be an engineer. I think we're trying to target a lot at university level, which is good, but I think we should do more work earlier on in … even primary school level and secondary school to try make engineering look sexy to young people."

Frank Whitsura (Honeywell Process Solutions)

With respect to technology-based solutions to the skills shortage, "The traditional way of doing engineering service for the last 100 years has been, 'you bring a person to where the problem is, either on a horse or a train or a plane'. With modern technology we can go a long way to mitigating some of the skills shortage by using technology tools like: remote collaboration, remote tools, embedding knowhow into systems, and shared knowledge databases. I don't think it can ever be the complete solution, but I think some of the technology tools we have available today can help mitigate that crisis."

For more information contact Honeywell ACS South Africa , +27 (0)11 695 8000, hsa@honeywell.com or www.honeywell.co.za

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T26

From the President's desk



Johan Maartens.

2012 is a year of great beginnings for the SAIMC. Firstly, I would like to congratulate the new branch committee members and thank them for their willingness to serve. I would also like to thank last year's committees for their valuable contributions and excellent leadership.

We have just signed a Memorandum of Understanding with the Automation Federation making us part of a world-wide automation group. The interesting thing is that they have also been working on bringing school children into our industry, as we are trying to do. As part of the strategy for improved delivery this year, the following committees were established at the first council meeting:

1. Training and development (incl. Grant Awards committee)
   Chairman: Solly Mabitsela
   Members: Johan van Jaarsveldt, Nico Marneweck, Kevin McElroy, Selvan Murugan
   Activities to report: Schneider award, Negretti award, bursary, list of training institutions, training schedule for publication.
2. Finance committee – not for publication on public forums
   Chairman: honorary treasurer
   Members: Johan van Jaarsveldt, administrator
   Activities to Report: financial figures for the AGM.
3. Public relations and project co-ordination committee
   Chairman: vice president
   Branding committee is now called the PR committee
   Activities to report: Branding progress.
4. Branch award committee (on LinkedIn and SAIMC forum)
   Chairman: president (or general secretary)
   Members: branch secretaries
   Activities to report: branch scores.
5. Innovations committee
   Chairman: vice president
   Members: branch chairmen to nominate
   Activities to report: 2012 goals, branding, membership value, website, bursary scheme, education & training.

COUNCIL ELECTION

It is that time of the year again. Just to make things clear, these are some rules that need to be followed:

1. Vice president takes over from president when the president steps down.
2. Council shall elect a new vice president at the last council meeting of the year out of the members serving on council.
3. New members can be elected by postal vote if necessary.
4. Current office bearers: Johan Maartens, Johan van Jaarsveldt and Debbie Scott.
   President, vice president and immediate past president, normally hold office for one year.
   Vivian MacFadyen and Peter Zietsman
   Treasurer and secretary hold office for two years or more.
5. Ordinary members: Vinesh Maharaj, Johan Grobler
   Each year three ordinary members shall retire, ordinary members hold office for two years, send nominations out by post 30 days before AGM, nominations received up to 7 days before AGM.
6. Branch chairmen: Ex officio members.
7. Retired council members: should wait two years before being elected on council again.
8. Voluntary association requirements: at least 50% of council must be registered with ECSA.

This last requirement can severely limit the choices of members being elected to council. If the branch chairmen are not registered with ECSA, it means that we need to make up for that by limiting council members to those registered with ECSA until the 50% target is achieved.

I would like to thank the current council for a wonderful job and appeal to all our members to nominate people to serve you at council level.

Secunda branch

The Secunda branch held its 2012 AGM on 19 January during which time the new committee was announced. Several members of the previous committee were re-elected and committed to continue serving the branch with the same level of professionalism they have in the past. The branch would like to congratulate all committee members on their new appointments.

Two long-standing committee members have handed over their respective portfolios: the branch would like to thank Fronnie Knoetze and Peter Zietsman for their valuable input and support through the years. Peter, who has been chairman for four years, will remain on the committee as an additional member to ensure the smooth transition of his responsibilities. Fronnie Knoetze has also made himself available to the branch should they require his services from time to time. These colleagues have added immeasurable value not only to the branch but to the SAIMC as a whole and they will be sorely missed. The branch would also like to thank Alpret Control Specialists and WIKA Instruments for the continued use of their facilities to conduct the committee meetings and Technology Evenings respectively.

TECHNOLOGY EVENINGS

On 2 February, the first Technology Evening of 2012 was conducted, hosted by Vega Instruments SA. Clint Viviers did a presentation on level measurement technologies and

highlighted the innovativeness of the various measurement principles, as well as the key points to remember during installation and commissioning. The branch is delighted by the number of new faces attending and looks forward to seeing you all on a permanent basis.

The new committee would like to invite all instrumentation and control engineering professionals to attend the Technology Evenings and share the wealth of knowledge they possess. Please also visit www.saimc.net where you will find the latest information on what is happening with the Secunda branch, as well as the SAIMC in general. Register yourself on the website if you are a member and please participate in the forum discussions.

Any queries regarding the Secunda branch can be directed to Kobus Reinecke at kobus.reinecke@sasol.com or Johan Maritz at johan.maritz@kentz.com.



The 2012 committee (l to r): Brian Edwards, Kobus Reinecke (chairman), Johan Maritz (secretary), Gerhardt Swarts, Thulani Shabalala, Peter Zietsman, Rickus Kriel, and Nick Strydom. Absent: John Phala and Dennis Hoffman (vice-chairman).

Durban branch

The Durban branch held its 2012 AGM on 1 February where secretary, John de Castro, ensured that proceedings ran without a hitch at this well attended event. SAIMC president, Johan Maartens, gave a brief overview of the goals of the SAIMC for the year ahead; branch chairman, John Owen-Ellis, delivered his report on the branch activities of the past year and treasurer, Dean Trattles, reported on the financials.

Forthcoming attractions: Save this date! The Durban SAIMC golf day is due to be held on 1 June at the Amanzimtoti golf course. As soon as the details have been finalised we will publicise them, for queries in the mean time please email Howard Lister on howard@nickzn.co.za



Johan Maartens flanked by vice chairman Vinesh Maharaj (l) and John Owen-Ellis.



For the second year, electronic voting took place in the weeks preceding the AGM and the new committee was duly announced. The new committee (front l to r): Kevin McElroy, Hennie Prinsloo (vice chairman), John Owen-Ellis (chairman), Dean Trattles (treasurer). (Back l to r): Rob Moxham (new member), Eddie Mohlmann, John de Castro (secretary), Jane van der Spuy, Howard Lister and Mark Calvert (new member).

www.saimc.net
Secretariat SAIMC, Tel: 08610 SAIMC, +27 (0)11 431 3802 (international callers only), Fax: 086 616 0141, admin@saimc.net

Johannesburg branch

The AGM for the Johannesburg branch was held on 8 February and attended by over 40 members and guests. The financial statements were presented followed by feedback from the 2011 Chairman, Debbie Scott. During 2011, the committee organised the following activities:

- February – AGM.
- March – Tony Rayner – Phoenix Contact – Lightning & surge protection.
- April – Malcom Heathfield – Adroit Technologies – Intelligence from data.
- May – Thierry Logeais - Nexans Cables – Cables in unfriendly environments.
- June – Gary Friend – Chairman FFSAMC– Overview of Foundation Fieldbus.
- July – Ralph Hauselmann – Buhler – Future of automation.
- August – Lita Mpahlwa - Endress+Hauser – Level measurement.
- September – Dr. Kelvin Klemm – Nuclear power.
- Oktoberfest – Brazenhead, Fourways.
- November – Barnyard Theatre, sponsored by the JHB branch.

A site visit to Unilever in Boksburg was organised in September and the annual Golf Day was held at Kempton Park in August. The SAIMC was represented at the Process Expo and the Control Roadshows in Windhoek, Walvis Bay and Vanderbijlpark. The organisation was a co-sponsor of the International Filed Networking Conference held in November.

The Schneider Awards were presented to Tjego Thejane and Bjorn Matthews from the University of Pretoria. Many thanks once again to Schneider Electric for their ongoing financial support.

The committee for 2012 is as follows:

- Alvin Seitz- chairman.
- Johan van Jaarsveldt – treasurer.



President of the SAIMC, Johan Maartens, with outgoing chairman Debbie Scott and 2012 chairman Alvin Seitz.

- Eric Carter – secretary.
- Andrew Bharath – Media liaison.
- Gary Friend – education.
- Roxanne Bredenkamp – events.
- Debbie Scott.
- Jay King.
- Syd Thoresson.

We look forward to welcoming you all to our Technology Evenings at 18h00 on the second Wednesday of every month at the Rivonia Recreation Club in Achter Road, Rivonia.

SAIMC

COHIBENDO PROFICERE

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Durban: **Vinesh Maharaj at vmaharaj@cubetech.co.za, Mobile: +27 (0)83 792 6518**
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Zambia: **Rodgers Kayombo at rodgers.kayombo@kcm.co.zm**
Council: **Ina at Admin@saimc.net, Mobile: +27 (0)82 440 8957**

Control 2010 Top 50 – Pinto summary



JIM PINTO

Jim Pinto is an industry analyst and commentator, writer, technology futurist and angel investor. His popular e-mail newsletter, JimPinto.com eNews, is widely read (with direct circulation of about 7000 and web-readership of two to three times that number). His areas of interest are technology futures, marketing and business strategies for a fast-changing environment, and industrial automation with a slant towards technology trends.

Once again Control magazine has done a great job on its published list of Top-50 Automation & Control suppliers – with separate rankings for global and N. America. I provide a review and brief analysis of the annual 2010 industry results from my own perspectives.

Here's a table-summary of the Industrial Automation Top-10 rankings, for Global and North America.

	Global		North America	
	Rank	$B	Rank	$B
Siemens	1	12.9	5	1.41
ABB	2	10.9	3	1.62
Emerson	3	6.5	1	3.33
Rockwell	4	4.4	2	2.23
Schneider	5	4.4	7	0.91
Mitsubishi	6	3.5	30	0.14
Danaher Industrial	7	3.2	4	1.58
Yokogawa	8	3.0	20	0.26
Honeywell Process	9	2.8	8	0.72
GE	10	2.2	6	1.15
Omron	11	2.2	22	0.25
FlowServe	12	1.8	10	0.61
PHoenix Contact	13	1.8	19	0.27
Invensys	14	1.8	11	0.53
Endress+Hauser	15	1.7	16	0.34
Ametek	16	1.3	9	0.67

Emerson, Rockwell and ABB retain their ranks at the top in N. America, with the same ranks as last year globally. Siemens drops to 5 in NA, behind Danaher. Honeywell Process has dropped from 6 to 8 in NA, while Schneider and Invensys retain their ranks in both NA and globally. GE and Ametek move up the ranks.

Here are some of the companies I keep an eye on:

- Omron is 11 globally but only 22 in NA.
- Yokogawa is still only 8 globally and 20 in NA.
- Phoenix Contact has advanced to 13 with $ 1.8B global, far ahead of its old rival Weidmuller.
- Endress+Hauser is 15 globally and 16 in NA.
- National Instruments is 15 in NA and 22 global with $0.87B.
- Beckhoff is 42 global but not on the NA top-50 - only honourable mention at $35M in NA. Note: Good growth in 2011.
- Opto-22 has grown to $75M.
- China's Hollysys is at $94M – could be a growth vehicle for China.
- OSIsoft grew from 35 to 32 in NA, but still not on the global list, with an honourable mention at $204M global.

In just the last couple of months, the automation majors have announced significant increases in bookings and profits. The automation business is doing well – most of the Top 50 indicate that their projected sales funnels are full, perhaps into 2014 and beyond.

Many of the majors are hoarding cash because they are worried about the uncertain future. The acquisition binge hasn't really started because of this uncertainty, but it will begin soon enough. Stay tuned.

PINTO 2012 AUTOMATION PROGNOSTICATIONS

I have a hunch that 2012 will not be another same-old, same-old year for industrial automation. Expect opportunity niches to arise around industrial wireless, Ethernet, mobile interface technologies. Here is a summary of my technology-picks for 2012, as published in Automation World, January 2012.

Industrial wireless: Standards are settling down into reality. Major suppliers are reporting revenues in the tens of millions of dollars. New markets and applications are generating revenues and market acceptance.

Industrial Ethernet: In the automation environment virtually everything is being connected to everything else via central networks and the Internet. Automation systems based on standard network architectures are spreading everywhere. Enhanced industrial Ethernet protocols such as EtherCat, ProfiNet, Modbus TCP and EtherNet/IP will continue to proliferate.

Pervasive Internet: Today's centralised controls and displays will give way to peer-to-peer input/output (I/O) and the other ingredients of decentralised automation systems. Industrial I/O products will become increasingly more autonomous in functionality, with vastly more robust operation in systems that include literally millions of I/O points.

Portable operator interface: The old-style operator control console interface is giving way to access via ubiquitous, compact WiFi equipment. More diagnostics and service functions can now be viewed via mobile phones, with cheap but effective two-way audio and video visibility. Most automation companies are now offering advanced features and functions using iPad, iPhone and Droid apps. This trend will expand quickly.

There are several growth opportunities hiding in plain sight and entrepreneurs – the risk takers – will see them first. Perhaps, some of the recently unemployed will be the ones to jump at the possibilities.



PSY Systems

VARIABLE FREQUENCY INDUCTION FURNACE CONTROLS UPGRADE AT ZIMALCO.

Zimalco, a division of the Zimco Group, is a leading South African producer of industrial and base minerals. Its products are used, amongst other applications, to dope bulk aluminium products to change the structure from brittle to soft. The product therefore needs to meet the highest industrial standards.

To ensure minimum losses when re-melting raw products, and for stirring requirements, Zimalco uses induction furnace heating based on resonant transformer action where the primary coil and capacitors are connected to a variable supply. The secondary coil is in fact the product to be heated and transformer configuration can be either in series or parallel. Induction furnace heaters have three main components: the furnace body, a capacitor bank to aid resonance and the power pack which supplies the furnace body with controlled power. These types of furnace have a wide power range; Zimalco's induction furnaces range from 650 to 1000 kW.

EFFICIENT OPERATION REQUIRES TIGHT CONTROL

According to PSY Systems' Paul Young, induction heating is an extremely efficient means of converting electrical energy to heat, but there is one golden rule that must be followed: the furnace must run very close to unity power factor. "To achieve this, the output frequency of the power pack has to run in tight control at the resonating frequency of the furnace load, commonly known as the tank circuit. This circuit consists of the furnace crucible, the product in the crucible to be heated and the capacitors across the furnace primary coil."

The existing controls required an exhausting number of operator settings to achieve the required control. In addition, production pressure exacerbated the fact that the operators were faced with the daunting task of overseeing multiple selector switches and control buttons. "We were approached to find a solution to the problem. In a nutshell, the team at PSY Systems needed to design a user friendly and efficient control system," says Young. "The power circuit architecture of Zimalco's furnace is current source inverter (CSI), whereas the incoming supply is rectified to DC via controlled bridge and feed to a large DC choke, which further supplies a single-phase inverter in a DC link configuration. The furnace coil and capacitors are parallel connected across the DC link as the load."

THE PSY SYSTEMS SOLUTION

Zimalco decided to keep all power circuits and only replace the controls. The new controls were based on the ABB AVANT AC800 series controller interfaced to two main control boards, one for the converter and the other for the inverter. "Due the maintenance records, which showed many failures from blow back into the control when any power device failed, it was decided to fibre link these control boards to the thyristers," says Young.

The two main control boards were designed and manufactured by PSY Systems. These boards also included all relevant I/O signals for their main functions, that is, converter board signals including phase synchronising, fibre SCR gate signals, current feedback, current limit set point and power signal set point. The inverter board signals included, fibre SCR gate signals, frequency to voltage converter, coil voltage feedback and frequency set point. Finally, both boards were designed to interface to standard process controller signals based on digital inputs, digital outputs, analogue inputs and analogue outputs.

Commissioning took place from two main engineering tools: a notebook computer with ABB's control builder software and an oscilloscope to monitor feedback and ensure tight control tuning. Process trails proved very successful with the furnace achieving record melt rates. Additional features were added such as the melt mode, which puts the furnace into high frequency and full power, as well as easy low-frequency selection for ultimate product mixing.

For more information contact Paul Young, PSY Systems, +27 (0)11 782 5449, paul@psysystems.com, www.psysystems.com

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SYSTEM INTEGRATORS

The full System Integrators tables can be found in the *2011 Instrumentation & Control Buyers' Guide* under the section Services (Industrial). Please contact Jane 031 764 0593, e-mail: jane@technews.co.za or Tina 011 543 5800, e-mail: tina@technews.co.za for more information on inclusion in this section.

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SYSTEM INTEGRATORS



Monitoring and optimising key performance indicators

Schneider Electric's Vijeo Historian process data solution improves production and reduces costs for optimised asset utilisation.

In today's challenging economic climate, the ability to capture real-time and historical operational data, and analyse and visualise that data in the right context, plays an increasingly important role in a company's success. With its ability to provide manufacturers with a plant's complete digital history, Vijeo Historian V4.30 – the information management layer of Schneider Electric South Africa's PlantStruxure – offers a powerful plant historian that supports the monitoring and optimisation of critical key performance indicators.

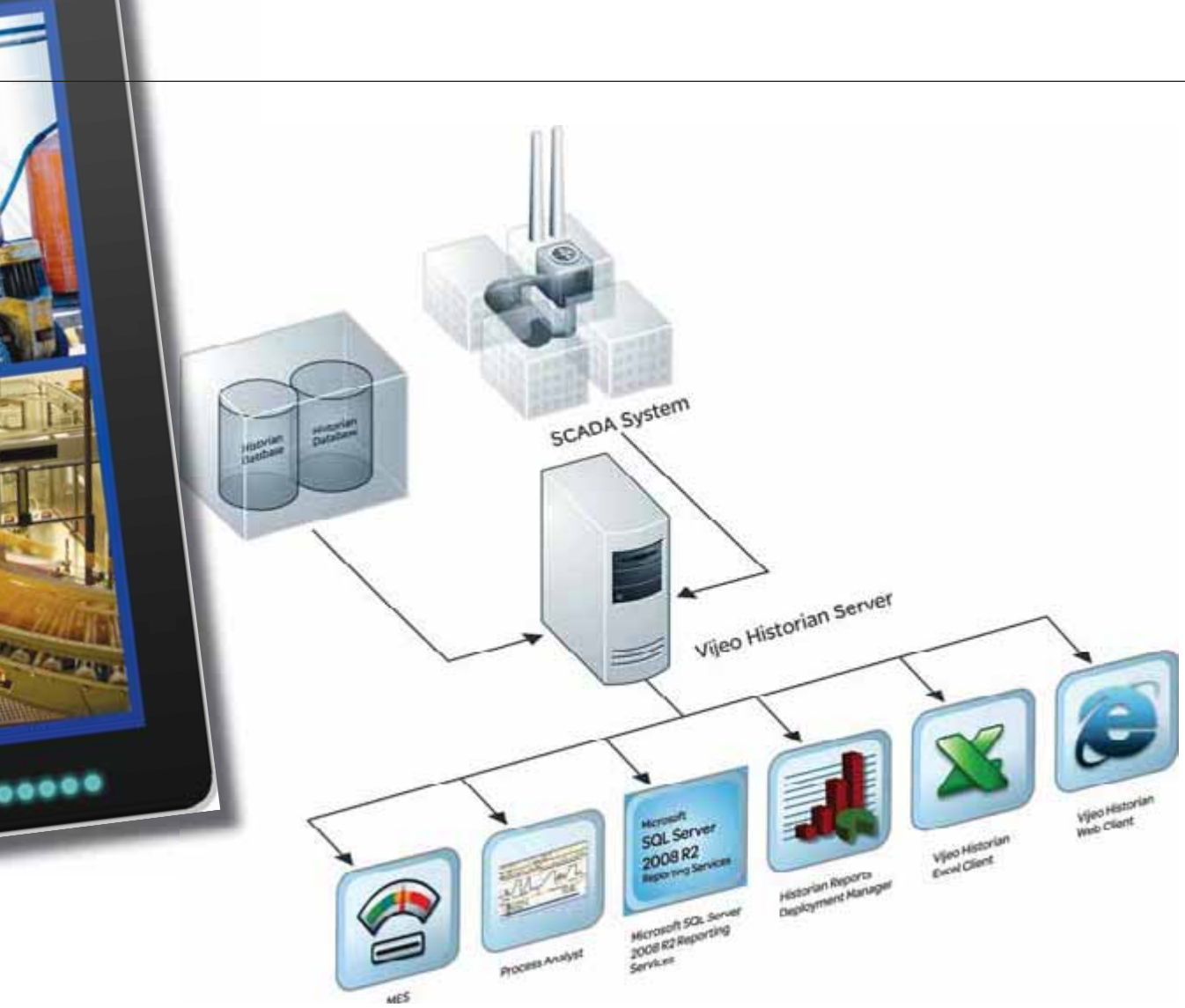
DATA ANALYSIS WITHIN THE PLANTSTRUXURE ARCHITECTURE

Cost reduction, meeting regulatory requirements and maintaining and improving energy efficiency are three areas in which industry is coming under increasing pressure. Quintin McCutcheon, Scada, MES and Telemetry commercial manager at Schneider Electric South Africa, points out that under these circumstances, industry is becoming increasingly dependent on plant historian technology and collecting process data is no longer a nice-to-have, but a necessity.

"Manufacturing and processing facilities that have invested in plant historian technology, are capturing large volumes of process information which becomes the main source of manufacturing intelligence data," says McCutcheon. "The simple to complex analysis of this information helps companies to make critical decisions on process performance management."

Schneider Electric South Africa is offering a robust, easy to use plant historian called Vijeo Historian V4.30, which enables industrial operations to view and analyse large volumes of plant data in real-time. Vijeo Historian constitutes the information management component of Schneider Electric's PlantStruxure, a collaborative system for process control and energy management.

McCutcheon says PlantStruxure integrates hardware and software components throughout the plant to deliver a complete process management solution. "This measured process and energy data can be analysed and the information used to holistically optimise the plant. New Microsoft platforms are increasingly being adopted



by hardware vendors and businesses and these enhance performance of plant historian software."

Vijeo HistorianV4.30 supports Windows XP, Windows Vista, Windows 7, Microsoft Windows Server 2003 R2, Microsoft Windows Server 2008 R2, SQL Server 2005 and the enterprise edition of SQL Server 2008 R2. The software is compatible with 32-bit and 64-bit operating systems and the enterprise edition also caters for data compression at about 43 percent on process data, which reduces the size of the database.

"The software offers insight into overall operational performance and improves business flexibility via integration with the latest technology platforms such as Windows 7," says McCutcheon. "It turns masses of data into meaningful information, giving plant management a complete picture of plant processes which enables improved return on assets and improved profit margin."

NEW FEATURES AND IMPROVED DIAGNOSTICS

The new version of the software provides enhanced reports using new chart controls available in SQL server 2008 R2 and offers Dundas-based charts, including advanced chart visualisations and other chart types. Plant-floor data can be presented in highly visual charts and customisable dashboards. McCutcheon says this ensures that each level of the organisation, from plant operators, production managers, engineers and quality assurers to upper management, have access to the information they need to make better decisions more quickly.

A historian web client feature offers a simple means of analysing alarms, events and trend data, providing a flexible means of comparing real-time and historical data in a single interface. The performance of the configuration manager feature has also been improved so that multiple reads are no longer required every time data is accessed.

From an energy management perspective, McCutcheon says plant historian software plays a key role in energy management. Capturing energy-related data allows users to better understand energy usage over time and determine if energy management procedures are being followed.

"Vijeo Historian is used to identify inefficient plant equipment and processes, diagnose anomalies in power consumption and make real-time adjustments to optimise energy efficiency and save power costs at equipment level, on production lines, across the entire plant and even on multiple sites."

For more information contact Quintin McCutcheon, Schneider Electric SA, +27 (0)11 254 6400, quintin.mccutcheon@schneider-electric.com, www.schneider-electric.co.za





Concentrated solar power

A KEY TECHNOLOGY FOR SUSTAINABLE POWER GENERATION IN AFRICA.

The average daily solar radiation in most African countries makes concentrated solar power (CSP) one of the most attractive renewable energy solutions to ensure a secure supply of clean electricity to power economic and infrastructure growth on the continent.

As a company, Siemens believes CSP is a key technology of the 21st century and is one of the world's leading solar power technology suppliers. "We are already the market leader in steam turbines for solar thermal power plants and, with the power block, we can offer a key part for solar power plants – the part that is responsible for power generation," says John Hazakis, head of solar and hydro at Siemens Southern Africa. Hazakis says that Siemens was the first steam turbine supplier to re-enter the CSP market in the 21st century, pioneering commercial solutions in the US and Spain. Today, Siemens is the world market leader in steam turbines for CSP plants, and has more than 20 years of experience with steam turbines for daily cycling.

With the acquisition of Solel in 2009, Siemens now boasts high-efficiency receiver technology and comprehensive expertise in the engineering and construction of solar fields. "We are able to offer the key components for the construction of parabolic trough power plants from a single source and to further enhance the efficiency of these plants, and this expertise is available for our customers in Africa," adds Hazakis.

"The roll-out of CSP in South Africa and the rest of Africa, is not only good for ensuring a sustainable source of electricity, it also offers huge potential for job creation and localisation," says Hazakis. "However, to create a sustainable industry more capacity needs to be allocated to CSP technology to ensure that the necessary investments can be made in local manufacturing and skills development."

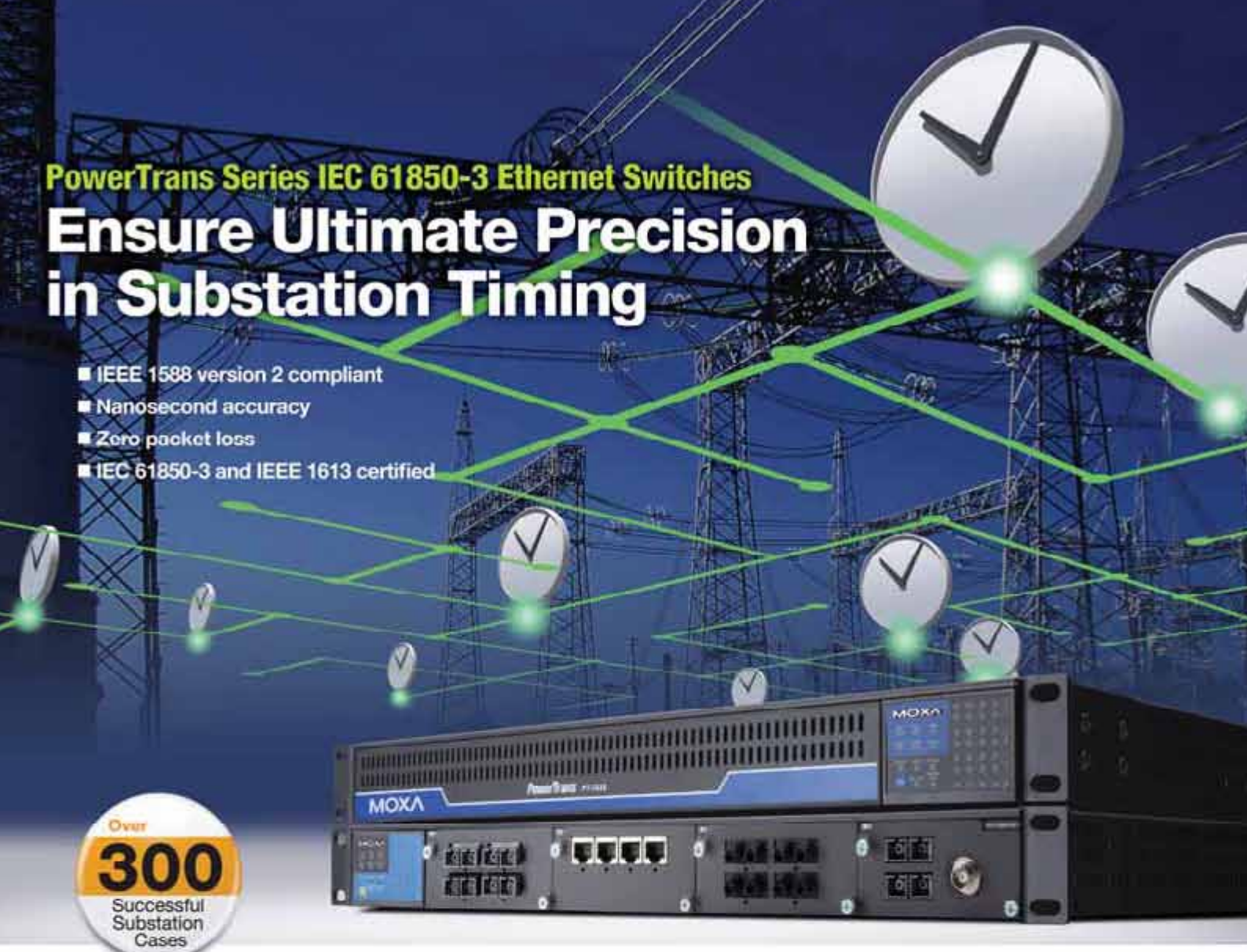
Siemens CSP power plants are very much like a standard power plant with an energy source, steam, and conversion of thermal power to electricity. In these plants the energy source is sunlight, focused by parabolic mirrors onto solar receivers located in the focal line of the parabolas. The entire parabola moves to track the sun and maximise the collection of solar energy. A fluid, typically a synthetic oil, flows through the receivers as a heat transfer medium. In the process the fluid is heated and its energy is transferred via a heat exchanger to water. The steam thus produced is used to generate electricity in a steam turbine-generator.

For more information contact Keshin Govender, Siemens Southern Africa, +27 (0)11 652 2412, keshin.govender@siemens.com, www.siemens.co.za

PowerTrans Series IEC 61850-3 Ethernet Switches

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- Zero packet loss
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The Moxa PowerTrans series is specifically tailored for power substations, and offers strong EMI immunity, a wide -40 to 85°C operating temperature range, and isolated redundant power inputs. The PT-7728-PTP series supports the IEEE 1588 v2 protocol to handle the synchronization requirements of future substation automation applications. The hardware time stamp allows highly precise time synchronization over the Ethernet network, delivering low jitter with accuracy exceeding 1μs.

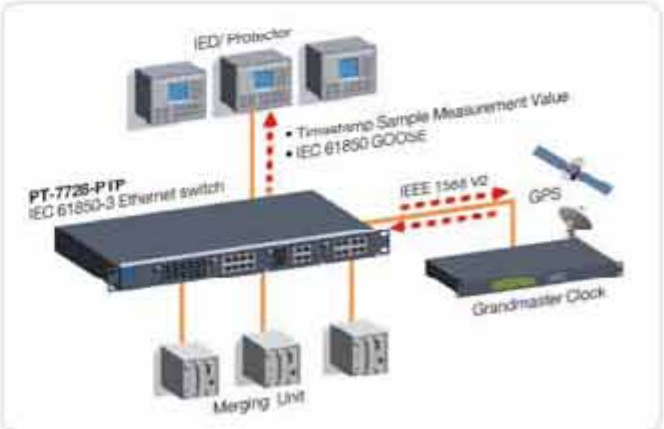
PT-7728-PTP Series

Up to 14-port IEEE 1588 v2 support



IEC 61850-3

Nanosecond Accuracy



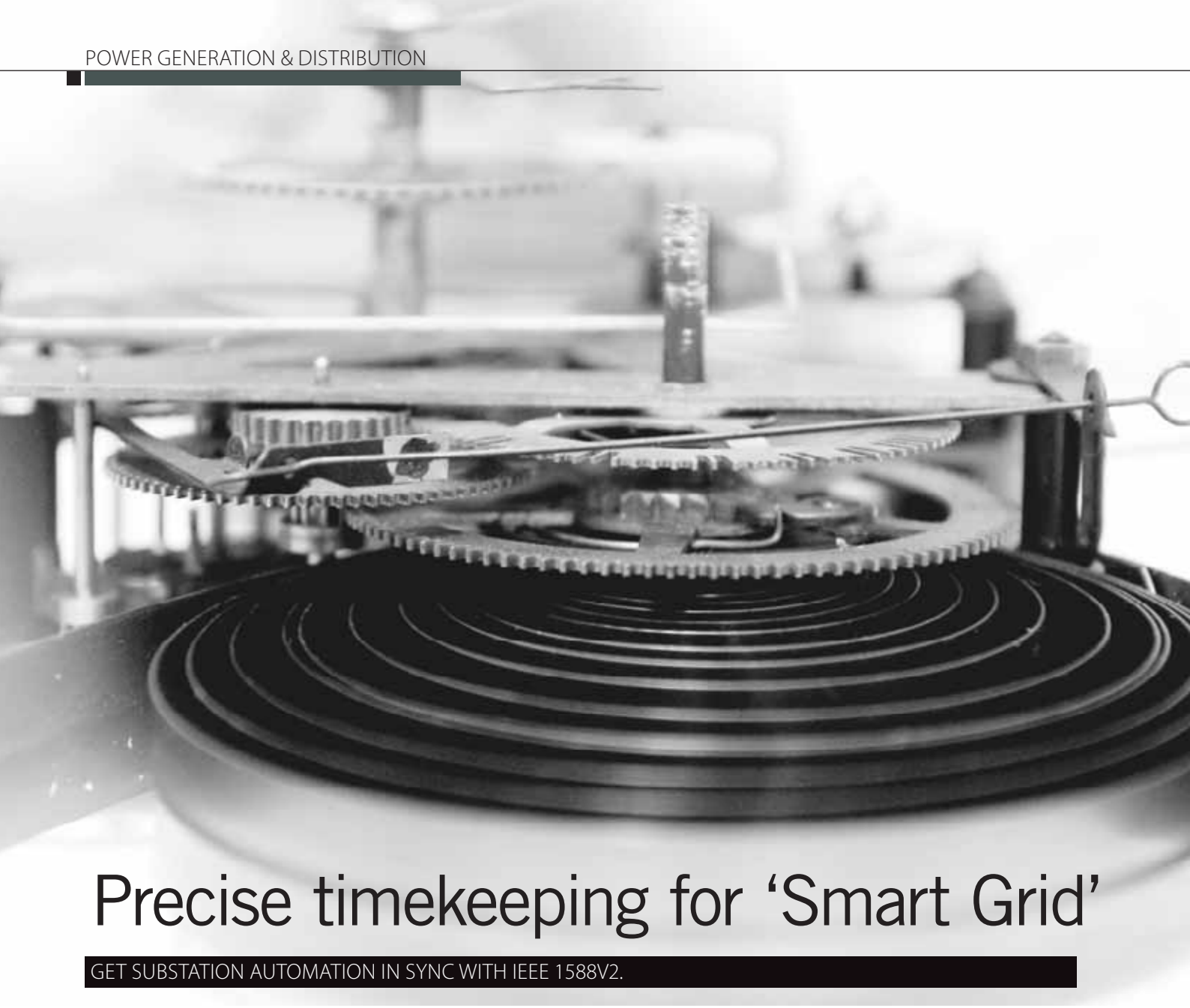
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Precise timekeeping for 'Smart Grid'

GET SUBSTATION AUTOMATION IN SYNC WITH IEEE 1588V2.

Justin Wu and
Bruce Chen, Moxa.

For industrial networks, accurate timekeeping offers advantages, this is especially true of power distribution networks, which need accurate timekeeping and network synchronisation in order to coordinate the activities of widely distributed equipment and regulate power transmissions. Higher levels of precision allow power grids to unlock their own revolutionary advantages: a 'Smart Grid' that can achieve new levels of efficiency, security, and reliability by autonomously and intelligently distributing power to end users in response to changes in demand.

This white paper explores some of the limitations that current industrial systems, especially power substations, face in synchronszing their networks. It also provides an overview of today's commonly used timekeeping technologies, such as NTP and GPS, and identifies how IEEE 1588 v2 precision time protocol (PTP) can transform how your current industrial network is run.

AN OVERVIEW OF HISTORICAL TIME SYNCHRONISATION TECHNOLOGIES

In an industrial data network, time synchronisation allows all of the different devices on that network to use a common clock to coordinate their activities. Network integrators currently have a number of different time synchronisation options available. Each has its own advantages and disadvantages, but not all of them are optimal for use in industrial networks.

Inter-range Instrumentation Group (IRIG): The IRIG standard defines a serial time code format for use with serial communications networks. First standardised in 1956, IRIG signals are a legacy technology used with older serial systems. IRIGB 205-87 is the latest update of this standard.

Network Time Protocol (NTP): NTP is a time protocol for data networks, first established in 1985. NTP relies on a hierarchical, layered system to promulgate the current time throughout the network. NTP imposes a hierarchical tree architecture on the network to avoid cyclical dependencies.

Global Positioning System (GPS): GPS satellites are highly accurate atomic clocks placed in orbit around the earth. Satellite signals carrying timekeeping information can travel at light speed to receivers on the ground. These

light-speed signals are also corrected according to the principles of general relativity, which gives each receiver on the ground highly accurate time information.

POTENTIAL TIME SYNCHRONISATION PITFALLS

Industrial systems, such as substation automation networks, rely on accurate time synchronisation in order to coordinate activity across many different subsystems and devices. However, many existing technologies are inadequate for the needs of industrial automation measurement and control systems.

Accuracy: For industrial networks every nanosecond counts, but most legacy technologies are simply unable to deliver that level of performance. For example, a substation automation network needs nanosecond-level accuracy on raw data sampled values in order to support mission-critical applications such as fault recording, remote monitoring and remote control. IRIGB and NTP are an order of magnitude too slow to achieve nanosecond accuracy. Even under ideal, local conditions, NTP's accuracy can be measured in the hundreds of microseconds.

Cost: The GPS network provides highly accurate time data measured by extremely precise atomic clocks, but in order to access that information the network must have a GPS receiver at each node. This is a prohibitive cost that is impractical for industrial networks where each device needs time information and its own GPS receiver. GPS would become more practical if there was some way to reduce the number of nodes in the entire network, or more efficiently use a fewer number of GPS receivers so that the entire network can benefit from the accuracy of GPS timekeeping.

A TIME PROTOCOL FOR INDUSTRIAL NETWORKS

NTP, GPS, and IRIGB are capable technologies that simply aren't suited for the requirements of substation operations. Fortunately, the IEEE 1588v2 Precision Time Protocol (PTP) is designed specifically for industrial networked measurement and control systems. In a network based on IEEE 1588v2, the grandmaster clock determines the reference time for the entire substation automation system. The Ethernet switch acts as the boundary or transparent clock, and additional devices (such as merging units, IEDs, and protection devices) are designated as ordinary clocks. All of these devices are organised into a master-slave synchronisation hierarchy with the grandmaster clock at the top. Exchanging PTP packets between master and slave devices, and automatically adjusting the ordinary clocks, effectively synchronises the entire network. Only the grandmaster clock needs a connection to GPS timekeeping; that data can be accurately distributed to the rest of the devices on the network.

An Ethernet switch that supports IEEE 1588v2 can guarantee time-stamping accuracy to within 1 µs, and be configured for master, boundary, or transparent clock functionality. To be truly precise, the rest of the network needs to support IEEE 1588v2 as well: in an industrial computing network, IEEE 1588v2-compliant computers fill the role of the ordinary clock that receives synchronised time data from the Ethernet switch.

When the entire network supports IEEE 1588v2, the system can coordinate operations down to the nanosecond level and still keep perfectly in sync. This level of coordination is especially valuable in power substation systems, which is why IEEE 1588v2 is part of the IEC 61850-2 standard specifying communications requirements for power automation networks. The IEC incorporated IEEE 1588v2 into the standard because more precise time synchronisation allows electrical substations and power automation networks to achieve the following benefits:

- Blackout prevention.
- Accurate fault recording and event loggers.
- More efficient use of assets.
- Demand response.

As part of a 'Smart Grid', highly synchronised substations are more efficient, more economical, more sustainable and more responsive. These advantages allow electricity providers to increase the profitability of their operations and decrease their impact on the environment.

THE MOXA ADVANTAGE: GETTING SUBSTATION OPERATIONS IN SYNC

Moxa's PTP-7728-PTP IEC 61850-3 Fast Ethernet switches support the latest version of IEEE 1588v2 technology to deliver precise time synchronisation for substation network and related applications. The PT-7728-PTP offers:

- Up to 14 100BaseFO (Multi-mode ST connector) or 100BaseTX, ports and 1 BNC connector, support for IEEE1588 v1 and v2 hardware time stamping on each port, and pulse outputs (pps) on one BNC port.
- 1 and 2-step for both transparent and boundary clock operation with accuracy under 1 µs in End to End mode.
- 2-step for both transparent and boundary clock operation with accuracy under 1 µs in Peer to Peer mode.
- Network clock synchronisation accuracy in the nanosecond range.
- Clock synchronisation to support large and distributed substation networks.
- Low-cost implementation in multicast messaging networks such as Ethernet.
- Fast re-synchronisation when system changes occur.
- Simple installation and maintenance.

For more information contact RJ Programmable Techniques, +27 (0)11 781 0777, info@rjprogtech.co.za, www.rjprogtech.co.za

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Instantaneous gas temperature from any area of the boiler

Acoustic pyrometers from Enertechnix.

The PyroMetrix acoustic pyrometer is a non-contact measurement device that obtains accurate instantaneous gas temperature data from any location within a boiler, which helps boiler operators reduce slagging significantly by maintaining optimal furnace exit gas temperatures. It also helps manage the critical temperature profiles that aid emission reduction and reduce tube damage by preventing hot spots and thermal shock. Using PyroMetrix to balance combustion can also improve heat transfer rates.

PRINCIPAL OF OPERATION

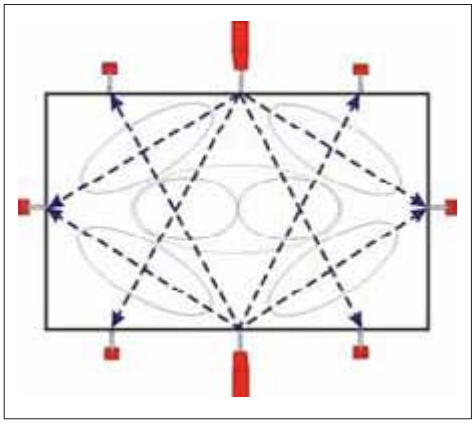
The system utilises the principle that the velocity of sound through a medium is related to the temperature of that medium. PyroMetrix generates a sound wave with a very sharp rise time and measures the time it takes to traverse the chamber. If the distance between the sound source and the receiver is known, then the average temperature between the two points can be determined. Instantaneous temperature data is spatially averaged to provide zoned temperature maps.

Typically the display is on an existing graphical page in a DCS or scada with each measured point represented by a simple 4-20 mA signal. By selecting the correct orientation, users can position zones that give fast reliable temperature readings i.e. place a zone over each burner bank and locate any burner alignment problems, position a centre zone and see if the fire ball is too high or too low.

ADVANTAGES

Accuracy: The system provides instantaneous, spatially averaged temperatures with less than 1% error.

Reliability: The system is designed for superior reliability in hot, dirty environments. Ports are cleared by periodic air blasts, and components are not sensitive to dust or dirt.



Flexibility: The system can be configured to suit monitoring needs. ASGs or ASRs can be added as desired. Boiler tube bends are not required to mount the receivers.

Leak detection capability: The system constantly listens to boiler noises, measuring against preset noise thresholds and time durations to identify persistent sounds indicative of tube leaks.

For more information contact R&C Instrumentation, 086 111 4217, info@randci.co.za, www.randci.co.za

Evolving IEC 61850 Smart Substations Through Trusted Automation Computers



Reliable Substation Computers with Domain I/O Modules

Designed to Act as Terminal Servers, Communication Gateways, Cyber Security Servers, and Networking Recorders in Substations

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- Supports IRIG-B, time-synchronization protocol for substations
- Supports fiber optic LAN ports and isolated serial ports for reliable communication in substations
- Remote Management Utility and event triggering relay to reduce down time



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EKI-4654R

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Measure – Manage – Save

Omnergy: a new solution for utilities management as a web enabled service.

Omniflex has developed technology to allow remote metering, aggregation and analysis of 'WAGES' (Water, Air, Gas, Electricity and Steam) and integrated this into an easy to use package called Omnergy.

Omnergy is a web enabled service that integrates through Omniflex remote data access gateway products with existing metering infrastructure to create a user friendly management tool for energy management and control. The result is that consumption patterns and cost information can be delivered across the enterprise allowing the reduction of energy consumption and an increase in profitability and sustainability. The system works in conjunction with existing metering infrastructure to provide:

- Utilities usage visualisation and control.
- Usage targeting and real-time alerts.
- Billing options, interface to accounting systems.
- Off-site secure historical data storage.
- No in-house computer infrastructure required.

TREND SCREENS – UNLIMITED NUMBER OF TRENDS

- Up to 6 data items on a single trend graph.
- Interactive visual zooming allows focus on a particular piece of data at a particular point in time.

USER DASHBOARD – LIVE USAGE INFORMATION ANYWHERE

- View consumption from any web browser.
- Gives users a real-time view of energy usage allowing instant savings.
- Clear performance indicators promote active energy consumption.

REAL-TIME ALARMS – ADVANCE NOTIFICATION OF TARGET BREACHES IN ADVANCE

- Usage logged every 30 minutes.
- Daily consumption logged.
- Monthly consumption target projected.
- Alerts sent by SMS or email.

REPORTING SERVICES

- Email reports automatically.
- Report on usage against target.
- Compare usage to prior periods.
- Report and track losses per group.
- Collate statistical data for mandatory reports.

BENEFITS

- Energy balance.
- Off-site administration.
- SaaS (Software as a Service).
- Live consumer access.
- Fully automated meter reading
- Immediate cost saving.

For more information contact Ian Loudon, Omniflex, +27 (0)31 207 7466, sales@omniflex.com, www.omniflex.com



Voltage detection in MV switchgear



Integrated voltage indicator with relay output.

The Vois R+ from Kries is an integrated capacitive voltage indicator according to the IEC 61243-5 standard for voltage detecting in medium voltage switchgear. The change from voltage present to voltage absent is indicated by a change in the arrow symbol on the front panel, from visible to invisible, and can also be monitored via the integrated relay outputs. The device needs no power supply to indicate the voltage, the energy necessary for indication is taken from the capacitive coupling and power is only required to drive the relay outputs.

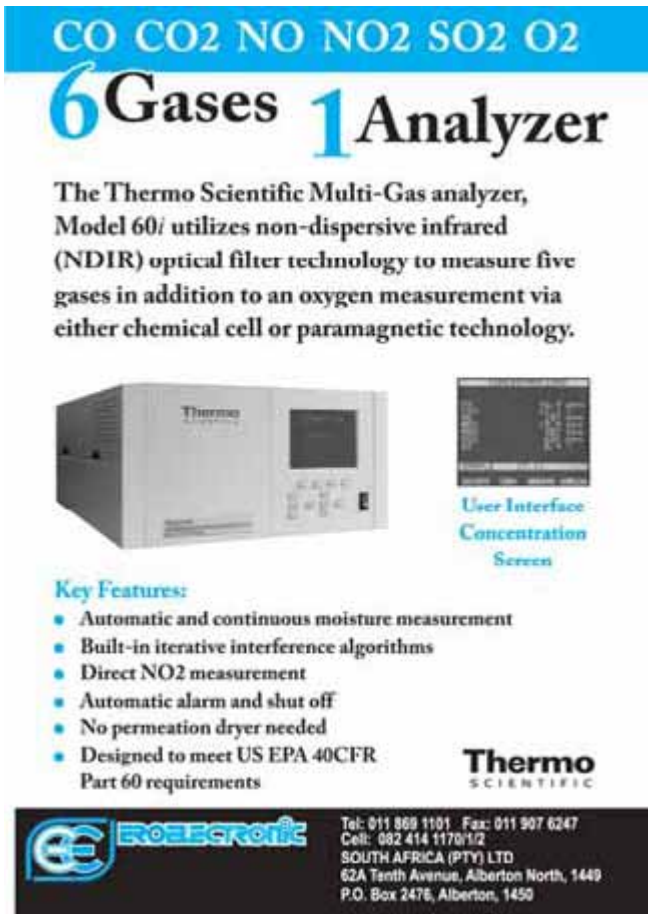
The safety rules and laws dealing with working on medium voltage equipment and the five safety rules according to DIN -VDE 0105 have to be observed. Independent of the indication on the Vois R+, the earthing switch must be switched on before opening the switchgear or working on live parts. Vois R+ is suitable for connecting to specified capacitive coupling electrodes. The coupling part which is connected to Vois R+ must meet the interface conditions according to IEC 61243-5.

For testing voltages higher than 15 kV and frequencies lower than 16 2/3 Hz, the three-phase connecting point on the front has to be short circuited. The three-phase connecting point is adapted to the LRM conditions according to IEC 61243-5. The connecting point is suitable for testing phase comparison, phase sequence, redundancy, and as an option, maintenance testing. For this purpose Kries offers the universal CAP-Phase combined voltage detector and phase comparator.

Vois R + is suitable for use in an operating temperature range from -25 to +55°C and meets IP54 protection requirements.

For more information contact Johan van Staden, Surgetek, +27 (0)11 792 1303, info@surgetek.co.za, www.surgetek.co.za





Omron South Africa completes first solar PV installation

Omron South Africa has recently completed its first photovoltaic installation using a 10 kW grid-tied 400 V inverter. The solar PV panel is roof mounted on a light weight aluminium structure designed to eliminate corrosion. DC power is fed into the panel mounted inverter, DC supply range is 225 to 850 V, with a nominal input voltage of 585 V. The AC power output is connected to the user's 400 V 3-phase supply. The system has Modbus communication on board allowing the values of voltage, current and power tp be monitored and logged.

With Eskom requiring a 10% energy saving across the board, this is a simple method that can be used to reduce consumption while still being able to supply sufficient power to the facility. One advantage of grid-tied inverters is that no alterations to the current electrical infrastructure are required, which means that there is no power down-time during installation. The inverter has a maximum efficiency of 97,5% and only consumes 1,5 W during dark times. This creates an ROI of just 5 years at the current prices of PV panels.

The trend across Europe, where the majority of Omron's PV inverters are installed, is that utilities buy back excess power that is generated by grid-tied inverters, thus using the grid as a storage system, which shortens the ROI even further.

For more information contact Terry McIntosh, Omron Electronics, 086 066 7661, terry_lynn_mcintosh@eu.omron.com, www.industrial.omron.co.za



Avoiding short circuits

In a motor circuit a short circuit condi-tion occurs when the insulation between the phase conductors fails and the load current level exceeds the normal locked rotor current level that the motor would draw during startup. The fault level of the motor feeder supply determines the current level during a short circuit and can be as high as 15 to 50 times the motor's full load current value.

A short circuit fault is a system level fault. This means that it compromises the stability of the entire network, resulting in total power loss to healthy drives and shutting down an entire process. The MCCB is a maintenance specific item that can appear to be functional even when it is not. This can result in a catastrophic failure on the next fault clearance.

NewElec's MA motor protection relay is able to protect against thermal overloads, locked rotor and running stall or jam. The MA also protects against short circuit faults and automatically takes care of trip co-ordination so that the main contactor is not used to disrupt high energy faults.

Additional features of the MA include real- time and date stamping recorded by the relay for the last four trips; a recording utility of actual RMS loads; thermal conditioning; mains phase voltage; line voltage; and status of digital inputs and output contacts. The user also has the option of expanding the unit for fieldbus communications via Profibus DP, Modbus RTU and Canbus protocols.

For more information contact Luc Dutrieux, NewElec, +27 (0)12 327 1729, sales@newelec.co.za, www.newelec.co.za

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SPS power supply from Manson



The SPS series power supply from Manson is a high power regulated, variable voltage, constant current switch mode power supply with voltage and current indication. The SSP series provides 25, 40 and 60 A units, which all have a voltage adjustment of up to 15 VDC. The products are light weight and compact in size and cater for a variety of applications such as telecommunications, 2-way radio and backup equipment and battery charging.

Digital voltage and current indications are built into the front panel, from where the voltage may also be adjusted. The unit has an output lock function which, when engaged, will supply 13,8 VDC only. Overload, over-temperature and over-voltage protection is standard, along with high RFI stability, efficiency and a power factor of 0,95. The products offer a floating ground, low ripple and have a built-in thermally controlled fan.

For more information contact Mantech Electronics, +27 (0)11 493 9307, jn@mantech.co.za, www.mantech.co.za

Surge arrestors for DC photovoltaic applications

The rapidly growing PV industry has placed excep- tional demands on the available SPD products intended to protect systems components such as inverters, arrays, or combiner box components from the effects of lightning caused surges and transients.

Responding to this demand, Dehn has introduced new products specifically designed to function under this DC voltage environment. The new DEHNguard M YPV and PV-SCI incorporate a switched fused circuit in parallel to the MOV discharge circuit, permitting arc free disconnection.

Dehn has recently expanded its initial product offering to include a broader range of voltages and product configuration. These unique SPD products are now available for 150, 600, 1000, 1200 and 1500 VDC applications and configured in both a three-pole configuration for floating systems and a two-pole design for grounded systems.

For more information contact Paul van As, Surgetek, +27 (0)11 792 1303, info@surgetek.co.za, www.surgetek.co.za





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Feedwater level measurement in the power station



Feedwater is a critical component of the circulatory processes in a power station as production of electrical power comes to a halt if insufficient is available. That is why measurement instrumentation is crucial to maintain the required levels of water in the feedwater tank, in case a fault results in loss of electrical power a gauge glass is usually integrated.



LEVEL MEASUREMENT IN THE FEEDWATER TANK WITH VEGAFLEX 66

Vega manufactures a guided microwave level measuring sensor that is unaffected by condensate as well as high process temperatures and pressures. The Vegaflex 66 guarantees high measuring certainty and reliability in the bypass and is able to operate in conditions from -200 to 400°C and 400 bar pressure.

As a part of the safety instrumented system (SIS), it can also be redundantly interconnected with the vibrating level switch, Vegaswing 61 up to SIL3. Safety-related implementation allows both overfill and dry run protection.

REDUNDANT DRY RUN PROTECTION WITH VEGASWING 61

To provide absolutely reliable monitoring of the minimum level in the tank, often two, or even three, Vegaswing 61 vibrating level switches are installed. As a single device it can be used as dry run protection in safety-related systems up to SIL2 and in case of redundant use, up to SIL3.

For more information contact John Groom, Vega Instruments SA, +27 (0)11 795 3249, john.groom@vega.com, www.vega.com



StruxureWare for Data Centres

TAKING DATA CENTRE EFFICIENCIES TO THE NEXT LEVEL.

For many years, as a country with the cheapest electricity in the world, South African businesses did not place the same focus on energy consumption and efficiency as companies situated in other countries. Now, we're finding that local CIOs have to work more closely with facilities managers to understand consumption and close the gap between facilities and business. So says Eben Owen, E&S manager at APC by Schneider Electric, who believes that these two areas, seen as very much separate in the past, are now bleeding together.

"One area under scrutiny for improved energy efficiency is the data centre, which historically fell within the responsibility of the facilities manager," he explains. "Increasing power densities and rapid changes in IT technology, coupled with growing energy costs and uncertain long-term plans for capacity or density have flagged the data centre as an area of concern from a consumption perspective.

"Add to this the requirement by cloud computing for applications to be available at all times and it becomes clear that now is the time to take data centre efficiencies to the next level."

Through the use of a solution like Schneider Electric's StruxureWare for Data Centres, companies can address all aspects of the data centre physical infrastructure needed to maximise availability and efficiency. "This offering empowers data centre managers by giving them access to all the data and tools they need to operate data centres that are more reliable, efficient, productive, safe and green," says Owen.

"Data centre environments traditionally rely on management tools that require the knowledge and experience of numerous IT and facility professionals leveraging specialised management dashboards to obtain the information they need. StruxureWare for Data Centres provides a toolset from which the data centre manager can access and manage across the domains of the IT room, building electrical power and power quality, facility and data centre cooling and physical security. This management software suite delivers comprehensive, accurate real-time data enabling them to take action based on the complete picture of the data centre.

"By using this type of approach, companies can gain better control over current inventories and dependencies, implement 'smart monitoring' for all critical devices and virtualise wherever possible. This will better prepare them to embrace the future of the data centre, where workloads are based on power and cooling requirements, excess server capacity is powered off when not needed and systems are being used as virtual desktops during the day and for batch processing during the night."

For more information contact Roz Shuping, APC by Schneider Electric, +27 (0)11 254 6400, roz.shuping@schneider-electric.com, www.apc.com

Corporate energy management

ALL DEPARTMENTS MUST BE INVOLVED IN DEVELOPING AND MAINTAINING AN ENERGY MANAGEMENT STRATEGY.

Deon van Aardt,
MD Invensys
Wonderware.

Energy management at the enterprise level is not only the responsibility of plant managers and engineers; it has now become the responsibility of corporate management as well. All departments within a company consume energy and add cost to a product or service, so everyone must be involved in developing and maintaining an energy management strategy that can be implemented to benefit the entire organisation's bottom line and to accurately allocate costs where they were generated.

Collecting data from meters is easy. Using meter data for business decisions is not. What's needed is a corporate energy management application that transforms energy data into the information energy managers need and which addresses real-time management of energy usage in an operation. This information becomes essential for energy consumption awareness and cost reduction.

Energy management is a continuous improvement activity. Achieving and sustaining improvements in energy efficiency require reliable, real-time and contextualised information that the entire organisation can use to prove that new equipment and procedures have a positive impact and continue to deliver results.

When deviations to plan occur, the contextualised information enables their root-cause to be determined. Once the procedural or equipment changes are made, the application is used to make sure the correction was effective and sustained. An example of non-contextualised information could be: packing line 2 tripped at 10H17. Adding context makes all the difference: packing line 2 tripped at 10H17 on changeover from product A to product B and when packing line 3 was being started for product C.

ENERGY COSTING OF OPERATIONS

Do you know the energy cost to make your products or produce your service? Clean your equipment? Heat and cool the facility? Does one product incur higher energy cost than another? Is shift 1 more energy efficient than shift 2?

To answer these questions, it stands to reason that any corporate energy management application must allow you to define energy usage in your terms. Energy events enable the capturing of combined energy usage for any fixed or variable period of time. Examples are:

- Production order, batch, lot.
- Unit operation, phase, segment.
- Occupancy, hot days, cool days.
- Demand response episode.
- High flow condition.
- Equipment state (start-up, idle, in production).

If you're a financial controller or planner, you should have the information needed to manage your energy as the variable cost it is – today and in the future (Figure 2). You should be able to manage energy as you do labour and materials and as such, energy costs should form part of the bill of materials for products and should be part of your variance reporting process. In other words, to reflect reality it's important to measure cost rather than consumption.

THE CORNERSTONE FOR CORPORATE ENERGY MANAGEMENT INITIATIVES

A vast array of equipment, techniques and consulting is available to corporations and agencies that embark on an energy management initiative. Governments and industries are adopting new standards to define and execute the programmes and the established standards follow in the steps of continuous improvement programmes for quality, safety and operational effectiveness. The corporate energy management application will be the one place where all members of the team come to determine if equipment, techniques and consulting are effective and sustainable.

The real-time and historical information provided will quickly become part of your organisation's operational culture as individuals can visibly see their contribution towards helping the company achieve its energy usage objectives.

SMART METERS – DON'T REPLACE, CAPITALISE

Energy providers world-wide are installing 'smart meters' on homes, businesses and industrial sites to enable automated meter reading, time-of-use price contracts and load conditions for grid operations. But that means replacing costly hardware with even more costly hardware – so why not add this functionality (and much more) to your existing metering hardware with a corporate energy management application that, thanks to its object-oriented technology, will turn each meter into a cost computer, applying real-time prices to energy usage and providing this information to operations, managers and cost accounting.

SMART GRID READY

Industrial, commercial and infrastructure facilities are constantly faced with challenges when managing the use of energy to produce goods and services. In times of high energy costs, the energy component of production cost is critical to the business. In the future, these facilities will be nodes on an intelligent energy supply chain network, also known as a smart grid.

A smart grid delivers electricity from suppliers to consumers using two-way digital technology to control appliances at consumers' premises in order to save energy, reduce cost and increase reliability and transparency. It will overlay the electricity distribution grid with an information and net-metering system.

Such a modernised electricity network is being promoted by many governments as a way of addressing energy independence, global warming and emergency resilience issues. Although the smart grid concept may not be implemented locally for some time in South Africa, the fact that the country will follow global trends can't be ruled out.

In the US, five to 10 years ago, industrial companies were typically able to sign contracts with their energy suppliers to lock in the cost of energy for a six to 12-month contract period. But with the opening up of the power grid system, those kinds of contracts have gone away. Today, it is not unusual for a company's cost of energy to change frequently, such as is the case for one Arizona copper manufacturer whose energy costs change typically about 48 times per day.

Managing the cost of energy will produce a competitive advantage over those who can't leverage the opportunities provided by demand response, self generation, exportable distributed generation and real-time market pricing for energy. An operation must understand its energy consumption characteristics to reap the benefits and avoid the penalties while meeting the needs of its customers and the environment. Operations will

Continued on page 44

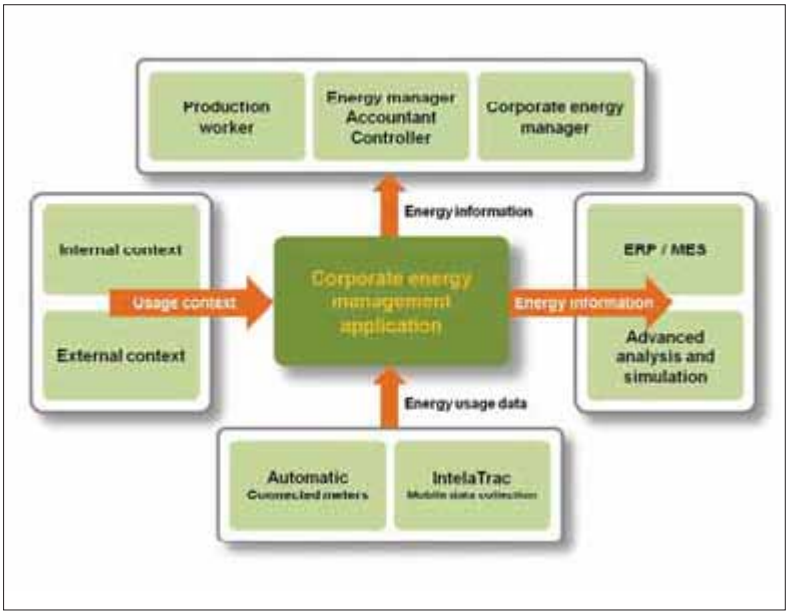


Figure 1. Context transforms energy data into energy information.



Figure 2. Energy costs have become a business variable that must be measured and controlled by manufacturers as part of a holistic approach to the problem of energy costing rather than consumption monitoring.

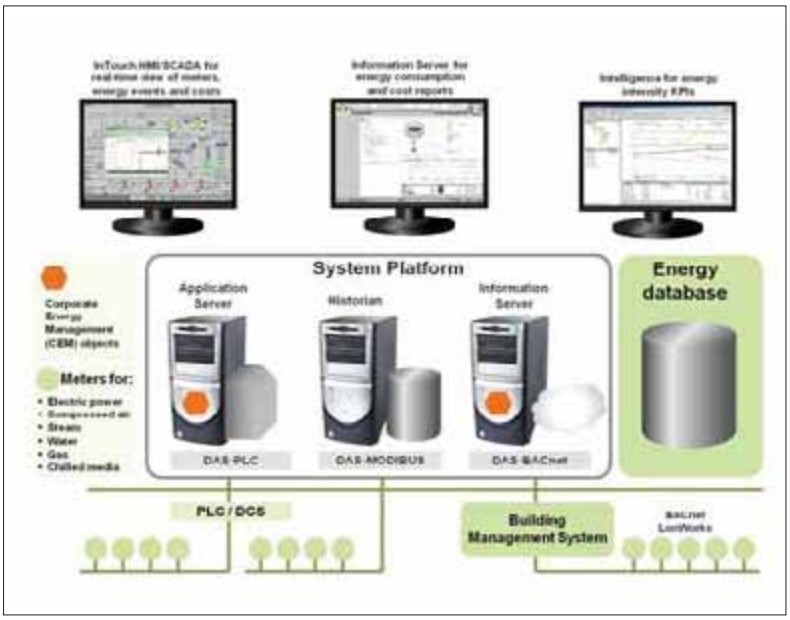


Figure 3. The Wonderware approach to corporate energy management.

Continued from page 43

incorporate new concepts such as 'green' scheduling as well as use-or-sell cooperative-generation capacity and load-shedding. In South Africa, we are already seeing new (although not national) players on the block of power distribution with large local companies willing to contribute their excess power assets to the national grid – but on what mutually-beneficial basis?

Undoubtedly, these concepts will be disruptive to normal operations and will affect industrial automation, maintenance, performance and asset management. In view of this, any corporate energy management application must help the organisation minimise the disruptions while enabling the opportunities and should be designed for the emerging requirements for smart grid connectivity and interaction.

INTEROPERABILITY

We now know that one of the most fundamental mistakes of the last few decades in respect of industrial information solutions was that each application was developed largely in ignorance of the role it had to play in the bigger picture of the unified enterprise and with other applications – after all, if you're a hammer, everything looks like a nail.

A corporate energy management application is no exception and should be designed to interact with and complement existing meters and automation assets while providing results and reports that can be easily integrated with those of other disciplines (e.g. facilities management, smart building management, etc.) in order to provide a more informative and knowledge-based platform for shaping decisions.

CONCLUSION

Because of its pervasive nature, energy management is a corporate initiative that demands the cooperation of all concerned. It now forms an important part of Enterprise Manufacturing Intelligence (EMI) and as such, its contribution should not be underestimated.

Note: The facilities, features and concepts in this article are all embodied in Wonderware's Corporate Energy Management application which is built on the Wonderware System Platform and fully integrates with the rest of the Wonderware suite of industrial automation, MES and enterprise integration/intelligence products.

For more information contact Jaco Markwat, Invensys Wonderware, +27 (0)11 607 8100, tech@invensys.co.za, www.wonderware.co.za





Sustainable, cost-effective solutions from Eaton

Forward-looking building technology saves energy, protects the environment and safeguards investment.

At Eaton, green building solutions encompass the inter-locking areas of power distribution, power management and power quality. From medium voltage to the mains socket, Eaton offers a comprehensive range of solutions for the design, layout, operation and evolution of a powerful and flexible building infrastructure – energy-saving, transparent, and lasting. Forward-looking systems for integrated measuring, monitoring and managing of energy in residential and functional buildings, together with smart building automation and distributed energy generation, ensure outstanding efficiency. Economical UPS systems enable a secure power supply without fluctuations, faults or outages and offer solutions for the optimised, energy efficient use of energy.

Climate-friendly – from medium voltage to the mains socket: protecting the environment and saving resources starts in the control cabinet at Eaton with environmentally friendly solutions for all buildings. Eaton's medium voltage systems dispense entirely with the potent greenhouse gas sulphur hexafluoride. The focus is on switchgear and components for application in power distribution networks (nodal points, substations and transformer stations) and complete power supply systems. The technically high-grade systems feature air and epoxy resin insulation and are commonly equipped with circuit breakers or load-break switching that include Eaton's vacuum interrupters. The integration of medium and low voltage distribution in one system saves not only space but also energy. For example, the compact design of the M2L distribution system combines medium voltage switchgear, the transformer and low voltage distribution in a single housing. This is why the transformation from medium to low voltage can occur locally, precisely and with low loss.

Building management – a profitable capital asset: whether used commercially or privately, electrical systems are not monitored transparently in many premises – the building technology lacks an overall perspective. This transparency is offered by Power Xpert monitoring, a system that manages the energy supply as a strategic asset. With the xComfort system, building systems functions from the heating and ventilation to the light and electric circuits are actively monitored and conveniently automated in the residential and light commercial environment. With xComfort, Eaton focuses on simple expansion and turnkey solutions for single family houses, apartment buildings and not least functional buildings. Eaton is currently expanding its xComfort product range with a further component, the Ethernet Communication Interface (ECI). With it, users can connect separate xComfort wireless solutions within a building via an Ethernet network. In addition, the ECI enables the simple expansion of individual components in functional buildings. The ECI is available in three versions: WLAN, LAN and LAN PoE. While WLAN and LAN require an additional power supply, LAN PoE is supplied via the Ethernet.

Photovoltaics – using solar energy efficiently and safely: with its mature and complete range of photovoltaic components, Eaton is setting trends in this sustainable, zero-emission field of energy generation. The portfolio ranges from the Sol30-Safety fireman's switch, which cuts the live line from the solar modules to the inverter, via surge protection modules specially developed for photovoltaic applications in a range of versions (also as a complete interface unit to IP67), all the way to DC load break switches which cut the voltage of an entire system.

Energy-saving solutions for reliable power supplies in buildings: power quality solutions from Eaton protect not only critical components of the building technology such as security systems, fire alarm systems, communications technology and all connected IT systems, they also make a valuable contribution to reducing the entire building's energy consumption. Eaton integrates networks, IT systems, building technology and automation in one intelligent network: the intelligent Power software makes transparent monitoring, reliable analysis and rapid reactions possible – even in virtual environments. And, as required, it supports the controlled and protected shut-down of affected systems.

Tailored service expertise: Eaton provides services that assist and support electrical building infrastructures, replacement services, retrofit services or lifecycle extensions. The objectives are protecting resources, reducing costs and extending service life.

For more information contact Rob Hare, Eaton Electric, +27 (0)11 824 7400, robhare@eaton.com, www.eaton.com

EtherNet/IP modules for industrial automation applications

16 DIGITAL CHANNELS WITH UNIVERSAL I/O FUNCTIONALITY FOR DIRECT CONNECTIONS BETWEEN SENSORS, ACTUATORS AND CONTROL SYSTEMS.



EtherNet/IP is one of the most important network standards in the field of automation. These two new LioN-M I/O modules from Lumberg Automation support the latest EtherNet/IP protocols and connect sensors and actuators directly to control systems. The I/O modules integrated dual-port switch facilitates a line topology as well as star topology. They also enable one to continue using existing communications structures, such as cable trays and energy chains, without modifying the physical layout, which leads to cost savings.

These two modules also facilitate a high availability ring topology: if the connection goes down, the device level ring protocol (DLR) immediately, switches to an alternative ring segment without interrupting machinery operations. Each provides 16 digital channels, in one version (16DI) the channels are used as inputs. The universal modules provide both 16 digital inputs and outputs – in any desired combination and without any need for individual parameterisation. Accordingly these modules offer a very high level of flexibility, not only at the planning stage, but also when making changes during commissioning or subsequent upgrades.

In addition, the 'easy diagnostic' concept with a display for each I/O port enables easy fault finding. The same information is also available via the Ethernet IP protocol, facilitating detailed analysis on a centralised control and display system without any need to configure the module. Thus, time-consuming error localisation is eliminated, minimising downtimes and maintenance periods, while increasing the availability of installation.

The colour coding of the individual plug-in connectors enables immediate recognition of the function of various sockets (fieldbus, power supply or I/O). The optimised arrangement of the M12 I/O sockets considerably simplifies the installation process, even T distributors can be installed with ease.

THE ADVANTAGES AT A GLANCE

- Connection of sensors and actuators to EtherNet/IP networks.
- Integrated dual-port switch (10/100 Mbit/s) facilitates line topology.
- Ring topology by means of the device level ring protocol.
- 16 digital channels, either as inputs or universally as inputs and/or outputs.
- M12 connections for network ports and I/O sockets.
- Fail-safe connections via colour coding of network ports.
- Rotary switches for easy IP address selection.
- Configuration via the control system's user interface.
- Ingress protection rating IP67.
- Temperature range -10 to 60°C.
- Flexible front or side mounting.

For more information contact Greg Pokroy, Jaycor International, +27 (0)21 447 4247, greg@jaycor.co.za, www.jaycor.co.za

RUGGEDCOM®
INDUSTRIAL STRENGTH NETWORKS™

Product Overview

RuggedCom's RX1500 Application Processing Engine (APE) is the ideal product to increase performance and functionality of the RX1500 multi-service platform. Leveraging the built in switching and routing platform of the RX1500, the APE is a utility grade computing platform that plugs directly into any member of the RX1500 family and enables customers to load and execute security and software applications directly on a RuggedBackbone™ device. From substation gateway, remote analytics and firewall to monitoring and control applications, the APE provides the computing power and reliability for mission critical applications. The RX1500 APE is designed to the RuggedRated™ specification, which provides a high level of immunity to electromagnetic interference (EMI) and electrical surges. With an operating temperature range of -40° to +70°C (-40° to +158°F), the APE can operate in harsh environments.



RUGGED APE™

RuggedBackbone™ RX1500 Application Processing Engine

Industrial Computing Line Module

RUGGED RATED

Benefits & Applications:

- Embedded design allows direct access to the RX1500 line module interfaces reducing the inventory of cables, devices and ports to manage
- Modular design for easy deployment and reduced maintenance
- Utility grade specifications for unmatched reliability and reduced outages
- Rack mount and DIN rail form factors available to fit specific application requirements
- Windows Embedded and most Linux distributions are supported

RuggedRated™ for Reliability in Harsh Environments:

- Immunity to EMI and high voltage electrical transients
  - . Meets IEEE 1613 (electric utility substations)
  - . IEC 61850-3 (electric utility substations)
  - . Exceeds IEC 61800-3 (variable speed drive systems)
  - . Exceeds IEC 61000-6-2 (generic industrial environment)
- -40°C to +70°C operating temperature (no fans)

RX1500 APE Specifications:

• Operating temperature:	-40 to +70°C
• CPU:	ATOM E660T
• MIPS:	5,000 @ 1.3GHz
• RAM[1]:	1GB DDR2 (up to 2GB supported)[2]
• Storage[1]:	8GB SATA SSD (up to 16GB supported)[2] (~30 MB/sec)
• Face Plate:	2 x USB DVI-D 1000BaseT RJ45 PWR BUTTON Status LEDs
• Backplane:	1GigE Port Access

Note 1: Placed directly on-board and not removable.
Note 2: Please contact your Regional Sales Manager for details.






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IS Ethernet applied in coal mining

THE USE OF IS PLCs AND AN IS ETHERNET SYSTEM PROVIDES ADVANCED FEATURES THAT ARE EASY TO ASSIMILATE UNDERGROUND FOR ONGOING BENEFIT.



Reliable control and portability are critical requirements for underground coal mining operations. A mine environment is an 'enclosed space' where explosive dust and methane gas can be present, often within areas many kilometres from the nearest exit. In such an environment, an intrinsically safe solution – not only for the 'at-face' mining equipment, but also the associated communications system – can provide significant economic advantage compared with conventional systems using other explosion protection techniques.

Daw Mill Colliery is the flagship deep mine of UK Coal. Set against a background of the earlier wholesale closure of mines and privatisation, UK Coal found that it needed to take a complete review of its control and monitoring to find newer and more supportable technology. Daw Mill decided to use intrinsically safe PLCs and intrinsically safe Ethernet systems on all underground production and conveying.

This environment has to be carefully managed and full control made available to ensure that production is constantly economically viable in the face of imported coal. Safety is paramount, in machine operation, transport, environment, ventilation and gas monitoring. This underground intelligence stems from the use of commercially available real-time scada software to supervise, monitor and attend to alarms. In addition, data is extracted from the mine's equipment that is used to provide real-time information for many management functions. Automation has been fully implemented at the mine with machinery now under the control of the IS PLCs.

SAFETY IN THE ARCHITECTURE

The existing systems architecture was inherently slow and prohibited effective control, so the decision was made to use IS PLCs with an IS Ethernet system to add networking communication with the surface. The PLCs are monitored and controlled at the surface control room by use of Wonderware's Intouch scada system and use this to manage the underground logic controllers. High value coal cutting machines and conveyors, as well as the environment underground, were thus brought under proper control. Safety, as already stated, is the primary concern and particular emphasis was placed upon monitoring airflow, methane and carbon monoxide. Alarms are set to shut equipment down at pre-determined levels and also initiate evacuation.

The PLCs control machinery including the coal-face coal shearing machines, ventilation and conveyors. Since the coal face at Daw Mill is currently 10 km from the control room, communication has to be fast and effective – fibre optic IS Ethernet is being used throughout the mine.

One of the unique differences found in mining, as opposed to manufacturing, is that the plant is under continuous redeployment. When a coal seam reaches a boundary all of the mining equipment may be moved to a position several kilometres away and have totally different operating conditions. It was therefore important to the engineers that the PLCs be capable of easy re-configuration and ongoing development.

Daw Mill has two shafts, one for the miners and one for equipment; but it is in fact a drift mine in that the coal is transferred to the surface by inclined conveyors. The conveyors have powerful motors to cope with the weight and loading changes in operation. These conveyor system motors are also monitored by PLCs.

Within the project period it was noticed that a motor was drawing higher than normal current, but examination of the fine resolution data showed that this was caused by external effects rather than

imminent motor failure. The fault proved to be a belt scraper that was pressing on the belt and causing it to slip – thereby explaining the high current.

BENEFITS

The financial benefit to UK Coal of accurate diagnosis that prevented a shutdown was about £500 000 (calculated from the cost of replacement parts and lost production). The diagnosis allowed the problem to be fixed in five minutes by clearing a blocked air pipe and was made possible by the availability of high resolution archive data that is continually updated from all controllers to the historian server. This example shows an immediate cost savings that was made possible by the system, but the benefits of real-time systems applied at the colliery are felt in day to day running by improved insight to events and ease of access to meaningful and useful information.

The system has become the backbone of the way that the colliery operations communicate; information that is required underground is readily available where and when it is required. This allows manpower to be allocated in an optimum way – with added certainty that the right team is being deployed for the right reasons. When distances underground are measured in kilometres, it is clearly very time consuming to be 'on the wrong mission'.

The IS Ethernet infrastructure allows the options of adding wireless access points to connect to IS mobile computers, cameras for operations, safety and security, RFID for asset and people tracking and VoIP telecommunications in the future.

The use of IS PLCs and an IS Ethernet system has benefited UK Coal by providing advanced features that are easy to assimilate for ongoing benefit. The system is subject to constant change and development according to variable operating conditions and the realisation of solutions that will directly affect productivity and cost – cutting edge technology delivering benefits in the depths.

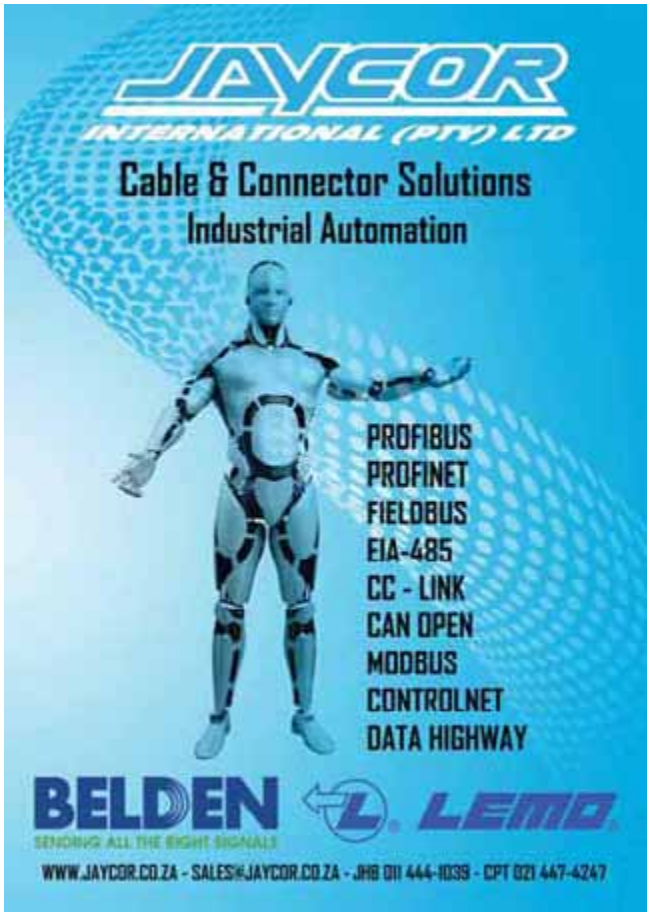
NETWORK PRODUCTS USED

Initially the Daw Mill used the 9260 range of IS Ethernet infrastructure components from MTL, more recent extensions have employed the 9466-ET 5-port IS Ethernet Managed Switch together with the option of Power-over-Ethernet to distribute the DC power.

IS Ethernet cards in PLCs throughout the mine are connected to a 'local' 9466-ET switch together with 9461-ET IS Ethernet serial gateways that are used to communicate with a range of IS serial devices. The 9466-ET switches are all interlinked with lengths of fibre-optic cable via 9465-ET IS Ethernet copper-to-fibre media converters, ultimately reaching the surface to link into the control system servers. The use of multimode fibre-optic Ethernet connections is invaluable for the long distances involved in coal mining, supporting 2 km distances at 100 Mbps or 5 km at 10 Mbps, while IS Ethernet over copper supports the standard 100Tx Ethernet distance of 100 m.

For more information contact Extech Safety Systems, +27 (0)11 791 6000, extech@jhbmail.co.za, www.extech.co.za





Cost effective monitoring from Endress+Hauser

WirelessHART makes it possible to acquire measurements from dense plant areas, remote locations and even moving vessels.

Wireless devices are the ideal solution for monitoring important process variables that are uneconomical to measure by conventional methods. What's more, wireless instruments support advanced asset management solutions, allowing information to be exchanged with the measurement sensor to aid proactive maintenance strategies across an entire plant.

GO DIGITAL

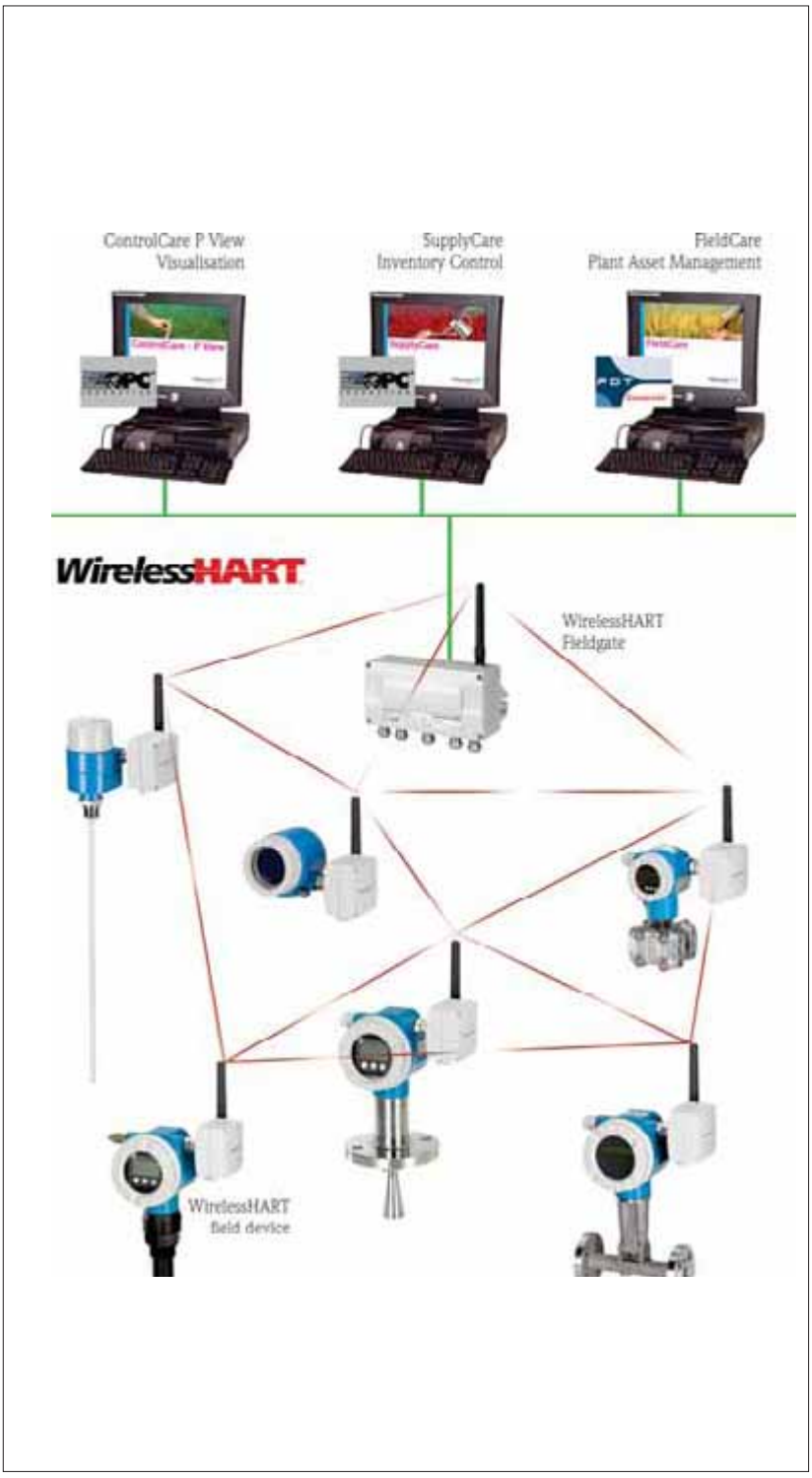
A WirelessHART mesh network protocol has built-in safeguards that ensure reliable and secure communication. The mesh network is self-organising and self-healing with redundant paths, so operation is reliable even when obstacles or interference interrupt a communication path. The WirelessHART Fieldgate collects the measured values at regular intervals and transmits the data, along with the device and battery status, to the plant network. The result is a cost-effective process measurement that includes signal status and device health information.

Like Profibus and Foundation Fieldbus, HART delivers more information than just the raw measured values – having access to the instrument status and diagnostics means greater confidence in the measurement and therefore optimised performance from an application. This in turn helps to reduce large on-going operational costs such as unnecessary plant downtime due to unscheduled or reactive maintenance.

SAVE POWER

As HART is a digital protocol, the field device can be set to operate in 'Digital Mode'. This means the power consumption is greatly reduced, operating at a fixed 4 mA compared to the normal 4-20 mA variable signal normally used to drive an analogue I/O system. Operating in 'Digital Mode' can give an energy saving of approximately 40-60% during typical use (mid span). Across a large plant, it offers an effective route to substantial energy savings.

For more information contact Lita Mpahlwa, Endress+Hauser, +27 (0)11 262 8000, info@za.endress.com, www.za.endress.com





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People for Process Automation

MultiHop radio with built-in I/O



A new radio from Banner Engineering combines wireless transceiver, power supply and I/O circuits in one compact housing. The DX80DR MultiHop Radio connects sensors directly to the network, eliminating need for dataloggers, power supplies, PLCs or HMIs at the sensor location. The radio is compatible with all industrial signals and also with specific instruments including moisture sensors, strain gauges, water quality indicators, flow counter, irrigation valves and temperature transmitters.

New applications

The enhanced radios expand the capability of Banner's SureCross wireless solutions in numerous industries including oil & gas, composting, grain handling, water monitoring, aquaculture, irrigation, parking, flood warning and facilities monitoring.

The increased capability of the radio allows real-time access to information that was previously logged at sensor sites for later download. I/O terminals are provided for thermocouples, digital and analog input and output, battery operated irrigation valve operation, soil and water sensors. RS485 Modbus RTU connection is built in.

The MultiHop system automatically repeats wireless messages as needed to extend distance capability to 20 km or more and allow signals to pass over hills or other obstacles. Radios can be powered by 10-30 VDC, battery or solar power supply, allowing wireless repeaters to be placed where no power is available.

The Banner system can provide wireless connections ranging from the simple point-to-point networks to those spanning hundreds of square kilometres and thousands of points.

For more information contact Rodney Topham, RET Automation, +27 (0)11 453 2468, rodney.topham@retautomation.com, www.retautomation.com

Ethernet switches for power distribution networks



Moxa has introduced the DIN-Rail PT-508/510 series IEC 61850-3 substation switches, the newest addition to its PowerTrans Ethernet switch series. Tailored for the power utility industry, the PT-508/510 series provide high reliability and availability for substation communications thanks to tough EMI immunity, -40 to 85°C operating temperature range and redundant features that tackle specific needs and requirements found in power substations. In addition, the switches support Moxa Turbo Chain, a highly resilient self-healing Ethernet technology, making them well-suited for wide, extended power distribution networks.

DESIGNED FOR POWER DISTRIBUTION NETWORKS

"System operators are always looking for reliable yet highly efficient networking connections to address the critical factors found in large-scale and complex power distribution automation applications," says Moxa's Leo Tsao.

The PT-508/PT-510 series switches feature Moxa's innovative breakthrough technology, called Turbo Chain, which provides a highly flexible self-healing Ethernet topology solution. Turbo Chain allows the creation of multiple redundant networks beyond the limitations of current redundant ring technologies, which makes it ideal for complex power utility network connections spread over multiple locations with correspondingly complex wiring requirements. The versatile redundancy technology liberates system builders from the costly limitations of conventional ring architectures. Compared with conventional ring topology, Turbo Chain eliminates the need for additional Ethernet ports to form ring coupling paths, as well as the extra cabling.

IEC 61850-3 CERTIFIED FOR POWER SUBSTATION ENVIRONMENTS

As with all of the Moxa PowerTrans products, the PT-508/510 series switches are specifically designed for substation environments. In addition, the switches have passed IEC 61850-3 and IEEE1613 certification, and are fully equipped with all of the features specified by the IEC standard. These include, an IP40-rated housing, -40 to 85°C operating temperature range, three or four optical fibre ports and isolated redundant power supplies.

For more information contact RJ Programmable Techniques, +27 (0)11 781 0777, info@rjprogtech.co.za, www.rjprogtech.co.za

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Modular industrial patch panel



Belden's new MIPP is a termination panel for cables that need connecting to active equipment such as switches. Thanks to the modular design, it can be linked to create a large single patch panel, to which both fibre and copper cables can be connected.

The MIPP is a new solution that provides the ideal connection between Belden cables and Hirschmann switches. Available in a choice of modules suitable for both fibre and copper cables, up to 6 modules can be connected to create a single panel that can accommodate both types of cable at the same time for maximum system flexibility. Using patch cords to connect to active equipment, cables can be terminated outside the cabinet in an organised and structured manner to ensure high reliability. The industrial design makes it highly suited for use in machine building, transportation, power generation & distribution as well as for general use in enterprise, buildings and other applications.

BENEFITS

The MIPP is designed for industrial use where its functionality and reliability can make a significant contribution to the uptime and availability of the system. Thanks to its modular design, installation is quick and easy saving time and reducing set-up costs, whilst at the same time contributing to greater reliability and easier and more efficient system management.

The unit has a lightweight aluminium structure for increased overall versatility. It easy to mount on DIN rail or it can attach at the sides of the cabinets using a wall mount plate. Thanks to its small housing, high port density and modularity, cabinet space is kept to a minimum.

For more information contact Greg Pokroy, Jaycor International, +27 (0)21 447 4247, greg@jaycor.co.za, www.jaycor.co.za



EtherCAT coupler series integrates digital I/Os

The EK18xx coupler series combines the functionalities of the EK1100 EtherCAT coupler with standard digital I/Os in a single housing. This results in a compact design with dimensions of just 44x100x68 mm that is especially suitable for applications with a small number of I/Os. Like the standard EK1100 coupler, the EK18xx couplers can be extended by all EtherCAT terminals. The digital I/Os are implemented with a single-wire technique, wiring can be implemented without tools using direct plug-in with single-wire conductors or wire end sleeves.

The following I/O variants are available:

- EK1814: 4 digital inputs/4 digital outputs.
- EK1818: 8 digital inputs/4 digital outputs.
- EK1828: 4 digital inputs/8 digital outputs.
- EK1828-0010: 8 digital outputs.

Beckhoff is consistently continuing the path towards miniaturisation of designs and cost optimisation with the EK18xx EtherCAT Coupler series. Tailored to applications with a small number of I/O points and cramped space conditions, the EK18xx coupler offers the user a precisely dimensioned solution.

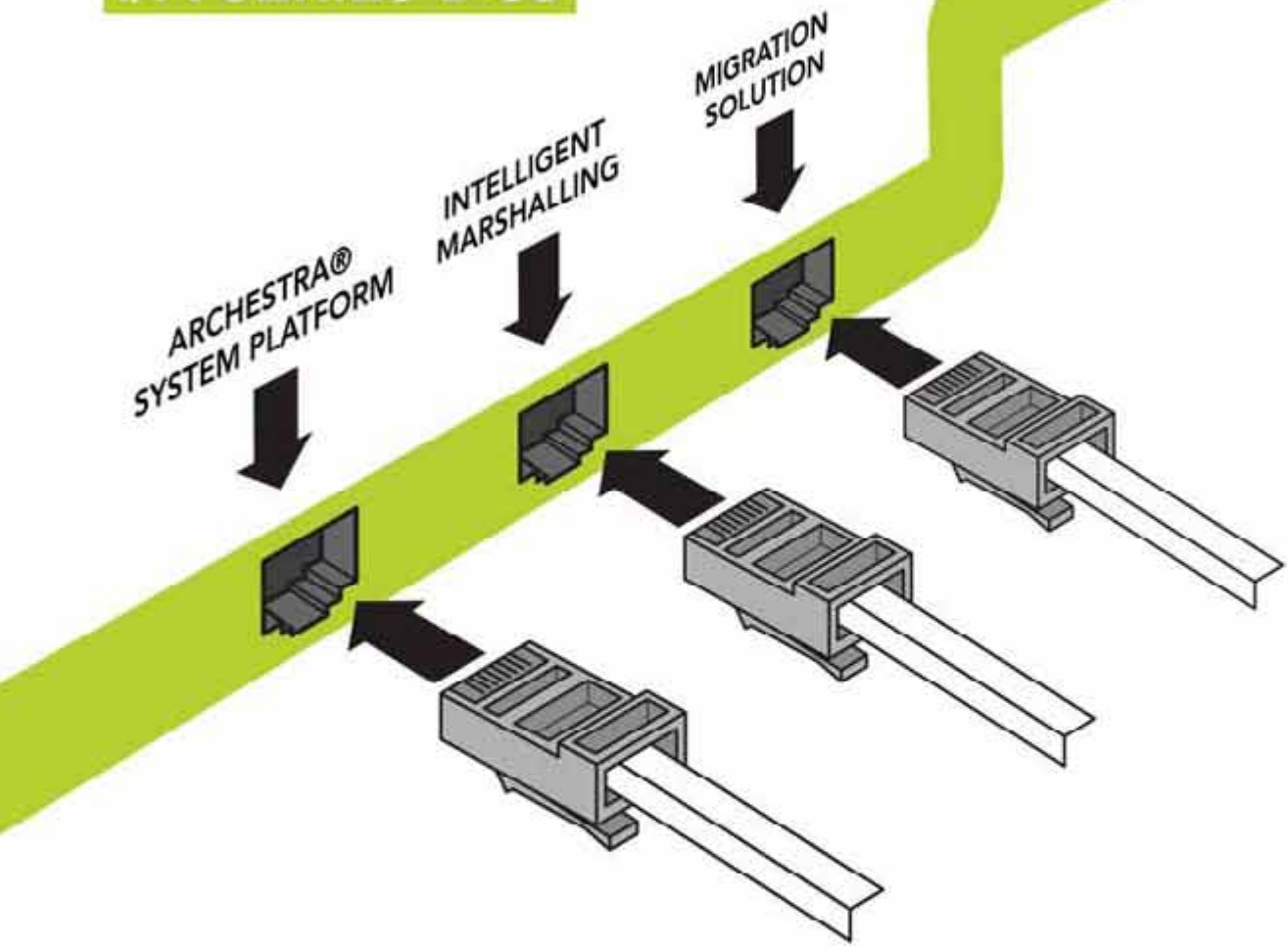
The EtherCAT terminal system is a modular I/O system consisting of electronic terminal blocks. EtherCAT's technological features are supported to the optimum, since the EtherCAT protocol is retained right down to the individual terminal. Suitable EtherCAT terminals are available for all common digital and analogue signal types encountered in the world of automation. Lower-level bus systems are integrated via local master/slave terminals.

For more information contact Conrad Muller, Beckhoff Automation, +27 (0)11 795 2898, c.muller@beckhoff.com, www.beckhoff.co.za

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Using VLANs in industrial networks

THE IMPORTANCE OF PLANNING, CONFIGURATION AND MAINTENANCE ON ETHERNET NETWORKS.

In recent years, Ethernet technology has grown in leaps and bounds and has become a networking technology of choice for running automation networks in industrial, utility, military and ITS environments. However, as the technology progresses and becomes more complex, so does network planning, configuration and maintenance need to improve to meet that complexity. Gone are the days of simple point-to-point serial communications running automation, where loss of communication is relatively simple to diagnose and repair.

BACKGROUND

Many Ethernet networks currently in place at these types of sites were not planned for correctly, especially when considering future expansion and the possible loads that would be placed on the network. This means that although the networks would work well when they were first installed, once Ethernet took hold as the network of choice, the networks were expanded to cover more and more of the site. This leads to network slowdowns end device communication failure due to untimely responses from master devices, and even complete network crashes, all due to the larger amounts of uncontrolled data travelling on the network.

For this reason, proper planning of a network is an extremely important part of the process of moving to Ethernet. The most obvious planning is to take into account the number of devices that will be on the network, as well as an idea of the traffic amounts these devices will impose on the network. However, at the same time care needs to be taken to make sure that the network will also be able to handle future expansion. Provided there was correct planning on the IP subnet structure on the network (The number of IP addresses available within each subnet to assign to devices) it is a relatively easy task to add on an extra switch at the edge of a network, however it is not such a simple task to upgrade the main backbone of a network. For this reason, planning and forethought is highly important to avoid future downtime due to required physical network upgrade.

However, planning for the physical network and the required bandwidth is only the first step. Even the most meticulously planned network may not operate in accordance with the user's requirements if not configured correctly.

NETWORK ARCHITECTURE

Due to the rapid expansion and evolution of Ethernet, it can often seem daunting to plan and configure features such as VLANs, routing, multicast control and so on. For this reason networks are often configured as a single flat network, with no multicast/broadcast control, routing or VLANs to try and separate and cut down on unwanted data reaching end devices. Although this may work for very small scale, non-mission critical networks, for a larger or more traffic intensive network, or a mission critical network where responses need to be prompt and

reliable, a flat network will not suffice.

There are several protocols built into Ethernet that can help with controlling the data on the network, and will largely improve the reliability and response times. One of the most important of these for any large, data intensive, mission critical network is VLANs (Virtual Local Area Networks). The use of VLANs, defined in IEEE Standard 802.3q, allows the network to be logically divided into sub networks, based on which device groups they need to communicate with regularly. This means that devices in one VLAN will not be able to communicate directly with devices in another VLAN. On a flat network, whenever a device sends out a broadcast message (a broadcast message is one that will travel to every device in the same subnet, and each device that receives that broadcast is obligated to inspect the packet to see if the device must respond), every other device on the network receives it. This means that PLCs will be receiving broadcasts from devices such as cameras, historian servers and any other network device sending broadcasts. The end devices will be devoting substantial amount of their time and processing power to inspecting packets that they have no need to see.

Using VLANs you can logically separate the network over the same physical cable, meaning that unwanted traffic in each VLAN, or group of devices, is eliminated. In the event that some devices in one VLAN do need to communicate with other devices in another VLAN, an Ethernet router can be introduced which will allow data to be routed between VLANs. For VLANs, pre-planning is essential, as too few VLANs and the broadcast traffic will still be travelling to devices it does not need to, but too many VLANs and routing between them can put unnecessary strain on the router in the network which could also cause problems.

MAINTENANCE

Ethernet networks are constantly evolving and expanding. For this reason even the most carefully planned and configured network still requires regular maintenance. Standard maintenance that is done on any electronic device is obviously a must, such as checking that devices are still getting correct power, are not overheating etc. For Ethernet networks, however, there is more logical based maintenance as well that is recommended in order to keep the network in a stable, reliable state (such as checking switches and routers on the network for any errors).

For instance a standard practice should be to check the log files and alarm lists of the Ethernet switches on the network, especially those making up the Ethernet backbone. This should be done on a regular basis, such as once a month or even every two weeks. This can prevent small problems, such as a failed redundant power supply, from become a much larger problem (Such as if the second power supply fails and brings the switch down altogether). Logs and alarm lists should also be backed up and then cleared once they have been checked and any concerns addressed, as this will greatly aid with future troubleshooting.

A network management package can greatly aid in this practice, as it can be set to automatically download log files on a regular basis, as well as backing them up and then clearing them off the devices themselves. Some can even be configured to check the firmware of the devices and upgrade them if necessary. If you are using a network management solution, you can even go a step further. By configuring SNMP (Simple Network Management Protocol) on the devices on the network you can allow the device to notify the network management station when they have an error, which in turn can be configured to notify the network administrator on duty. Again this will allow problems to be identified and corrected in advance, rather than waiting for the problem to escalate.

Another maintenance step that can be a great preventative of network issues and crashes is by having the network regularly audited by a team of Ethernet network support specialists. Using hardware and software tools, along with specialised skills, a team such as this can analyse the traffic on the network at various times, and using this can identify problem devices and incorrect configurations before they cause critical network failures.

In today's age, with Ethernet networks being used for mission-critical networks for everything, from mining to production to substation automation, network engineers and administrators cannot afford any network failures. For this reason, correct planning and configuration, as well as regular preventative maintenance are essential to having the peace of mind that a stable, reliable communications network can bring to the end user.

For more information contact H3iSquared, +27 (0)11 454 6025, info@h3isquared.com, www.h3isquared.com



Variable area flowmeters support Foundation fieldbus

Krohne has introduced the H250 M40 variable area flowmeter featuring Foundation fieldbus functionality. This guarantees complete integration into and interoperability with Foundation based control systems. This also applies to previously installed devices: the modular design of the H250 M40 launched in 2009 also allows for retrofitting to Foundation fieldbus devices. It is fed completely from the fieldbus and offers much more than previous stopgaps with converters which only converted the 4-20 mA signal to a digital value on the bus.

One of the first buyers of the new variant is a major German manufacturer of turbo compressors: the manufacturer ordered 240 devices in nominal sizes DN 15 and DN 25 to monitor the multi-stage gas seals on compressors used to liquefy natural gas. With the FF functionality, the status of the seal on the shaft end is transferred directly into the control system. This also makes it possible to set the limit values of the flowmeters, realise PID control using the control system and read the integrated totaliser. This type of monitoring is a typical application for the variable area flowmeters, which can still reliably display the measurement even in the event of a power outage.

For more information contact John Alexander, Krohne SA, +27 (0)11 314 1391, johna@krohnesa.co.za, www.krohne.com

Resinkem improves shift log information with FlexiLOG



Resin manufacturer Resinkem, a 50% joint venture between AECI and US-based Georgia-Pacific Resins, has implemented ApplyIT's operations information management system, FlexiLOG, to manage shift handovers, tighten reporting, track history and ensure accurate data for efficient decision making.

Part of ApplyIT's operations performance OpSUITE, FlexiLOG is designed to provide a user friendly electronic method to log and process shift information simply and accurately. Prior to the FlexiLOG implementation, Resinkem relied on a manual system to manage its shifts using hand written reports logged in A4 hardcover books. This process was not only time consuming but unreliable with readings for critical processes such as tank levels often not recorded accurately. Resinkem operations manager, Manoj Ram, says the old logging process was cumbersome and erratic. The logbook was not always accessible between shifts and, being manual, it had no back-up in the event of loss or damage, the risk of data loss was high.

BUILDING ON THE EXISTING PLATFORMS

The operations team at Resinkem were already enjoying the efficiencies of ApplyIT's OptiRUN for managing and planning plant maintenance activities, and SmartSURE for capturing, reviewing and reporting SHEQ-related activities. "FlexiLOG was an obvious choice, both from a system quality perspective and for ease of use," says Ram.

The system went live two weeks after implementation: "Good communication between the ApplyIT team and Resinkem ensured the transition to the new system went smoothly and we were able to go live quickly with very few changes," explains Ram. "FlexiLOG now manages information on all plant operations at Resinkem, including shift handover logs, batch sheets and downtime records. Having critical data readily available allows the plant supervisor management to utilise resources more effectively and management make timeous decisions."

TEMPLATES ADD STRUCTURE

"The template-based format of FlexiLOG requires the input of information in a structured manner, promoting the communication of critical information that directly affects production. The templates compel the foreman to provide information of specific details about the shift, this requires precision in reporting, ensures accuracy and provides traceability," says Ram.

The quick trending facility in FlexiLOG tracks data from the previous 10 inputs and monitors process patterns based on comparisons of performance and output. The information provided by the system forms a basis for generating reports, managing key performance criteria and diagnosing and resolving faults.

The operations team has also used FlexiLOG to create an instruction log into which the production manager inputs requirements for the foreman in a structured format pertinent to specific plant areas. ApplyIT product director Scott Bredin says FlexiLOG has complemented Resinkem's use of OptiRUN and SmartSURE through their operating and reporting processes. "We have configured a section in FlexiLOG that summarises all work orders (through OptiRUN) and outstanding incidents (through SmartSURE) per shift. This creates an awareness of maintenance requirements and incidents during and between shifts, so the foreman can make decisions based on current information."

Bredin summarises the benefits FlexiLOG offers: "The name says it all – it is a flexible operator log book solution. What makes FlexiLOG unique is that it also supports workflow processes and so provides an audit trail."

Future plans for using FlexiLOG at Resinkem involve empowering each of the plant's operators to write their own shift log, with the foreman inputting a summary of the day's shifts.

For more information contact Scott Bredin, ApplyIT, +27 (0)31 514 7300, bredins@applyit.com, www.applyit.com

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Streamlined CNC production using Sinumerik

SIEMENS REDUCES PROGRAMMING EFFORT AND SHORTENS CYCLE TIMES WITH SHOPMILL AND SHOPTURN SOFTWARE.

Kottmaier CNC Drehund Frästechnik, headquartered in southern Germany, specialises in machining stainless steels and high-alloy tool steels. To enable it to carry out the time-critical process of producing sample and prototypes in line with customer requirements, the company has installed ShopMill, or ShopTurn respectively, on all of its CNC machines. The advantage of this arrangement is that workers can use several programming stations simultaneously, without the need for additional hardware solutions and office space.

Both solutions accurately reflect the operator's view of the work pieces and work processes by enabling an intuitive approach to programming and providing swift access to the CNC functions on the machines. Employees can therefore create parts programs themselves after just a short induction period.

PROGRAMMING IN GRAPHICAL OR ISO MODE

Although 3+2-axis machining of complex, non-prismatic parts makes up a high percentage of the order volume, the manufacturing company does not require especially trained CAD specialists and programmers. Company owner Helmut Kottmaier attributes this largely to the software's user-friendly user interface: "Graphical parts programs can be created quickly. Even for thread repairs, the programming of the thread cycles takes just a few seconds. It used to take at least several minutes."

Most of the employees now use graphic-aided program creation. For those who prefer the traditional ISO programming process, this option is available at any time in ShopMill and ShopTurn. The ISO code can be combined freely with the cycles used by the shop floor software. It is possible to switch from the graphical interface to the ISO view at any stage of the work process. This allows staff to continue using their CNC expertise. The operating concept of both software modules is also the same for both turning and milling. Staff can be dispatched flexibly for operation of both the turning and milling machines used in production.

NO MORE MANUAL RETOOLING OF WORK PIECES

Besides companies from the automotive industry, one key customer of the metal processing company is the robot manufacturer Kuka, whose order specific prototypes and sample parts often include complicated gearbox covers or flanges. To ensure that development and production of the prototypes and pre-production models makes steady progress, Kuka needs the parts as quickly as possible. "With the contour path editors in the shop floor software, we can create the design quickly from the drawing and manufacture it directly," explains Kottmaier.

The two turning machines from Spinner, recently procured, have also been equipped with ShopTurn as a user interface. A Sinumerik 840D is used as the control system. As there is no perceptible difference between the user interfaces and controls on the new machines and those on the machines used to date, staff were able to get to grips with the new machines straight away.

ShopMill and ShopTurn help not only with the production of individual parts and small volumes, but also in serial production of parts that are in frequent demand. Thanks to the software, the strengths of the individual technologies and machines can be utilised to optimise series production. Through clever use of the software, a part that was previously manufactured on the milling machine from parts cut in five mounting processes can now be produced on the CNC turning machine without the need for manual retooling. Labour time is minimised, which is particularly important for serial parts where the profit margin is often just a few cents.

SIGNIFICANT INCREASE IN OVERALL OUTPUT BY COMBINED USE OF MACHINES

With complicated turned parts that require the use of many tools, the combined use of a turning and milling machine is often a faster alternative, according to Kottmaier. "Producing a gearbox cover often requires the use of 32 different tools. This exceeds the capacity of the tool changer on many turning machines. Milling machines can hold considerably more tools.

"Instead of manufacturing a part completely on the turning machine and continually swapping the required tools manually, it is quicker to switch to the CNC milling machine for milling, drilling and thread cutting. Despite retooling, the parts are ready in a shorter amount of time and the overall throughput rate of the turning machine is considerably better. Precise retooling is also simpler and faster thanks to the shop floor software's setup function."

For more information contact Keshin Govender, Siemens Southern Africa, +27 (0)11 652 2412, keshin.govender@siemens.com, www.siemens.co.za



Analogue data-logger for power/voltage signals

Instrotech has introduced the German manufactured data-logger, the Ebro EBI 2, specifically designed for the recording and logging of power/voltage signals for use in laboratories, process monitoring, quality control, production and development.

The EBI 2 is accurate to within 0,1% of measuring range (+1 digit) at a programmable sampling rate of 1 to 8 s and a memory capability of 10 000 readings per channel. The highly-accurate, 4-channel EBI 2 is compact and handy and has an IP rating of 40. The unit is not dependent on AC power supply and battery has a life of approximately five years.

Programming and evaluation is via a PC using Winlog 2000 Pro software. Accessories include interface cables and a lockable wall mount bracket. Units are delivered with a factory calibration certificate.

For more information contact Pieter Deysel, Instrotech, +27 (0)11 462 1920, info@instrotech.co.za, www.instrotech.co.za

Personal electronic dosimeter

The EPD Mk2 electronic personal dosimeter from Thermo Scientific alerts personnel of potentially dangerous ionising radiation when working in radioactive areas, together with its range of software solutions, the EPD Mk2 is suitable for all dosimeter applications. Easy to use, the unit, together with its advanced radiological performance and multi-detector technology, provides an excellent response to gamma, beta and x-radiation.

The small, lightweight unit is immune to electromagnetic interference and provides loud and visual configurable alarms for dose and dose rate. The packaging meets IP55 protection for dust and water ingression to the level of low pressure jets from all directions. A single alkaline AA battery powers the unit continuously for approximately seven weeks. An interrogation software package incorporating infrared communication up to a distance of 1 m is also available. The EPD Mk2 is capable of retaining data up to 10 years without a battery; display and function control is via a single button on the front of the unit.

Over 1 500 units are in use in South Africa in applications ranging from laboratory through to nuclear power station safety.

For more information contact Mauritz van Niekerk, OEN Enterprises, +27 (0)11 675 4447, mauritz@oenenterprises.co.za, www.oenenterprises.co.za





New thermal imagers from testo

Sophisticated ergonomics, innovative features, outstanding image quality and the most modern technology.

The new thermal imager pro generation, consisting of the testo 885 and testo 890 are designed to address the most demanding applications. According to product manager, Sabine Hinkel, "In the forefront of the development were the varied measurement tasks of our customers.

Innovative features which make thermography easier and safer, in combination with the excellent image quality, the possibility of analysing thermal processes in detail over time and an intuitive operation are, in this combination, a benchmark on the world market for portable thermal imagers."

FOCUS ON BEST IMAGE QUALITY

The combination of first-class optics with a light, wide angle or telephoto lens, 320x240 (testo 885) and 640x480 (testo 890), pixel detector, an NETD (<30 mK) testo 885 and (<40 mK) testo 890 and high-quality system components, means exceptionally sharp focus and meaningful thermal images. In addition, a fourfold resolution can also be achieved, made possible by the patented SuperResolution technology. The SuperResolution takes multiple images within less than half a second, incorporating the natural movement of the hand. These images are then combined to create a much higher definition image.

If the depiction of the measurement object is not possible with a single image, the testo cameras offer a new panorama image assistant, which makes one total view out of several individual images – with full attention to detail and a full overview. The patented SiteRecognition technology can be used to support the efficient completion of inspection routes. This automatically takes care of the recognition of the measurement sites and the storage and administration of the resulting thermal images. The possibility of taking measurements in temperature ranges up to 1200°C allows the new series to offer the measuring range for every application.

FOR THE MOST DEMANDING USER

For the analysis of heat development in microelectronic components, the resolution of structures up to 115 µm is possible at a focus distance of only 10 cm. Using the fully radiometric video measurement (testo 890), thermal processes can be recorded in real-time, and valuable analyses for R&D carried out on a PC. The exact temperature values of all pixels are available at any time. Additionally, reliable thermographic evaluations can be carried out, even on extremely distant measurement objects, thanks to the telephoto lens and the excellent image quality.

The 885 and 890 cameras come equipped with the free testo IR Soft, an intuitive and easy to use analysis package with broad functionality.

ARCELOR MITTAL STEEL – SALDANHA DECIDES ON TESTO 885

Arcelor Mittal Steel in Saldanha chose the testo 885 thermal imager after its panel of engineers had reviewed a variety of different thermal imager brands and models on the market. Subsequently, supplier unitemp was chosen based on the 885 testo thermal imager's specifications as well as the training, support and competitive pricing. Saldanha Steel will be using the testo thermal imager to detect problematic areas in both the electrical and mechanical applications.

Mechanical friction results in heat that produces infra-red which is undetectable to the human eye. In industrial settings, the testo thermal imager is able to identify worn bearings easily, allowing the necessary maintenance to be done before a breakdown resulting in costly production downtime.

On the electrical side, the testo thermal imager can determine which conductors are overheating, caused either by overload, a loose connection, hot electrical switchgear components and connections or malfunctioning steam traps, allowing correction of unbalanced loads over a three phase supply.

The unit discovers problems like insulation voids, hidden wet areas from leaky seals in pipes and many others. In fact, testo thermal imagers easily identify most heat related problems.'

For more information contact Natalie Liddle, unitemp, +27 (0)21 762 8995, testo@unitemp.com, www.unitemp.com

NEW

Thermal Imagers

testo

IR Cameras for the highest demands



	testo 885	testo 890
Detector size	320 x 240 px	640 x 240 px
SuperResolution	640 x 480 px	1280 x 960 px
Thermal sensitivity	temperature resolution < 30mK	temperature resolution < 40mK
Field of view	30° (standard lens) & 11° (telephoto lens)	42° (standard lens) & 15° (telephoto lens)
Video measurement	3 points & video	3 points & fully radiometric video
Extras (for both)	Panorama image assistant • Site recognition technology • High temperature up to 1200°C	

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Leuze sensors in a modern logistics centre



Leuze electronic sensor systems have been installed in one of Europe's most modern logistics centres in Weilerswist near Cologne. The new dm-drogerie markt logistics centre covers a total area of 50 0000 square metres including the order picking warehouse for pallets and small parts as well as the high–bay warehouse. This massive facility boasts 18 000 pallet spaces, 5000 order picking spaces, 13 high–bay storage devices and some 3500 metres of pallet conveyors and has the capacity to handle up to 390 pallets per hour.

On top of this, there are some 140 000 storage spaces in the small–parts area in which 52 high–bay storage devices are used to warehouse, supply items in the order picking zones and retrieve the order containers used here. Small parts and boxed goods still on pallets can be temporarily stored in 30 metre tall high–bay warehouses in 14 000 additional pallet storage places. To facilitate order picking, up to 163 controlled telpher line vehicles or 'caddies' travel through the aisles over a total length of more than 2000 metres.

The complexity and high level of automation give insight into the demands placed on the sensor systems: specialised Leuze sensors are used to direct the flow of goods over extensively branched pallet and container handling technology; to position high-bay storage devices and telpher lines; to ensure the functions of machines such as robots, stretchers and turners and to transmit data and provide protection in safety areas. Some 1800 pallets carrying over 15 000 different articles are delivered to the dm markets every day from the new distribution centre.

QUICK IDENTIFICATION, SWITCHING, DATA TRANSMISSION AND STORAGE

The Leuze sensors used in the new dm–drogerie markt drugstore chain distribution centre are a key component of the state-of-the-art system technology. From the simple switching sensor to high-end solutions for barcode identification of pallets and boxes, Leuze solutions are playing a major role in ensuring the accurate and productive performance of the centre.

By using scanners for the precise positioning of high-bay storage devices and order picking telpher lines, as well as systems for wireless data transmission, Swisslog has created a gigantic logistics solution for the dm – drogerie markt in Weilerswist and as far as efficiency goes has broken new ground in Europe.

Some 700 barcode readers are located at each branch of the extensive material flow system allowing reliable and fast product identification. These devices were selected according to the necessary scanning distances and reading field sizes. Leuze positioning systems are used for the positioning of the high-bay storage devices and the CaddyPick telpher line vehicles, offering a high level of functional reliability and redundant data polling as well as minimal maintenance.

The transmission of the position data in the 75 metre long aisles is performed by DDLS 200 data transmission photoelectric sensors which permit contact and wear free optical communication without interference at a transmission rate of up to 2 Mbit/s over distances of up to 500 metres.

For more information contact Gerry Bryant, Countapulse Controls, +27 (0)11 615 7556, clive@countapulse.co.za, www.countapulse.co.za



Print marks yield to pattern

New detection method from Sick makes it possible.

Packaging without the need for print marks is precisely what the ML20 markless sensor from SICK Automation is designed to do. Used like a print mark sensor, however it does not require any optical marking on the package, instead it detects a pre-taught design pattern. The sensor features more flexibility and design freedom and needs less set-up time, thus avoiding unnecessary use of material and therefore wastage.

PATTERN INSTEAD OF MARK

The ML20, which can be used for controlling cutting edges, employs a novel method for contrast pattern detection. The technique involves teach-in of a pattern sequence that can be detected during processing at a scanning speed of up to 7 m/s and resulting in the output of a switching signal. Even in the case of complex designs, tolerances in the track course of the continuous material and high machine speeds, the ML20 delivers convincing performance due to reproducibility of up to 0.6 mm and extremely precise and stable pattern detection.

The ML20 opens up new possibilities for packaging processes. From the perspective of design engineers, the sensor offers a high degree of ruggedness as well as ease of integration and commissioning. For instance, the patterns to be recognised can be taught-in, saved, and then retrieved using the sensor's operating panel, the control cable, or Ethernet. The stable and process-tolerant detection of even complex image patterns ensures high availability and minimises downtimes and set-up times when changing formats are extremely short.

When changing label or film rolls, the sensor avoids unnecessary cutting loss and waste since pattern detection is active immediately. Moreover, in the case of procedures for labelling bottles, it is possible to save additional material that used to be dedicated exclusively to the mark. The ML20 is an economical and sustainable solution.

For more information contact Mark Madeley, SICK Automation, +27 (0)11 472 3733, mark.madeley@sickautomation.co.za, www.sickautomation.co.za

SILEC PROXIMITY SWITCHES

GST
GUDUZA SYSTEM TECHNOLOGIES

Silec proximity switches are very robust and designed to work in harsh environments and are used for detection and control of moving bodies. The switch consists of two changeover contacts and has a large operating gap for instances where there are big tolerance constraints.

Switch	Magnet and Maximum Operating distance
1919PS (Latching) 1919RA (Auto reset)	1927A36 N/S 60mm
	1922 60mm
1919PS (Latching) 1919RA (Auto reset)	1927A50 N/S 110mm
	1895 180mm

Main Features
- Operating gap 60mm to 180mm
- Very robust enclosures
- Operate in high temperatures

Features
- Maintenance free
- Reverse voltage protection
- Control Intrinsically safe circuits

Applications
- Level detection
- Positioning of moving bodies
- Door closing control mechanism
- Travelling crane - limits
- Vehicles - position and limits
- Skips and cages
- Monitoring and control



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Superior metal working performance from Roland Electronic

Thickness measurement and double sheet detection from one unit.

Some customer specific applications call for innovative thinking and new solutions. The new capacitance double sheet and in-line thickness control system, C100-S from Roland Electronicc, has been developed to accommodate just these and is now part of its standard product range. The machines capacitance principle brings some unique advantages when compared to sensor principles such as magnetism, eddy current and induction.

THE MEASUREMENT IS INDEPENDENT OF THE TYPE OF ALLOY

The capacitance double sheet detection system for metals is entirely independent of the type of metal, alloy or the type of heat treatment applied. For all metals, the system always measures the same thickness regardless of whether it is steel, aluminium or any kind of alloy. This is of special significance for many of the new materials used in automotive lightweight bodies, for example, the so-called duplex and TRIP steels. These and many other new materials often have vastly differing electrical and magnetic properties, which require measurement equipment to be calibrated taking into account many different metallurgical properties. The unique features of the capacitance principle lead to significant and cost effective simplification of the measurement platform and its integration into higher order control systems.

THICKNESS MEASUREMENT AND DOUBLE SHEET DETECTION IN ONE UNIT

When a metal sheet is moved into the sensor gap the system determines in-line the absolute thickness as an analogue value, which is then made available to higher order control systems for evaluation. This information is needed to ensure that the material is of the right thickness for further processing, e.g. in straighteners, slitters, cut-length lines or blanking presses. Alternatively, the absolute thickness can also be used as a nominal value to generate the lower and upper switching thresholds for double sheet detection. The capacitance system is therefore an excellent substitution for the mechanical double sheet detectors commonly used by manufacturers of press brakes, bending cells, turret punches or laser cutting machines.

COMPACT DESIGN

The system consists of a C-measurement frame and a control unit, in each part of the frame there is an electrode with a temperature probe. The C-frame is attached to the enclosure of the control unit C100-S and the measurement value is supplied to the control circuit as an analogue voltage, it is also visible on the LCD display. The integrated double sheet detector issues the output signals 0-1-2 sheets and for reliability, it has integrated air channels for periodic cleaning of the electrodes with an air purge. Metallic brushes at the sensor gap provide the necessary electrical ground contact to the sheet.

For more information contact Evanet,
+27 (0)11 802 3320,
evanet@evanet.co.za,
www.evanet.co.za

WORLD-BEAM® QS30ELVC
Clear Object Sensor



A Clear Choice for Clear Object Detection



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- Count clear bottles or vials on high-speed packaging machinery
- Detect glass tubes in light manufacturing
- Sense film web break
- Detect flat panel LCD screens
- Distinguish between single or double sheet of clear materials
- Detect semiconductor wafers





Popular WORLD-BEAM® QS30 for dependable detection of clear objects

The QS30ELVC Clear Object Sensor is an easy-to-use, high-performance retroreflective sensor optimised for detection of clear, translucent or opaque objects. The QS30 Clear Object Sensor utilises an advanced optical design to allow the sensor to see translucent materials, ensuring reliable detection of clear targets including PET bottles and glass containers. Additionally, the optical design prevents the sensor from being tricked by specular reflections, allowing the sensor to detect optically engineered surfaces such as mirrors, LCD glass with polarising films and semiconductor wafers.

- Reliable detection of clear, translucent or opaque targets—including mirror-like surfaces
- Auto compensation algorithm for adjustment to dust or contamination on sensor or reflector, and for ambient temperature changes
- Intuitive sensor setup with 3 selectable thresholds based on type of object being detected
- Push buttons or remote wire for easy sensor configuration
- Rugged IP67 (NEMA 6) housing for harsh environments and 1200 psi washdown rated per NFMA PW12
- Range of 100 mm to 2 m with a fast 0.5 millisecond response time
- Highly visible bargraph display plus LEDs for easy configuration and status monitoring during operation
- Two corner-cube reflectors included
- Extensive line of reflectors, brackets and cordsets for any installation requirement

Powerful capabilities with simple setup

- Compact housing with 30 mm threaded barrel and integrated side mount
- Single 1 SET programming of sensor
- Visible red LED sensing beam for simple alignment
- 10 to 30V dc with bipolar NPN (sinking) and PNP (sourcing) output
- Operating temperature range of -10° to 55° C
- Bright LED indicators for power and output status
- 2 m or 9 m attached cable, or 5-pin Euro-style integral or pigtail quick-disconnect

For more information visit:

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B63

Baumer's resistant inductive sensors



The inductive outdoor sensors IFRR (proximity switches) and IWRR (distance -measuring sensors) from Baumer meet the requirements of the highest corrosion category: C5-M. Both sensors are ideally suited to use in demanding ambient conditions at sea or on the coast, e.g. in wind power stations.

Digital inductive sensors are mainly used to detect nacelle position and the rotor speed. Measuring sensors are applied for monitoring the brakes and the shaft deflection. To meet the requirements of C5-M, the sensors were subjected to a 60 day salt spray test followed by a 720 hour water resistance test in accordance with ISO 6270. The results show that these inductive sensors meet the highest corrosion resistance demands and Baumer guarantees long service life from the products.

Thanks to the unique proTect+ concept, the sensors are leak proof in accordance with protection class IP69K even after numerous temperature cycles. This means that the sensors are especially low-maintenance, with an average service life (MTTF) of over 100 years. The wide temperature range of -40 to +70°C also ensures constantly reliable measured values and detection results, even at extreme temperatures or great temperature fluctuations.

For more information contact Temperature Controls, +27 (0)11 791 6000, sales@tempcon.co.za, www.tempcon.co.za

Attention-getting equipment status signals



Banner Engineering has introduced the new EZ-Light TL50 beacon tower light – a hybrid of its popular TL50 tower and K50 beacon lights – to its existing line of lighting and indicators. Like other models of the TL50, the new beacon tower is segmented and may be ordered with 1-4 different coloured LED lights, thereby allowing several messages to be conveyed at once. The new TL50 also features the ability to have any of the segments flash or rotate the extremely bright signals in addition to displaying a solid 'on' indication.

"We see tower lights as an important solution for so many industrial applications, and our new TL50 is one of the most versatile indicator lights we've developed," says Chuck Dolezalek, product development manager at Banner Engineering. "Not only does it work for both indoor and outdoor applications because of its brightness, but we also have audible models, which opens the door to the number of applications even wider."

The extremely bright output of the LED beacons provides easy-to-see indication from all directions when multiple lights may need to be on simultaneously. As a result, operators have the ability to monitor equipment status from a distance. Audible models that have easily adjustable sound intensity are also available, as are a number of bracketing options to enhance functionality and placement options. The TL50 has a wide 12-30 VDC operating range, which enables it to be used in battery-powered mobile applications.

For more information contact Rodney Topham, RET Automation, +27 (0)11 453 2468, rodney.topham@retautomation.com, www.retautomation.com



Presence detector with dimming

CP Electronics' new passive infra-red (PIR) presence detector provides automatic control of lighting loads with optional manual control. The EBDHS-DD is a high sensitivity detector suitable for high bay applications, such as warehouses and factories where high detection sensitivity is needed. It is designed for ceiling mounting and all functionality is fully programmable using an IR handset.

Output channel 1 comprises a mains voltage relay capable of simple on/off switching, while channel 2 provides dimmable control of either DALI or DSI type ballasts. Functioning as a presence detector, the unit can turn lights on when a room is occupied and off when the room is empty. Optional settings allow lights to be turned off in response to ambient daylight, or to implement a luminance daylight harvesting system.

The flexibility of having two channels and two switch inputs allows the following example scenarios: dim an outside row of luminaires whilst internal fittings are switched; provide absence detection for two separate channels; maintain an illuminance system with manual up/down controls.

The PIR sensor detects movement within the unit's detection range, allowing load control in response to changes in occupancy while a light level sensor measures total light level in the detection area.

For more information contact Chris Huxham, Shorrock Automation, +27 (0)12 345 4449, sales@shorrock.co.za, www.shorrock.co.za

Inductive sensors for the mining industry

In all automated and mining processes inductive sensors are absolutely necessary to provide the PLC with information. The sensors supply the necessary signals on positions, limit or serve as pulse pick-ups for counting tasks or for monitoring rotational speed.

Compared to mechanical switches, inductive sensors offer ideal conditions such as non-contact operation free from any wear and tear; high switching frequencies; and accuracy. In addition, they are insensitive to vibration, dust and moisture. Inductive sensors detect all metals without contact.

The efector m inductive sensors from ifm electronic feature an increased 4-22 mm sensing range. The sensors are suitable for almost all application areas and are resistant to aggressive oils and coolants. They are weld field immune, use the same sensing range for all metals, and offer ferrous only detection. Due to their universal connection technology, the sensors can be used as 2-wire or 3-wire units. The integrated LED provides quick and fast adjustment, reduces mounting time and ensures utilisation of the increased sensing range. All units have a permanent laser etched type label. This allows clear identification of the units even after many years of use.

The sensors have threaded cylindrical brass housings with diameters ranging from M8 to M30 and they offer a choice of NO/NC PNP/NPN outputs.

For more information contact Chris Cronjé, ifm electronic, +27 (0)12 450 0370, chris.cronje@ifm.com, www.ifm.com/za



Spacepak SP 2000 series



The SP 2000 series consists of a self-contained transmitter SPT and receiver SPR, which are to be used in thru-beam mode, an SPP for diffuse proximity, SPRR for retro-reflective, SPPR for polarised retro-reflective and SPBS for background suppression. All are offered with sensitivity adjustment via integral potentiometer with either cable or rotatable plug connection.

The complete series is available either as a 4 wire, NPN/PNP transistor output with 10-30 VDC supply voltage or as a 5 wire, relay output with a 12-240 VAC/DC supply voltage, both offering switch selectable light or dark function and potentiometer adjustable 0-10 sec on/off time delay.

The test input in the 10-30 VDC, SPT is intended to be used for disabling or enabling the transmitter power temporarily for test purposes or for multiplexing applications. The DC series is protected against reverse polarity of power supplies, test input and output signals. The output is protected against short circuit and inductive loads.

For more information contact Gail Norton Instrumentation, +27 (0)31 701 4861, telco@telcosa.co.za, www.gailnortoninstrumentation.co.za

Ethernet-compatible CCD Scanner

German based Wenglor has introduced its Ethernet-compatible CCD Scanner that recognises barcodes at distances of up to 320 mm. Housed in rugged aluminium, the CCD Scanner checks labels for completeness and quality of information and reads barcodes which are printed or laser etched directly onto glossy materials. The use of red light even enables barcodes with minimal contrast or poor printing quality to be reliably detected.

Scanned codes and the status of inputs and outputs can be viewed at a single glance in several languages via the OLED display, which aids intuitive operation. Inadvertent changes to selected sensor settings are prevented by password protection. The green 'Good Read' LED mounted directly on the scanner provides the user with clear feedback on the operating status.

In addition to the graphic display, the CCD Scanner can be optionally configured with the help of website setup specially designed for this product. Worldwide access is made possible by a web server so that data can be read out quickly and easily from anywhere. Remote maintenance is also enabled via this function.

For more information contact Anastas Schnippenkotter, ASSTech Process Electronics & Instrumentation, +27 (0)11 708 9200, info@asstech.co.za, www.asstech.co.za





Comtest introduces P3 series thermal imagers

Comtest Distribution has introduced the Fluke P3 Series thermal imagers that deliver practical performance for a wide range of diagnostic applications. Based on the award-winning Ti32, the P3 Series adds four additional products providing high-performance imagers for any budget.

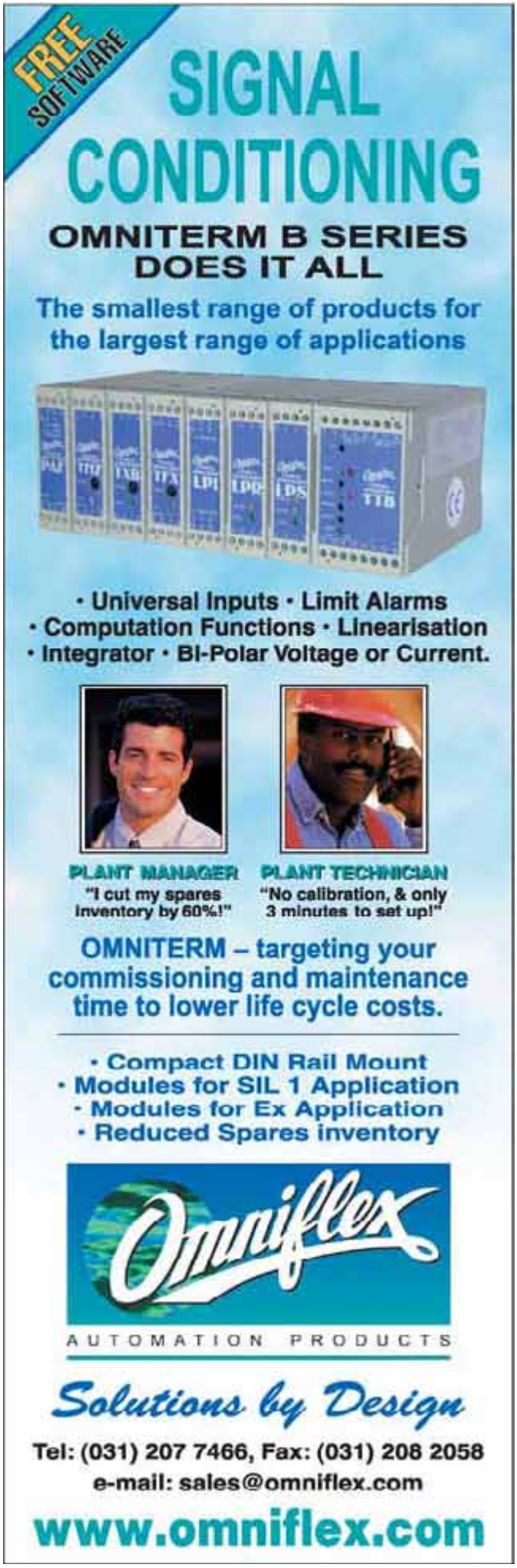
The P3 Series includes the Ti32 and TiR32, plus the new Ti27 and Ti29 models for industrial, electrical, mechanical, and process applications; and TiR27 and TiR29 models for building inspection, energy audit and building maintenance applications. All models are specifically designed to work in even the harshest environments, providing superior image quality.

The new family of imagers feature industry-leading thermal sensitivity, large total pixel image count and superior spatial resolution that, combined with the full VGA-colour LCD displays, provides sharp, clear images to make it easy to spot even the slightest temperature difference. They measure temperature from -20 to 600°C on Ti models (-20 to 150°C on TiR models) and feature a high temperature colour alarm on the industrial models and low temperature or dew-point colour alarm on the building models. They all allow voice annotation.

Fluke SmartView professional IR analysis and reporting software is included with each thermal imager with unlimited licences and free lifetime upgrades. The modular suite of tools for viewing, annotating, editing and analysing infrared images enables users to edit images in five viewing modes and generate customised professional reports in a few steps using its report wizard. The software fully supports Fluke IR-Fusion technology and features Fluke 3D-IR™ three-dimensional viewing that makes hot or cold spots pop out of the background for easy viewing and analysis.

The P3 series are supplied complete with a 2GB memory card, two rechargeable batteries that provide more than 4 hours operating time each, AC charger/power supply, adjustable hand strap and Fluke signature hard and soft transport cases.

For more information contact Val Verwer, Comtest, +27 (0)11 608 8520, info@comtest.co.za, www.comtest.co.za





Locked in real-time profitable safety

IMPROVING SAFETY AND ENVIRONMENTAL INTEGRITY IN REAL-TIME CAN IMPROVE PROFITABILITY.

Peter G. Martin, vice-president and Invensys Fellow.

Safety of people, equipment, facilities and the environ-ment is becoming increasingly important in industrial operations. This is not just because of highly publicised accidents or even for altruistic reasons. Rather, it has become evident that the safety of an operation is directly linked to profitability. Although it is true that industrial executives are very concerned over the safety of their operations. It's the increased profitability potential that can be realised through more effective safety management that is really starting to turn heads and to truly bring safety and environmental integrity into the mainstream of industrial business processes.

For decades, industrial professionals have recognised that the cost of unexpected events, such as fires or explosions, can be very high in terms of injury, death, equipment damage, facility damage, environmental damage, business interruption, and insurance. Industry has responded to this high potential cost through functional safety programmes, such as installing safety instrumented systems to detect pending unsafe conditions and automatically taking the correct response. These systems have proven to be very effective at the avoidance of predefined unsafe events and certainly represent a huge step forward. But cost avoidance of unexpected events represents just a small piece of the overall potential profitability impact that may results from an effective safety control solution. The 'cost avoidance' perspective tends to be too easy for business executives to turn a blind eye to, because they are lulled into believing that unsafe events will hit other businesses and not theirs. Unfortunately this tends to be an all too human thought process for all kinds of similar events, which seems to have served to inhibit capital spending on safety improvement initiatives.

All executives have a keen interest in is profitability and most are willing to invest in approaches that measurably improve profitability. What may not be totally clear is the impact operational and functional safety control can have on profitability even in the absence of costly unsafe events. It has often been difficult for Environmental, Health & Safety (EH&S) leaders to get funding for the capital projects they propose because the 'payback' is not obvious. But the payback for a systematic approach to operational and safety management and control can be huge. Industry needs a new way of thinking about, measuring and improving safety.

REAL-TIME VARIABILITY AND PROFITABILITY

Over the past decade, key industrial business variables, such as the price of energy or raw materials, have transitioned from highly stable over long periods of time to high variability within a day or even an hour. This shift to real-time variability of industrial business variables presents daunting challenges to industrial executives. A number of key components of the profitability of an operation, such as production value, energy cost and material cost are fluctuating more and more frequently with time. These three variables can be considered the real-time vector components of profitability. The primary object would be

to maximise the production value of the operations while simultaneously reducing the energy and material costs by as much as possible.

Balancing these three variables to maximise the profitability of the operation is a fairly classic control problem, which is why it is referred to as real-time profit control. To solve such a control problem requires a classic control approach.

THE IMPACT OF SAFETY ON REAL-TIME PROFITABILITY

There are constraints on the profitability that limit the length of the vectors and the overall profitability of the operation. For example, the installed manufacturing equipment itself will provide some limits, such as the capacity of a pump, the size of a vessel or pipe, and the efficiency of a unit. Most constraints are fairly easy to identify and are fixed. But there is one constraint set that tends to fluctuate in real-time. That constraint set is safety (people, process, environmental). The actual real-time profitability model should include the safety constraint as there is a tight relationship between operational safety and real-time profitability.

Since many industrial operations are very safety focused, the safety constraint is typically a tighter constraint on the profitability of the operation than other constrains, such as physical equipment constraints. Also, since safety risk is seldom directly measured, most industrial plants assume the risk is greater than it may truly be and operate in a more conservative manner than necessary, limiting profitability. Therefore, if the safety constraint functions can be expanded, the operation can be driven harder in order to increase profitability. One key to improving operational profitability, therefore, is the opening of the safety constraint on the profitability.

OPENING THE SAFETY CONSTRAINT

The key to being able to determine where the safety constraint truly is and how to free it up is to measure accurately the safety risk of a piece of equipment, process unit, plant area or plant in a real-time, continuous manner. With ongoing real-time measures of safety risk, plant operations can determine how hard they can safely drive the plant. Without the real-time safety risk measures, when circumstances around the operation combine to increase the safety risk, that increased risk often goes unnoticed. This results in not only a less profitable operation, but also a less safe operation.

The deployment of continuous, real-time safety risk measures is the key to controlling the safety constraint. This can be accomplished by developing a dual view of safety risk and combining this dual view into composite safety risk measures. The first view of safety risk, operational and compliance safety risk, involves following the processes and procedures with respect to inspections, audits, and reviews determined during the design of the plant. These audits should be accomplished on a defined schedule to ensure that the operational and compliance risk are as low as possible. History has demonstrated that operations that follow the compliance audits judiciously tend to have much lower safety risk.

The second view of safety risk, conditional safety risk, involves online condition monitoring and measuring. Unexpected events and conditions that cannot be discerned during an inspection can develop to drive up safety risk. The information necessary to discern increases in the probability of an unexpected, unsafe event should be readily available for any plant with a reasonable operational history. The aspects of the conditions that might lead to an event can be monitored to detect any suspicious changes. Automatic workflows can be triggered that further assess the situation, increase the conditional safety risk measure according to the probability and severity of a potential event, and advise the operating personnel.

Since having two separate safety risk views and corresponding measures increases the operational difficulty of assessing the actual safety risk and could lead to confusion, a composite safety risk measure for the operation should be developed. This composite measure should take the operational and compliance safety risk measure and conditional safety risk measure as inputs and perform appropriate analytics on the two measures to establish the actual current safety risk. The composite measure should be heavily weighted to the higher of the two input safety risk components. It might be easy to merely take the higher of the two component risk factors, but this approach may be too simplistic for complex production operations. Some operations will require an analysis of both component measures against each other to develop the composite safety risk measure of the operation. With this real-time safety risk factor for every operation in the plant and for the plant as a whole, the operating personnel can make decisions that improve plant profitability while understanding the impact of their decisions on the safety risk in the plant.

REAL-TIME SAFETY RISK CONTROL

For operational and compliance safety risk issues, controls can be applied by keeping the operational safety risk measurement visible to the leaders of the EH&S team responsible for the safety compliance audits, and, if significant violations of expected processes are encountered, to automatically notify plant management of the violation. Automatically triggered workflows can also be developed in the system that trigger workflows to notify and guide the compliance processes and to keep management in the loop when the processes are correctly followed as well as when there may be an issue. This level of feedback sets up a manually operational and compliance safety risk control system that will certainly help to reduce the safety risk of a plant.

A similar approach can be taken to the conditional safety risk of the operation, although conditional risk is typically less dependent on human processes than it is on unexpected failures and other unanticipated events. For the conditional safety risk a predictive conditional safety system can be developed using automatic workflow to monitor for a potential safety condition, identify the condition and take

Continued on page 74

Continued from page 73

the appropriate corrective action. All components of this conditional safety risk control system can be set up for automatic operation.

Setting up the correct safety conditions monitor involves a detailed understanding of the history of operation of the plant and of other similar plants. Historical data must be analysed to identify the lead indicators of an unexpected event and must set up the safety condition monitor to trigger from these lead indicators. A corrective action function must be set up to take various corrective actions based on the circumstances identified for the potential event. This is not unlike the approach to safety taken with traditional batch service logic for decades, but this perspective must be taken for the plant as a whole.

REAL-TIME PROFITABILITY CONTROL

Opening the safety constraints on a production operation does not, by itself, lead to increased profitability. It merely means that there may be potential to increase profitability safely. Actually driving increased profitability requires the application of control theory to the three variables of real-time profitability. This can be accomplished through a manual feedback control system driven by contextualised profitability dashboards for each operator. The issue of tying the safety risk measure and the profitability controls together is as simple as providing a real-time safety risk indicator on the operation profitability dashboards. This visible real-time safety risk value will provide immediate feedback to the operators of the increased or reduced safety risk resulting from any actions they take. This enables operators to take appropriate profit improving actions while simultaneously mediating safety risk. In a sense, simultaneously balancing safety risk, production value, energy cost and material cost this is tantamount to solving a multiple objective optimisation problem. Mathematically solving this type of problem is very challenging and time consuming, but experience has demonstrated that an operator with a moderate educational level and reasonable experience can learn to solve such a problem quite effectively over time.

SUMMARY

Traditionally, safety and profitability have appeared to be diametrically opposed concepts for many manufacturing and production operations. Since the primary objective for any manufacturing or production business is to drive profitability, the safety of people, plant and environment have often been viewed as somewhat necessary evils. As such Environment, Health & Safety (EH&S) teams in industrial companies have typically been viewed as adjunct organisations that are not part of the main stream business. Such positioning has made their job more challenging. Fortunately, the state-of-the art in control theory is leading industrial companies in a different direction with respect to safety. The safe operations of industrial plants are now understood to be a direct factor in the profitability of those plants. ES&H can be turned into a mainstream business process of industrial companies and new levels of both safety and profitability can result. Real-time safe profitability is no longer a dream – it is a reality!

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Multi-sensing technology in gas lift applications

Yokogawa application guide.

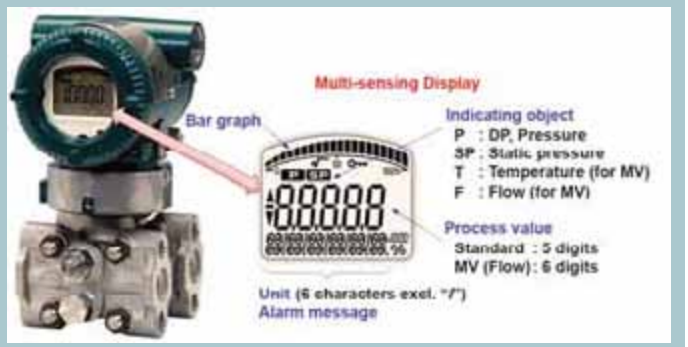
Cost saving with EJX pressure transmitters.

In a typical gas lift system, a compressed gas is injected into the production tubing to reduce the hydrostatic pressure of the fluid column. As a result, oil or gas can be produced at higher flow rates. In this application, the flow volume of injection gas pressure and produced oil/gas are measured differentially using in-line gauge pressure transmitters; compliance with the latest safety standards such as IEC 61508 is becoming increasingly important on these types of production line.

Particularly challenging is unexpected overpressure from underground being generated in the line. The process static pressure of the injection gas line is very high (approximately 1,7 MPa), so In order to secure accurate and reliable measurement the stability of the transmitter in the process is critically important.

THE SOLUTION

Yokogawa's EJX110A multi-sensing transmitter delivers the two process variables of differential pressure and static pressure simultaneously. This enables end users to save the cost of a gauge pressure transmitter for monitoring the line pressure.



The DPharp silicone resonant digital sensor exhibits superior stability and repeatability even under overpressure or high static pressure condition. In addition, EJX series guarantees long-term stability, 0,1% of URL for 120 months, under real-world process conditions.

With its unique fail-safe feature of DPharp sensor, EJX series transmitters, except fieldbus communication types, are certified by RWTUV in compliance with IEC 61508 as standard. The EJX delivers reliability for plant safety while optimising operational uptime.

For more information contact Johan van der Westhuizen, Yokogawa SA, +27 (0)11 831 6300, johan.vanderwesthuizen@za.yokogawa.com, www.yokogawa.com/za

Cables for safety valves in oil pipelines

Since the devastating oil disaster in the Gulf of Mexico, safety and environmental protection have become critical issues for the oil and gas industry. Lapp Muller SAS, part of the Stuttgart-based Lapp Group, is particularly involved in this. Through its many years' experience and excellent reputation, Lapp Müller recently won a contract from the Italian company Breda Energia for a project in conjunction with Gisma and Seacon.

The contract involves the development and production of an underwater cable (umbilical), made up of high-pressure hoses and control cables for opening and closing underwater valves in oil pipelines at depths up to 50 metres off the Mediterranean coast of Egypt.

The cable has been produced in a single length of 900 metres with a tensile strength of nine tons and a total weight of almost six tons. The construction of the cable is impressive. It has a diameter of 71 mm and is filled with two high-pressure hoses for up to 210 bar, as well as special four-wire screened control cables stabilised by a steel cable in the centre of the underwater cable. All components are surrounded by a three-part sheathing layer made up of a polyurethane sheath, extremely strong aramide fibres and an outer polyurethane shielding.

The control cables open and close the giant valves using the high-pressure hoses. The cables are also connected to the controllers and drives that control the position of the valves. The internal logistics alone brought major challenges for Lapp employees as the project had to be completed in less than three months. To complete the job, additional machines had to be rented along with a special crane for loading the three cable drum. Before the underwater cable was approved and delivered, it was tested for a week for safe functioning and reliability by specialists from Seacon.

For more information contact Lapp Group Southern Africa, +27 (0)11 201 3200, info@lappgroup.co.za, www.lappgroup.com





New Profibus indicators for hazardous and safe areas

Beka has introduced five new eight variable Profibus indicators with large digits to compliment its well-established Profibus displays that are already in use in petrochemical and pharmaceutical plants worldwide. The new range comprises economical field and panel mounting intrinsically safe models, a field mounting type 'n' instrument for Zone 2 applications, plus field and panel mounting indicators for use in safe areas.

For maximum visibility, the new models feature a large, easy to read 5 digit display plus a 31 segment bar-graph. One of up to eight pre-configured process variables is displayed, the displayed variable being selected by scrolling-through the preconfigured variables using the indicators up and down front panel push buttons.

Each Profibus indicator may be configured as a fieldbus listener or as a fieldbus node.

When configured as a listener, the indicator is not visible to the fieldbus host; may not be subject to a node licence fee and is configured and controlled via the instrument's push buttons. As a fieldbus node the indicator is configured via the fieldbus host, but the indicator's fieldbus address may be entered via the instrument's push buttons, which may also be used to return operator acknowledgements to the host.

The field and panel mounting intrinsically safe models have ATEX, IECEx, FM and cFM certification permitting installation in most gas hazardous area, the field mounting indicator also has dust certification. For applications in Zone 2 or 22, the BA444NDF has Type nL, nA and tD approval, permitting installation without the need for Zener barriers or galvanic isolators thus significantly reducing installation cost. South African IA certification will be available.

The field mounting models have a robust IP66 GRP enclosure with an armoured glass window suitable for installation in harsh industrial and offshore environments. A separate terminal compartment with forward facing terminals simplifies installation and maintenance. Accessories include pipe-mounting kits and stainless steel tag plates that can be supplied etched with customer specified tag and application information.

The panel mounting indicators have a 144 x 72 DIN enclosure with an IP66 front panel and a gasket to seal the joint between the instrument and the mounting panel which allows them to be hosed-down. Accessories include pipe mounting kits and display escutcheons printed with customer specified application and tag information; alternatively for applications requiring a stainless steel tag, the transmitter can be supplied with a custom etched stainless steel legend plate secured to the front of the instrument.

For more information contact Extech Safety Systems, +27 (0)11 791 6000, extech@jhbmail.co.za, www.extech.co.za

Equipment control in heavy duty environments

Guduza System Technologies introduces Silec magnetic switches.



A problem in the heavy equipment industry is the reliability of the sensitive electronic equipment operating in these harsh environments. Proximity switches, which play an important role in detection, are subjected to an environment where moisture, temperatures as well as mechanical shock and vibration puts endurance to the test. Productivity dictates the need for reliable and durable ancillary equipment that can 'take the punches' over a long period of time.

The Silec proximity switch series has been designed to work in harsh environments. Housed in thick metal enclosures, the magnets are potted in preparation for these rough environments. The design makes provision for an air gap between magnet and switch of between 60 and 180 mm to accommodate equipment with large mechanical tolerances.

The two types, namely automatic and latching, with their associated magnets are designed to cover most applications in the mining industry, including the control of skips and cages, positioning of conveyor belts, controlling of door locks, positioning of cranes as well as providing limit stops.

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www.gst-guduza.co.za

Easy to read
Profibus
Indicators and
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PROFI BUS



PROFIBUS PA INDICATORS

> 8 variables selected by front panel push buttons
> Easy to read 20mm high 5 digit display
> 31 segment bargraph
> Configurable as Profibus Node or Listener
> Push buttons can return operator acknowledgements
> Intrinsically safe, Type n, nonincendive and safe area models
> FISCO and FNICO compliant
> Field and panel mounting

PROFIBUS PA DISPLAYS

> 8 variable display
> Choice of 11 standard display screens, some with bargraphs
> Backlight
> Push buttons to return operator acknowledgements
> 6 optional outputs
> Intrinsically safe, nonincendive and safe area
> FISCO compliant
> Field and panel mounting

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Leuze's Rotoscan RS4 safety laser scanner

PROTECTING AND POSITIONING IN ONE DEVICE.

In-plant logistics solutions increasingly use self-driving shuttle systems, known as side-tracking skates, between the conveyors. They permit goods being conveyed to be shifted perpendicular to the direction of transport in an automated fashion.

More and more areas are being automated where products were traditionally moved manually or by forklift truck to intermediate buffers or transported to packaging lines. This level of automation with shuttle systems, generally known also as side-tracking skates or distribution cars, provides a solution which guarantees greater flexibility and the major advantage that these areas remain practically open, allowing people to walk through unhindered or free moving corridor supply vehicles to drive through.

However, with this advantage comes the need to ensure the protection of workers in the area, which is typically done using safety laser scanners on these vehicles or systems.

LEUZE OFFERS THE SOLUTION

Leuze's Rotoscan RS4 safety laser scanner is considered highly suitable for this type of application as it has the ability to measure distance and detect a vehicle position with absolute accuracy.

Gerry Bryant, managing director of Countapulse Controls, cites as an example a solution implemented in the paper processing industry where the guillotined paper is placed onto pallets and then shrink-wrapped for protection. The pallets are then transported via roller and belt conveyors and the sideways distribution of the pallets to the various plant areas is implemented via side-tracking skates.

"Firstly, in this type of application it is necessary to protect personnel in the areas where the side-tracking skates pass and secondly it is necessary to track and determine the exact positioning of the side-tracking skates at all times," says Bryant.

The Rotoscan RS4 Safety Laser Scanner fulfils both requirements in one device and has an integrated interface for the optimum connection to Profibus DB (Profisafe). One of the main features is the intelligent MotionMonitoring function, which makes the devices predestined for use in mobile applications such as side-tracking skates. Apart from reliable protection for the transportation path, this function permits robust and accurate measurement of speeds and distances: up to six speed values can be monitored.

If necessary, protective field adjustments can be applied in the form of extended protective fields at higher speeds. Moving personnel do not interfere with the robust distance measurement in this case. The current measurement value is given a 'quality' rating and the respective rating or measurement quality of the data determines the subsequent behaviour of the Safety Laser Scanner.

Bryant explains that in practice this lets people move through the sensor's measuring field without triggering a stop signal, depending on the distance and speed of a side-tracking skate. In-plant movements on the transportation path thus do not interfere with the operation of the shuttle – without restricting the safety of persons in the path.

The robust distance measurement can also be used to position the vehicles, and the new measuring procedure of the Rotoscan RS4 permits measurements of the distance to defined reference planes. This can be used to determine positions precisely.

Just as for the side-tracking skates on which pallets with paper stacks are transported to the packaging machines, the conveyor segments of the vehicles can stop at the loading and discharge stations with millimetre precision to transfer the pallets. The braking, stopping and loading at the roller conveyors is also monitored.

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Retrofitting machines affects safety

When does a substantial and thus CE-relevant transformation apply?

SICK Automation

The EU Machinery Directive, MD 2006/42/EC, is addressed to all manufacturers and distributors of machines, including operators that assemble or retrofit existing machines if they implement substantial modifications.

WHEN IS A TRANSFORMATION SUBSTANTIAL?

The EU guideline for the use of the machinery directive defines a substantial transformation as a transformation that has resulted in a machine that can be considered as 'new machinery'.

In the Blue Guide of the EU Commission, the aspect of risk is addressed: "if the risk assessment leads to the conclusion that (after retrofitting the machine) the nature of the hazard or the level of risk (emanating from the machine) has increased, then the modified product (machine) should be considered as a new product (machine)."

The party responsible for retrofitting must carry out the entire CE conformity procedure if, through the substantial modification, the retrofit machine becomes a new machine. The GPSG interpretation guide entitled 'Substantial transformations of machines' helps to classify the modifications in terms of legal and safety related aspects. It follows from this that a hierarchy of criteria must be met in order for a retrofit machine to become a new machine. The first criterion is the existence of modifications concerning performance, function and safety technology. This is followed by the assessment of whether new danger or increased risk exists. If examination of further criteria reveals that no sufficient technical safety measures on the machine are possible, protective guards cannot be used, the danger of substantial personal injury and damage to property exists, and the probability of an accident is high, new proceedings to evaluate conformity must be carried out. In this connection, it is important that users do not have to take existing safety technology out of service automatically.

What may be used depends on a variety of factors: the safety experts from SICK Automation, an accredited inspection centre, are available to help.

For more information contact Mark Madley, SICK Automation Southern Africa, +27 (0)11 472 3733, mark.madley@sickautomation.co.za, www.sickautomation.co.za



From the world's largest safety portfolio

The compact evolution of the S3000 and S300: the S300 Mini safety laser scanner.
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SICK
Sensor Intelligence.



Easy integration of IS serial measurement devices

INTRINSICALLY SAFE ETHERNET GIVES ACCESS TO ZONE 1 AND ZONE 0 HAZARDOUS AREAS.

MTL Instruments has announced the release of the 9400 series intrinsically safe (IS) Ethernet range and the launch of an enhanced power supply, 9491-IS, which incorporates MTL's patented Fast-Switch technology.

The deployment of Ethernet into process automation has long been hindered by the hazardous area classification in which the end device is often installed. Consequently the majority of industrial Ethernet installations have been in the factory automation environment. Additionally, in process automation it is preferred to use a single cable, providing power and communications to the end device. The 9400 Series delivers Intrinsically Safe 'Power over Ethernet' (PoEx), allowing live connection and disconnection of the end device in Zone 0 and Zone 1 hazardous areas, typically reducing the cost of installing the LAN in hazardous areas by 40%.

The 9400 series comprises a wireless LAN access point, managed Ethernet switch, copper to fibre media converter and serial gateway, all certified to be mounted in a Zone 1 hazardous area with connectivity into Zone 0. Additionally, Zone 2 mounting IS Ethernet isolator and associated IS power supply complete the product range. The 9491-IS power supply uses MTL's patented Fast-Switch technology to allow live power connection/disconnection without the need for a gas clearance certificate.

The gateway gives legacy IS equipment Ethernet connectivity by allowing conventional serial communication port equipment to be connected to an Ethernet network. This enables easy integration of a wide range of IS serial measurement devices including analysers, weighing systems, dust monitors, displays, I/O etc. into today's control systems.

The IS wireless LAN product is a multi-functional module that can be used as an 802.11a/b/g/h access point, wireless bridge or wireless repeater. This provides hazardous area infrastructure for the mobile operator using IS PDAs anywhere on the process plant. The IS-managed Ethernet switch allows the interconnection of intrinsically safe Ethernet networking components via its five ports. The Ethernet switch can also distribute power to compatible devices connected to each of its five ports, thus providing IS PoE via the RJ45 Cat5e cables. The fibre optic media converter allows an Ethernet network to be extended over a greater distance.

This technology has been proven in use in many mining applications and with the new ATEX and IECEx certification the 9400 series delivers the same benefits for all process industries. The 9460 series has South African IA certification as required by ARP0108:2007.

For more information contact Extech Safety Systems, +27 (0)11 791 6000, extech@jhbmail.co.za, www.extech.co.za

Level switch for the petro-chemical industry



ENDRESS+HAUSER INTRODUCE THE LIQUIPHANT WITH STAINLESS STEEL HOUSING.

Proven in more than three million applications worldwide, the Liquiphant is a market-leading liquid level switch. Now its application possibilities have opened up further with the introduction of the dual compartment stainless steel housing specially designed for the demands of the petrochemical industry.

The new heavy-duty stainless steel housing addresses the need for a more robust device for the rigours of petrochemical applications, offering ATEX Ex d certification on its dual compartment housing. Liquiphant is the ideal replacement for maintenance-intensive float switches as it's simple and cost-effective yet offers real benefits in terms of its self-monitoring function and SIL2 and SIL3 ratings. This gives Liquiphant a big advantage in safety-critical applications as it offers outstanding overspill protection. What's more, Liquiphant's process connections are now available in a C22 highly resistant nickel-chromium alloy with a 3.1 certificate for all wetted parts.

Able to withstand high temperatures (up to 280°C) and pressures (up to 100 bar), this level switch has been made-to-measure for petrochemical processes. The patented piezo drive offers improved immunity to both foam and plant vibration. Able to be mounted in any position, its small process connections make it ideal for applications where space is at a premium such as offshore rigs and downstream installations. Even better, Liquiphant is maintenance-free and operation is unaffected by flow, turbulence, air bubbles or build-up, making it the perfect choice for high level and low level alarms. Its integral function check means that Liquiphant can be remotely tested without being removed from the process, improving safety procedures and saving valuable time.

For more information contact Jan Gerritsen, Endress+Hauser, +27 (0)11 262 8000, info@za.endress.com, www.za.endress.com



Fire protection for substations

PYROGEN AEROSOL GENERATORS LOCALISE THE FIRE SOURCE AND MINIMISE COLLATERAL DAMAGE.

For industry the need to protect critical processes is crucial to minimise risk exposure that might be catastrophic, take lives and cost a company millions in lost production and recapitalisation. Substations are a key component in many critical processes mining, providing power to winders, crushers, conveyors, pump stations, etc. Any damage to a substation could result in severe or crippling losses to a mine therefore, fire protection is seen as a valuable requirement in the construction and maintenance of any substation as well as forming a vital thread in the overall fabric of any mining company's health and safety policy.

The principles behind fire protection of an enclosure are to:

- Have an early warning fire detection system.
- Have an effective fire extinguishing system.
- Detect and extinguish any fires.

LIMITATIONS OF TRADITIONAL METHODS

Gaseous extinguishing systems operate in conjunction with a smoke detection and control system. Typically, a smoke detection system will comprise optical and ionisation point smoke detectors located along the ceiling or roof section of a substation. While this will detect smoke, it is less than optimum for fire protection purposes, because it does not adequately meet the need of an early warning smoke detection system in the context of high tension electrical cabinets inside an enclosure. Testing has shown that should a fire occur inside a pressurised IP55 cabinet, the fire would have to be well into its final stage before there is enough smoke emitted to reach the point smoke detectors.

It gets worse when operators consider the temperature inversion due to the effects of the sun on the corrugated iron roofs. Due to the difference in density levels between the air near the roof and the air below it, it requires a lot of smoke for Brownian motion to penetrate this layer and reach the smoke detectors. Heat detectors are not an option either since they will react at a much later stage.

NEW TECHNOLOGY ALTERNATIVES

For protecting substations, cutting edge technology introduced by Alien Systems & Technologies ensures that all the principles of fire protection are met for cabinet fires that occur indoors, and more particularly, inside the electrical control cabinets within substations. The introduction of Pyrogen aerosol generators expands the choice of fire extinguishing mediums and advances the available detection and actuation methods.

The Pyrogen range of fire extinguishing aerosol generators, from 20 gram to 17 kilogram, has major advantages over gaseous extinguishing systems that use point detection systems. One is the ability to be situated internal to the volume to be protected, ranging from engine compartments in vehicles, aircraft, and marine vessels, to electrical and electronic equipment. The various methods of activation, plus the nature and composition of the post activation residue give the Pyrogen unit the advantage inside the cabinet.

Activation of the Pyrogen units includes electrical, thermal cord and self, and is described more fully below:

- Electrical, from a conventional fire alarm panel.

- Thermal cord attached to the Pyrogen unit, initiation at a pre-set temperature.
- Self-activation in the event that the two previous methods fail.
- The post-activation residue has a 24 kV insulation property, enabling immediate re-use of electrical equipment following replacement of the components which created the fire.

These design and construction features of Pyrogen enhance the normal 'failure to safety' required in fire extinguishing systems, resulting in fire detection and extinguishing systems employed in electrical equipment being a fail stage autonomous detection and fire extinguishing system. The system can be configured to require no external power or input fire signals, on standby 24 hours per day, for the ten years installed life of the Pyrogen unit, all with minimum maintenance.

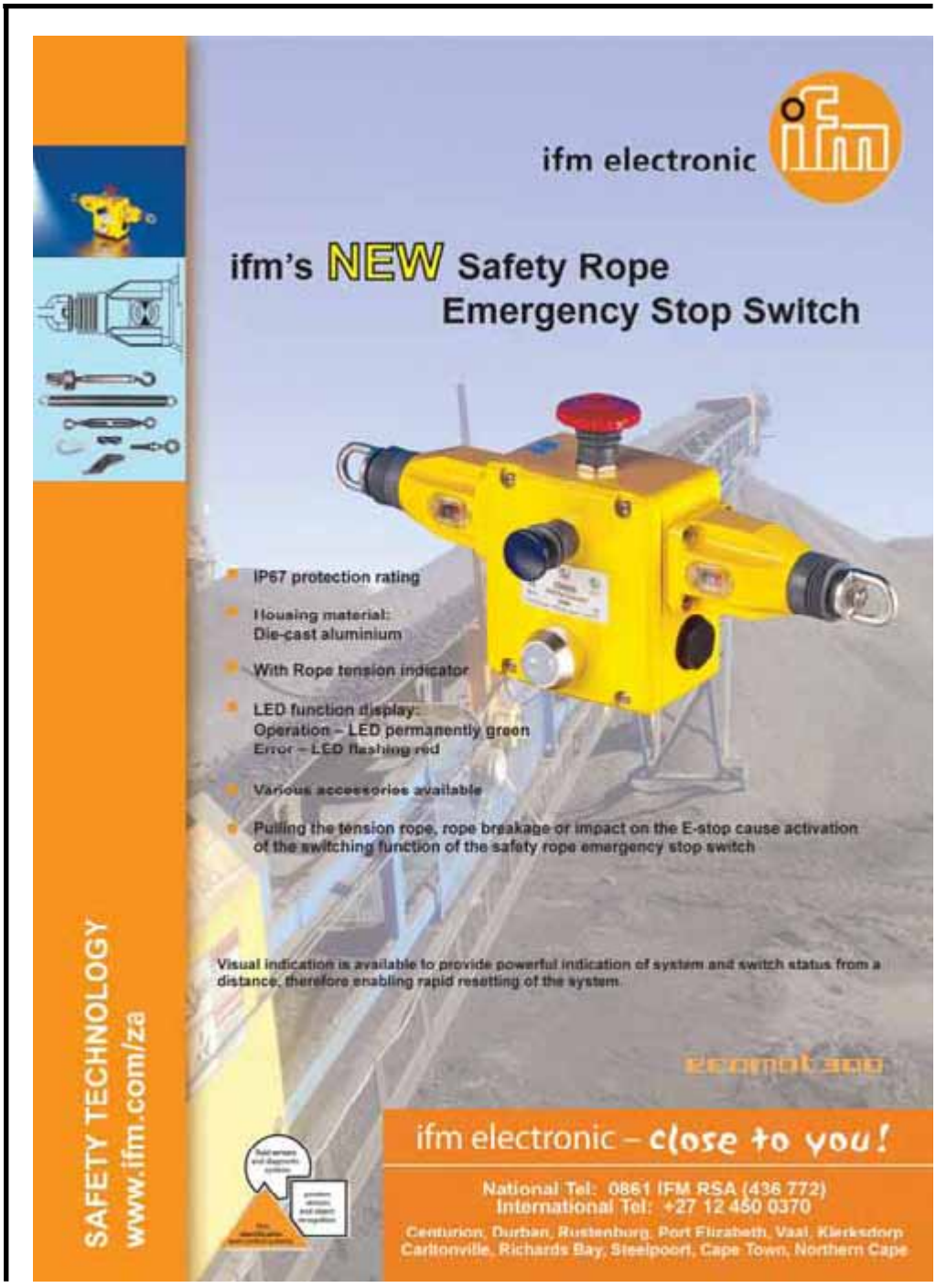
ADVANTAGES OF FIRE DETECTION AND EXTINGUISHING SYSTEMS FITTED INSIDE THE CABINET

Electrical equipment, by requirement, is constructed to minimum IP55 protection i.e. gasketed doors, covers and fitments. This precludes the egress of smoke from any fire or electrical fault within any compartment of the switchgear being protected. This is especially true when the substation is ventilated by positive pressure fan systems. The fire would be of major proportions before the smoke detectors in the substation would activate and initiate the release of the fire extinguishing medium. When activated, the substation is flooded with the extinguishing medium having no means of ingress to the switchgear compartment. The result is burn out and destruction of the electrical switchgear panel in which the fire started.

The point smoke detection system, in addition to rigid servicing requirements, is susceptible to extraneous fumes being drawn into the substation by the substation fans resulting in the activation of the gaseous fire extinguishing system. As with a real fire, the system cylinders would have to be dismantled, refilled and re-assembled, during which time the substation is left with no fire extinguishing system.

The in-cabinet Pyrogen system, being impervious to extraneous fumes and smoke, would only activate on an actual fire within the cabinet being protected, thus requiring only one Pyrogen unit replacement, the remainder of the adjacent switchgear being left with full fire protection. Pyrogen aerosol generators localise the fire source and minimise collateral damage.

For more information contact Alien Systems & Technologies, +27 (0)11 949 1157, sales@astafrica.com, www.astafrica.com





Conveyor control system from ifm electronic

Chris Cronjé, ifm electronic SA.

In the mining industry there are many conveyors that transport the raw material from underground facilities to the processing plant. The mineral bearing rock has to be moved over a long distance via robust conveyor systems. These conveyors can be longer than 2 km and have to be very flexible due to the different locations of the mining fields. ifm electronic has developed a conveyor control system using AS-i (Actuator-Sensor interface) which allows connection of all the stations of a conveyor system e.g. safety rope switches, emergency stop switches, belt alignment switches, temperature sensors, speed sensors and start-up alarms.

SYSTEM OVERVIEW

The AS interface is a hot-pluggable alternative to conventional wiring technologies, it is easy to use and extend. Speed, reduction of installation, maintenance costs and high availability are features of this system. The interface is ideally suited for the robust transmission of small quantities of data under rugged industrial conditions. It has proven itself and has become accepted as the standard for the networking of sensors and actuators in many application areas. The two-core cable for data and energy transmission, the connection in the clamping technology, the mapping in the PLC as an I/O component assembly and the extensive diagnostic possibilities all lead to significant savings.

Designed for the lowest level of the automation hierarchy, the AS-interface offers an easy, reliable, and fast transfer with optimal price-performance ratio. Higher-level systems such as Profinet, Profibus DP, Ethernet/IP, Modbus, DeviceNet or CANopen are connected via gateways. Decentralised solutions relieve the control system, depending on the extent of the setup this permits cycle times of 5 to 10 ms for one AS-i branch. The system can be extended, may be commissioned in partial areas only, reduces the communication on higher-level field buses and offers an increased failure resistance through autonomous sub-areas.

SAFETY IN THE SYSTEM

Based on the same technology and the same protocol, safety-oriented components such as safety rope switches, emergency stop devices, opto-electronic protective devices and safety guard interlocking devices can also be integrated. This requires only the installation of one safety monitor and some safe slaves on the branch. A mixed operation of both safe and non-safe AS-interface slaves is also easily achieved; the safety monitor monitors the data communication on the AS-interface line. For the safe slaves, dynamic code sequences (8x4-bit data sequence) that are stored in each slave are transmitted, the safety monitor 'learns' these during commissioning.

During operation, the safety monitor compares the expected with the actual sequence in each cycle and carries out a safe shut-down within 40 ms if there are any deviations, e.g. as a result of device failure, communication problems or the like. The time for re-activation is 100 ms. Safe field and switch cabinet modules are available as slaves, including intelligent safety sensors and safety command devices with an AS-interface chip. The system can be used up to control system category 4 according to EN 954-1 or up to SIL 3 according to IEC 61508 [10] and may be used for stop category 0 and 1 according to EN60204-1.

The Safety Rope Emergency Stop switch detects both pull and slack rope conditions and allows integration of safety signals into the plant diagnostics. Visual indication is available to provide local indication of switch status. Steady green – machine running and flashing red – machine stopped.

The conveyor control system consists of an AS-i master, AS-i power supply, a safety monitor/relay unit, AS-i cable and various I/O modules (safe and non-safe). The main benefit is a safe control system that allows for fast and flexible expansion in a cost effective way. The status of safety rope switches and emergency stop switches is displayed locally by the AS-i master and also remotely in the PLC or on a scada system.

ifm electronic SA offers a two day AS-i training workshop including topics such as installation, maintenance, fault finding for mine system integrators and the maintenance team. The company has vast experience with applications in the mining industry and can offer assistance with AS-i network design, commissioning assistance, etc.

For more information contact Chris Cronjé, ifm electronic SA, 086 143 6772, chris.cronje@ifm.com, www.ifm.com/za



Gas detection for refrigeration

Leak detection and contamination prevention in cold-room applications.

A critical factor affecting the quality, safety and shelf life of agricultural and food products is a constantly cool temperature while transporting and processing food produce. A small leak in a large refrigeration system can wreak havoc, costing the producer millions in profit. MSA Africa offers a comprehensive range of fixed and portable gas detection devices, designed to detect harmful refrigerants escaping into the atmosphere and to protect people from the potentially toxic effects.

MSA's gas detection devices can detect a variety of gasses down to one part per million and can be configured to detect a specific refrigerant, or a group of refrigerants. This makes it applicable in various industries and applications, such as refrigeration compressor rooms, ice maker plants, plate freezer areas, cold storage plants and research laboratories.

"Apart from constantly monitoring the levels of refrigerant gasses required to keep temperatures at its optimal level, our gas detection products protect and save lives," says Robbie Taitz, marketing manager, MSA. "This is achieved by providing an early warning system that forewarns employees of potentially fatal gasses leaking into the working environment and causing intoxication or an explosion."

One such product is the MSA ChillgardRT Refrigerant Monitor that detects leaking refrigerant compounds long before the air is contaminated, or the industrial refrigeration units break down. The system utilises stable and highly selective photo acoustic infrared technology to detect refrigerant gasses. It features a high immunity to interfering conditions and substances commonly found in traditional electromechanical detectors, such as turbulence, humidity, cleaning agents and solvents.

While traditional, non-dispersive infrared units have inherent zero drift, which leads to erroneous gas readings, the MSA ChillgardRT PIR instruments have a more solid zero reading, resulting in precise results each time. Units are easy to setup and use and can be configured to monitor up to eight remote areas, and detect up to six different refrigerants at a time. There are also many other options available in the MSA refrigerant and gas detection range to meet individual monitoring needs.

"The MSA ChillgardRT has become the industry standard for refrigerant leak detection monitors, providing economical, low-level monitoring of gasses used in most refrigeration systems," concludes Taitz.

For more information contact Robbie Taitz, MSA Africa, +27 (0)11 610 2600, robbie.taitz@msanet.com, www.msaafrica.co.za

Rockwell Automation's Safebook 4

Keeping industry updated on matters of machinery safety.



2012 heralds changes to the Machinery Directive that will impact on machine builders and suppliers worldwide. Effective 1 January, the European Machinery Directive no longer references EN 954-1 as a safety standard. Anyone using EN 954-1 to show conformity may have to rectify their Declaration of Conformity.

Rockwell Automation has launched an updated version of its Safebook series. Titled 'Safebook 4 – Principles of Machine Safety: Legislation, Theory and Practice', this publication tackles the latest safety directives and gives readers up-to-date information relating to all facets of machinery safety.

Safebook 4 now incorporates more information on two standards – EN ISO 13849-1 and IEC/EN 62061 – which can be referenced instead of EN 954-1. Application examples with performance level calculations offer guidance on the principles of machine safety, relevant standards and methods of safety implementation.

In a handy A5 size, it gives an overview of many safety facets, including:

- International and regional regulations.
- Relevant standards for machinery safety.
- Safety strategy.
- Risk assessment.
- Protective measures and complementary equipment.
- Safety distance calculation.
- Prevention of unexpected power-up.
- Structure of safety-related control systems.
- Functional safety of control systems.
- System design according to IEC/EN 62061.
- System design according to EN ISO 13849-1:2008.
- Sistema calculation example.

This booklet forms part of the total safety offering that Rockwell Automation presents to industry. Recognising the impact that the change in the directive will have, as well as the opportunities it presents, the company is putting significant budget and resources behind a strategy aimed at assisting companies to address the new legislation.

From white papers, videos and technical publications, through to machine-level software and Sistema-compatible library files, Rockwell Automation offers a one-stop solution for all matters relating to machine safety.

For more information contact Jeff Sandison, Rockwell Automation, +27 (0)11 654 9700, jksandison@ra.rockwell.com, www.rockwellautomation.co.za



Precise measurement of forces

High-precision measurement of compressive and tensile forces is possible with the S9M force transducer from HBM. The transducer works with strain gauges (SG) and the S-type geometry gives an accuracy class of 0,02, which is an improvement on its predecessor the S9, which had an accuracy class of 0,05. Compensation for parasitic influences, such as lateral forces or bending moments is excellent, and measurement reliability improves accordingly. The transducer is also insensitive to the effects of temperature and temperature gradients. The S9M is available with different nominal (rated) force ranges, from 500 N to 50 kN.

The strain gauges are hermetically integrated into the measuring body for safe protection. The stainless steel housing provides protection to IP68 for use in adverse ambient conditions. Mechanically, the transducer is fully compatible with its predecessor the S9 so there is no problem with compatibility of accessories and the older model can easily be replaced with the new S9M.

For more information contact Elton Murison, Esteq Engineering, +27 (0)12 809 9500, e.murison@esteq.com, www.esteq.com

Weighing solutions for wash down

Low profile BM24R and BM6G load cells now with OIML approval.

Richter Scale has announced its newly approved BM24R and BM6G load cells. The BM24R cell is an excellent solution for the most harsh wash down and hazardous environments, in the food, chemical and process industries. The load cell has been awarded the OIML R60 C3, C4 and C5 approval. Capacities from 60 kg to 60 t are available with a complete set of mounting accessories. Other features include:

- Stainless steel housings.
- Hermetically sealed IP68 protection class.
- High corrosion protection.
- Laser welded diaphragms.
- Glass to metal cable entry.
- Low profile.

The BM24R tank-mount together with the BM24R product family provide a low profile weighing assembly suitable for process control, batch weighing, silos and belt scale applications.

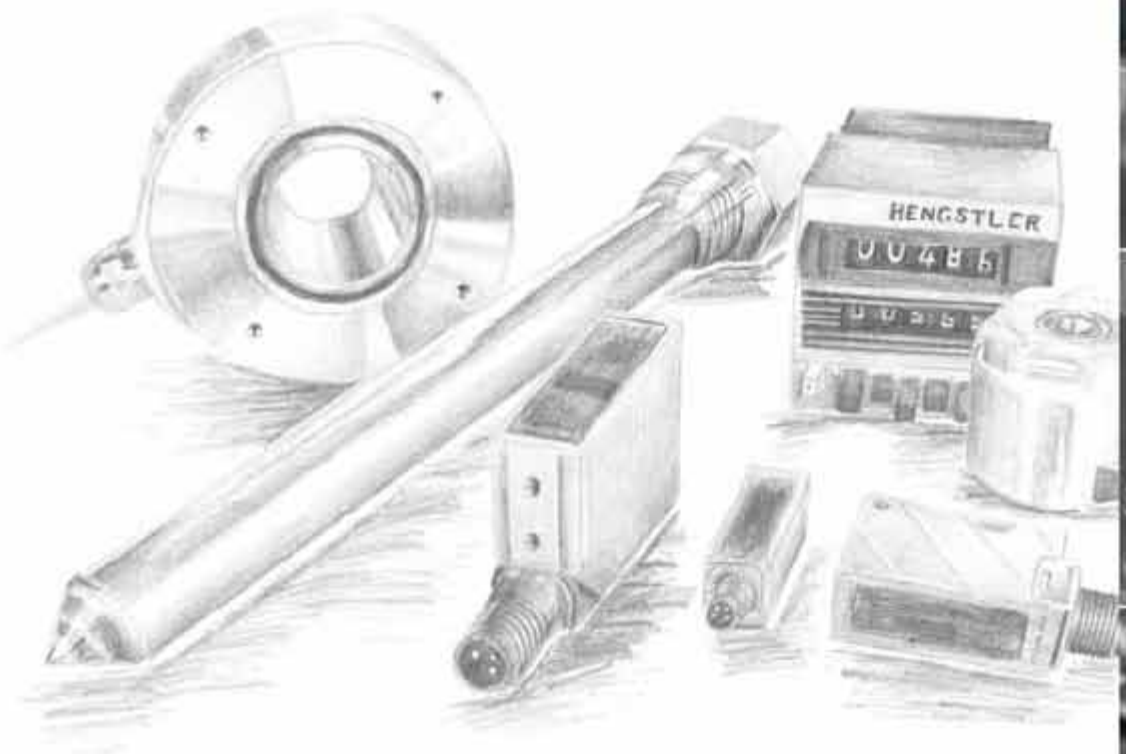


The stainless steel design with rocker-pin load introduction guarantees long-term reliability even in the harshest environments. The mount design tolerates controlled movements in all directions and eliminates the need for check rods unless major load movement is anticipated. The design allows the load cell to be placed or replaced after installation of the mount.

The single point load cell BM6G from Zemic is now also OIML R60 approved and capacities from 100 to 500 kg are available. Samples are also being prepared for a future OIML approval for the capacities 10 to 50 kg. The BM6G has been corner corrected to fit platform sizes up to 800x800 mm for 100 to 500 kg.

The unique selling point for this load cell is the combination of stainless steel and IP69K. (IP69K is the classification for devices that can be cleaned by high pressure water jets.) This combination makes it suitable for application in weighing devices used in the food industry. The BM6G has several other approvals amongst them EMC and ATEX.

For more information contact Richter Scale, +27 (0)12 386 2060, loadcells@richterscale.co.za, www.zemic.co.za















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Optimass 2000 gets enhanced corrosion resistance

Krohne has announced the launch of the Optimass 2000 in Super Duplex. Aimed primarily at the food and beverage and oil and gas industries, it is well suited to bulk measurement of products such as syrup, molasses and raw chemicals. Known for its unique 'compact envelope' and innovative straight tube design, the Optimass 2000 will provide accurate measurement of volume, mass, density, and concentration. This design can be fitted in tight spaces with a minimum of headroom without compromising accuracy or performance.

Now available in a version that utilises Super Duplex material for all wetted parts, providing superior resistance to corrosion and additional strength and making it an ideal product for sour environments. The new version can also claim higher pressure performance, operating up to 180 bar at 20°C and 130 bar at 130°C.

The meter is also available with a PED approved secondary containment of 150 bar at 20°C and 100 bar at 130°C. This ensures that in the event of tube failure the product will remain contained within the secondary casing.

For more information contact John Alexander, Krohne SA, +27 (0)11 314 1391, john.alexander@krohnesa.co.za, www.krohne.com

Silo weighing the ezee way

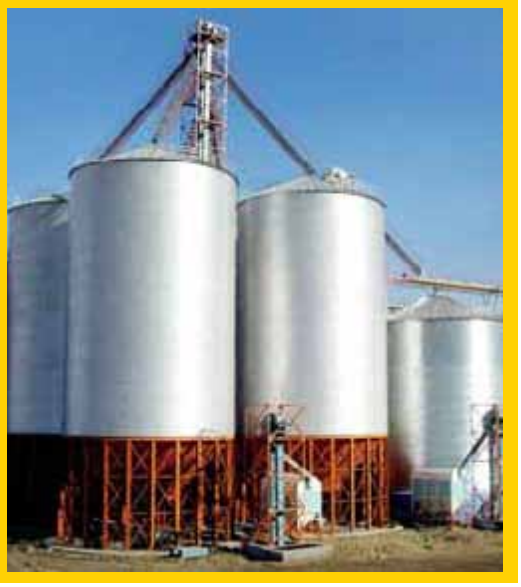
Silo weighing with the Ezee-Mount load cell solution measures the actual weight of the material in the silo. Ezee-Mount load cells connected to a load cell transmitter, measure the actual weight, rather than the level, to give an accurate indication of the amount on hand. It is more than a high-end level sensor, for example it can display the weight of: cement, sand, aggregates, flour, grains, mineral ores, coal, coke, chemicals, plastics and even liquids and liquefied gases. This means that it is easy to monitor what goes into and out of the silo.

Ezee-mount load cells mounted below each leg of the hopper have anti-lift bolts that make it safe from wind and other external forces. Load cells are robust and available invarious ranges from 1000 to 50 000 kg.

WHY SILO WEIGHING?

- The Ezee-Mount load cells give a better accuracy than level measurement. Typically below 0,1%.
- Level and density change as material packs down, reducing accuracy of measurements.
- Weight does not change, even if material bridges in a discharge cone or sticks to the sides.
- Simple to install and service – all parts are at a convenient working height.

For more information contact John Harrison, Loadtech, +27 (0)12 661 0830, sales@loadtech.co.za, www.loadtech.co.za





2-wire Coriolis mass flow measurement

Endress+Hauser has introduced Promass E TB2, a 2-wire Coriolis meter for easy integration into new and existing plant. By using state-of-the-art low power electronic components, the need for 4-wire technology, with its separate signal and supply wires, has been overcome. The advantages are reduced installation and wiring costs and easy integration into existing systems.

As a tried and tested multivariable sensor of the Proline flowmeter family, Promass E simultaneously measures the mass flow, fluid density and temperature, allowing users to monitor and control their processes more accurately. In addition to liquids such as acids, alkalis, and solvents, the unit is also suitable for gases. The application range extends to a nominal pressure of PN 100 and a process temperature of 140°C.

Operator, maintenance and expert menus can be called up at the touch of a button with guided parameterisation to aid the user during configuration. In the event of an error, remedy information is immediately displayed. Promass E TB2 provides data management and back-up, with configuration data transfer to other measuring points via the local display. In the case of servicing, data back-up via the HistoROM enables swift exchange of the measuring electronics and automatic reconfiguration without the need to be recalibrated.

In the chemical and petrochemical industries, 2-wire measuring devices are in demand as intrinsic safety is important in hazardous areas. Promass E TB2 meets standards in the process industry such as NAMUR, HART, and SIL. Both self-monitoring and error diagnostics, for example, are strictly in accordance with the specifications of NE107 (NAMUR). This guarantees a high degree of safety and optimal system availability.

For more information contact Frans van den Berg, Endress+Hauser, +27 (0)11 262 8000, info@za.endress.com, www.za.endress.com



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Capable of measuring flow rates from 0.3 kg/h to 2 300 000 kg/h and temperatures from -195 °C to +230 °C, the highly acclaimed OPTIMASS range of Coriolis mass flowmeters from KROHNE can provide a solution to almost any application.

As well as measuring mass flow, the OPTIMASS range can measure density and temperature with options for volume flow and concentration by mass or volume. It can even handle high viscosity fluids and particulate matter with ease, and maintains excellent accuracy even in the presence of entrained air. Offered in 4 materials including Tantalum, the OPTIMASS can also provide corrosion resistance for almost any process fluid.

KROHNE's patented 'Adaptive Sensor Technology' (AST), intelligently 'tunes' each meter independent of external forces and fluid density, producing an inherently balanced meter, which remains stable even in the most demanding processes.

KROHNE – Process engineering is our world.

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MICHAEL BROWN

Michael Brown is a specialist in control loop optimisation, with many years of experience in process control instrumentation. His main activities are consulting, and teaching practical control loop analysis and optimisation. He gives training courses which can be held in clients' plants, where students can have the added benefit of practising on live loops. His work takes him to plants all over South Africa, and also to other countries. He can be contacted at Michael Brown Control Engineering cc, +27(0)11 486 0567, michael.brown@mweb.co.za , www.controlloop.co.za

Control loop case history 124

No control problems in a perfect process plant!

Some years ago I gave a talk at a control engineering conference on the control loop problems that exist in process plants and how most people in those plants are unaware that problems exist. A member of the audience, the chief instrument and control engineer of a large chemical plant, told me afterwards that they were using an absolutely fabulous auto tuning package that eliminated all problems and that the controls in their plant were all running beautifully in automatic.

The gentleman moved to another job a few years later and his successor called me in to look at some of the loops, which he didn't think were working well at all. Naturally I was very curious to see just how good a job that incredible auto tuning package had done, but what I found was that the controls in the plant were probably amongst the worst I have ever encountered anywhere on that type of plant. The loops were full of problems that had not been recognised. I did not find a single loop that had even close to reasonable tuning, and, although most loops were running in automatic, when the plant was performing at steady state conditions the operators controlled the plant with controllers in manual to change operating set-points or recover from load upsets. The loops were tuned so abysmally slowly that they could not deal adequately with changes.

This article will show a few of the really interesting things I came across during my work in that plant.

FLOW LOOP TESTING

Figure 1 shows an open loop test on a flow loop that had always been running in manual, even their amazing tuning package had been unable to come up with tuning parameters for this loop.

The test shows that the Process Gain is over 20. This is the ratio of the step in PV to the step in PD (controller output) which caused the PV to respond. This ratio for flow loops should ideally be unity. If it is greater than one it usually means the valve is oversized. If less than one it generally means the transmitter span is too wide. In this case a ratio of 20 means the valve is 20 times oversized!

Just looking at the test one can see that a movement of 0.2% on the PD caused the flow to move by close on 5%. This makes control almost impossible as the slightest movement on the controller's output can jerk the valve around, it is impossible to get fine control. Valves just aren't good enough to handle this; they haven't got sufficient resolution.

Out of interest, the smallest proportional gain that can be inserted in this client's DCS system is 0,1 (or 1000% proportional band). When one has high Process Gains in a loop, it is necessary to reduce the proportional gain in the controller. In this particular case the tuning called for a proportional gain of 0,007 (or 14 173% proportional band)! This was one of the most oversized valves I had ever come across.

The following day we investigated another flow loop where the operators were complaining that they couldn't control the flow at all, even in manual. Initial tests gave no real clue as to what was happening; the flow seemed to be doing its own thing irrespective of the value of the PD (controller's output). A field inspection showed the valve was following the PD reasonably well, although one could see that there was an overshoot on step changes on opening, probably due to poor positioner tuning. We then tried blowing out the impulse lines to the transmitter. This helped a lot and we could see things responding to changes properly, but to our amazement we found the Process Gain of this loop was an unbelievable 150! All records of the previous day were broken. I can't believe people are trying to control with a valve 150 times oversized – really amazing. A step from the open loop test is illustrated in Figure 2. If it were possible to control with this valve, which of course it isn't, then one would need to insert a proportional gain of about 0,003 (or 33 000% proportional band) into the controller. This would make the *Guinness Book of World Records*.

AND NOW FOR LEVEL

Figure 3 is a closed loop test of a level control loop. This level is in fact uncontrollable, mainly due to an excessively long deadtime. All attempts to try and get the level into balance to allow one to do normal tests in manual so as to determine the process dynamics, and hence to get tuning parameters, failed. Basically the level either ran away upwards or downwards irrespective of whatever one did. The operators did not mind if the loop cycled, their requirement was that the level should not go too high or too low. We determined that if we used a high value of gain using P only control, effectively deliberately increasing the instability, then one could keep the cycle small enough to satisfy the requirements. One should really call it a type of 'ON/OFF' control.

This is not a satisfactory solution, but time did not permit further investigation into why the deadtime was

so long and if alternative solutions could be found to achieve better control. However, it certainly worked, even though I can't say I liked it.

THUMBS DOWN THE AMAZING TUNING PACKAGE

The third and last example in this article is actually of tuning and it shows how poorly the so called amazing tuning package performed in practice. The loop in question is a fuel oil pressure control loop, the secondary cascade to the temperature control of the furnace. The operators were terribly nervous about letting us work on these loops as a huge downstream plant section was dependent on the temperature being held especially constant. Also if the pressure dropped too low, the whole furnace and downstream plant would trip.

Figure 4 shows a closed loop 'as found' test using the original tuning parameters: P (gain) = 0,06 and I = 0,05 minutes/repeat.

The performance was terrible. The control was virtually never at set-point with the pressure drifting around it all the time. Frequent load changes were also taking place in the oil pressure upstream of the valve and the controller was completely unable to maintain the desired downstream pressure. In the diagram a small set-point change of only 2,5% was made – it took the control nearly 5 minutes to get it close to the new set-point!

Correct, stable and scientific tuning resulted in parameters of: P (gain) = 0,3 and I = 0,03 minutes/repeat.

As can be seen in Figure 5, the response with these parameters to a similar change resulted in the PV reaching set-point in less than a minute. Now, the process can easily deal with the load changes.

It is important to note that it is essential for cascade secondary loops to operate much faster than the primary loop. (Refer Loop Signature Article P1-17 for a detailed discussion on cascade control.) If, as in this example, the secondary is not controlling properly, then there is little chance that the primary loop (which is the important one) will control properly. In this case the primary was the vitally important temperature loop.

More interesting examples taken from the same plant will be given in future articles, including the tuning of the primary temperature loop.

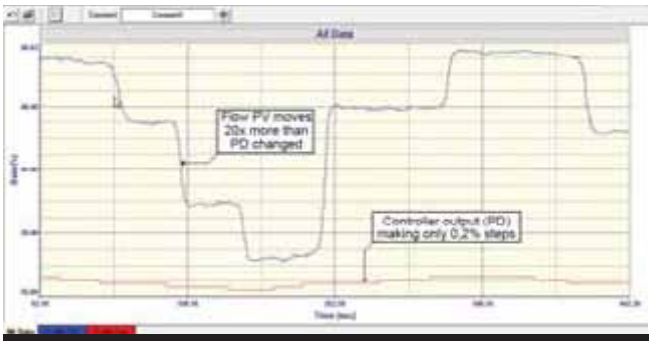


Figure 1.

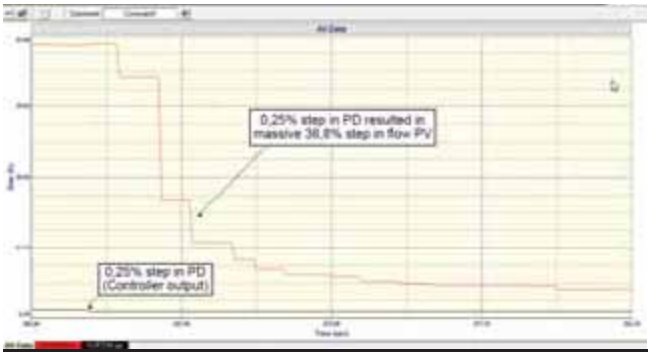


Figure 2.

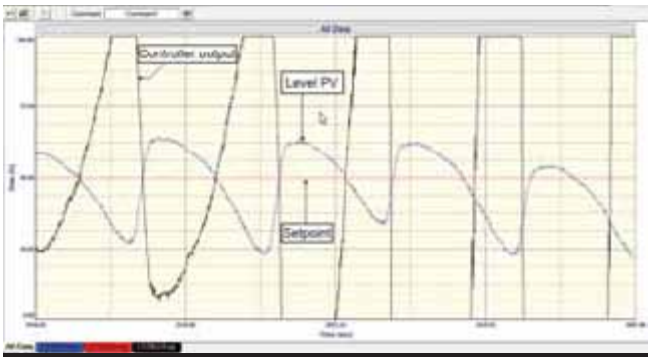


Figure 3.

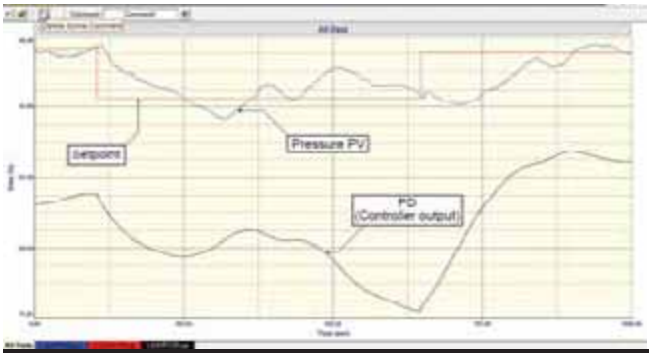


Figure 4.

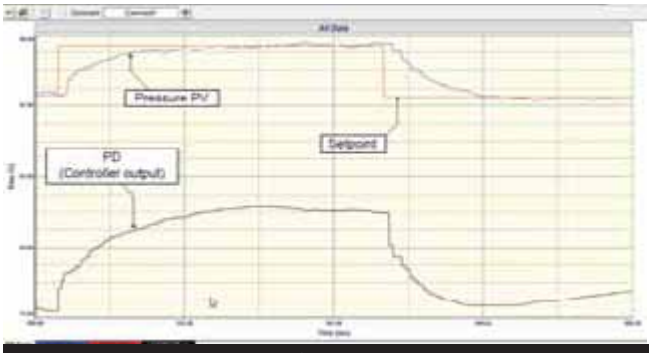


Figure 5.

Compact PC control from Beckhoff



Multi-optional interface enables integration with all common fieldbuses and industrial Ethernet systems.

Beckhoff Automation's CX9020 is a DIN rail-mountable controller with ARM Cortex A8 CPU and microSD card with large storage capacity and very compact design. The Embedded PC integrates a CPU, numerous IT connectivity solutions and optional interfaces, a power supply and a direct I/O interface for Beckhoff I/O systems, all in a compact housing. The unit automatically detects the connected bus or EtherCAT terminals and switches to the respective mode for more convenient system management.

The CX9020 CPU has access to 1 GB of internal DDR3 RAM. It comes with a 256 MB microSD card for the operating system. A free microSD card slot can be used for additional operator data. In addition, 128 kbyte of NOVRAM are available for data storage in the event of a power failure.

The basic version features two 10/100 Mbit/s Ethernet ports, four USB 2.0 connections and a DVI-D interface. The multi-optional interface enables integration of an additional interface, e.g. for EtherCAT (slave only), Profinet, Ethernet/IP, Profibus, CANopen, DeviceNet, serial interface or audio. In combination with the TwinCAT automation software platform from Beckhoff, the system becomes a high-performance PLC and Motion Controller that can be used with or without visualisation.

The extended operating temperature range of -25 to 60°C enables application in demanding environments covering a wide range of manufacturing facility types and outdoor applications, including wind turbines and solar power plants. The operating system is Microsoft Windows Embedded Compact 7.

For more information contact Conrad Muller, Beckhoff Automation, +27 (0)11 795 2898, c.muller@beckhoff.com, www.beckhoff.co.za

CERTEC MEASURING CELLS FOR PHARMACEUTICAL APPLICATIONS



Damaged or cracked metal diaphragms are one of the most frequent causes of failure in pressure transmitters. The result is that the isolating diaphragm liquid, which serves as a transmission medium between the separating diaphragm and sensor element, leaks into the process. The fault may not even be noticed by the sensor electronics, since the product itself also exerts a measurable pressure on the sensor. The exact time of damage can therefore not always be determined.

The isolating diaphragm liquids used are generally subject to FDA regulations and are not harmful to health. For the plant operator, however, they represent a product quality risk. The implemented transmission oils do not mix well with the water-based pharmaceutical products – serious contamination is sometimes the result.

The pressure transmitters with ceramic-capacitive Certec measuring cell from Vega offer a solution for these quality and validation processes. The ceramic measuring cell works without transmission liquid and ideally satisfies all requirements of the pharmaceutical industry. The measuring cell is dry, self-monitoring and registers both damage to the diaphragm and moisture leaking into the sensor.

The diaphragm, which consists of sapphire ceramic, is 10 times harder than high-grade steel and easily resists abrasive media. The diaphragm and ceramic body are joined together with a glass solder, allowing constant product temperatures up to +150°C. The design is completed by a sophisticated sealing concept, depending on process requirements a double seal, or a hygienic form, can be implemented. High surface finish provides optimal cleaning and sanitation.

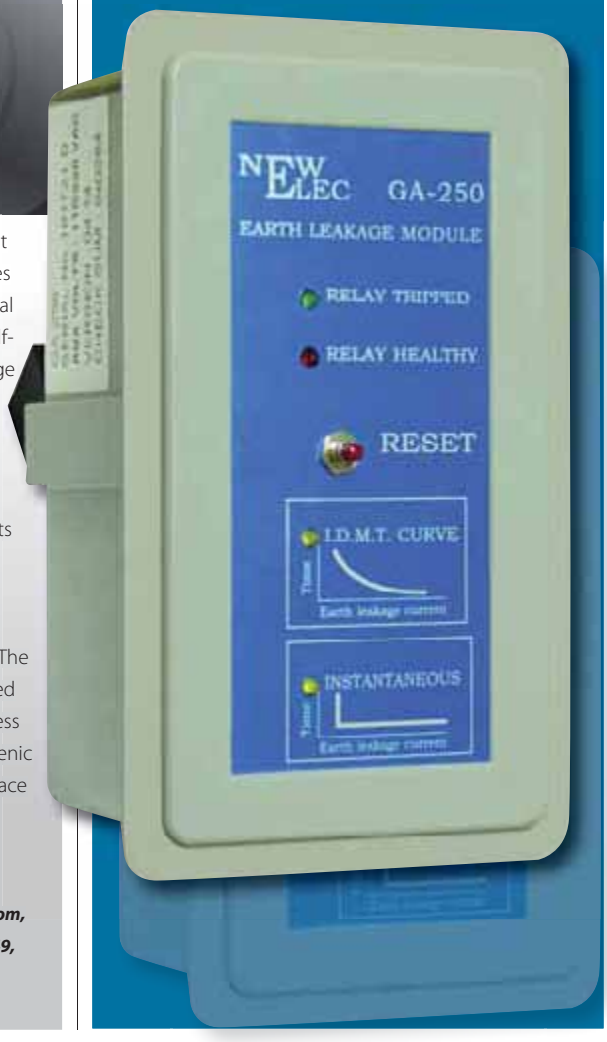
For more information contact John Groom, Vega Instruments SA, +27 (0)11 795 3249, john.groom@vega.com, www.vega.com

SENSITIVE EARTH PROTECTION RELAY WITH NO TRIPPING

NewElec's GA earth leakage protection relay enables more sensitive earth leakage protection of high kilowatt motors started direct-on-line, as well as variable speed drive applications, without nuisance tripping. This is achieved through the use of harmonic filters. The GA is available in diverse configuration options. These include latched; non-latching main trip contact; fail-safe; non fail-safe; and 110 – 220 V or 380 – 525 V AC auxiliary supply voltages.

Two models are offered: the GA with a standard 74 mm x 134 mm cut-out; and the GA-ELX with an adapter plate to fit a 71 mm x 94 mm cut-out. The relays provide harmonic suppression and isolation to IEC 255-5 App A. The front control panel is a door mounted unit and clearly displays the user tripping curve selected as well as the relay trip status by means of suitably coloured LEDs.

For more information contact Luc Dutrieux, NewElec, +27 (0)12 327 1729, sales@newelec.co.za, www.newelec.co.za



SITRANS P 500



Increasing requirements and ever more complex applications also present measuring systems with significant challenges to which they must respond with expanded functionalities – such as improved accuracy and long-term stability. The Sitrans P500 with a new measuring cell and additional static pressure sensor meets the high demands.

The P500 has been developed to meet the expectations for measuring accuracy, ruggedness and user friendliness, and it merges seamlessly into the SITRANS P family. Users can expect differential pressure measurement at the highest level: it guarantees an accuracy of 0,03% as well as outstanding values for the influence of static pressure and temperature on the measured result. In this way, a total performance of 0,09% up to a turndown of 5:1 is achieved using only one measuring cell for different measuring ranges. This means inventory costs can be significantly reduced without sacrificing measurement accuracy for losses in the quality of the measured result. The high long-term stability of the transmitter also reduces maintenance costs thanks to extended calibration intervals, and ensures that you can always rely on the measured results on the long run.

An innovative sensor assembly enables direct connection with process temperatures of up to 125°C even without a remote seal system. With an improved 3-push button operator input concept for setting the device, the Sitrans P500's host of diagnostics options can be used conveniently without software. Even curve displays and trend diagrams can be represented on the display for user-friendly process monitoring.

For more information contact Keshin Govender, Siemens Southern Africa, +27 (0)11 652 2412, keshin.govender@siemens.com, www.siemens.co.za

RACK CONTROL AND MONITORING UNIT

Mean Well has launched new peripheral control and monitoring units to collocate with RCP-2000 rack power, RKP-CMU1 and RKP-1Ux-CMU1 for more complete rack power solutions. Up to 32 units of RCP-2000 can be monitored by one RKP-CMU1 and the data pertaining to output voltage, current, internal temperature, operation status, series number and firmware version of each unit can be shown on the LCD front panel. Using the buttons on the front, devices can be turned on or off directly and the output voltage as well as current limit point of each RCP-2000 unit can be trimmed. In addition, RKP-CMU1 is equipped with USB, RS-232 and Ethernet interfaces for PC connection and can also operate remotely via external GSM modem. Features include:

- **Control and monitor RCP-2000 rack power system.**
- **LCD front panel and control buttons.**
- **Remote warning via external GSM modem.**
- **Data and event log functionality.**
- **Accepts 4 external digital inputs.**
- **4 built-in user programmable relay contacts.**
- **Window-based monitoring software.**
- **Built-in LED indicators.**

For more information contact Ettienne van der Westhuizen, Current Automation, +27 (0)11 462 4253, ettienne@rectifier.co.za, www.rectifier.co.za

CAPACITIVE LEVEL SENSORS FOR THE MINING INDUSTRY



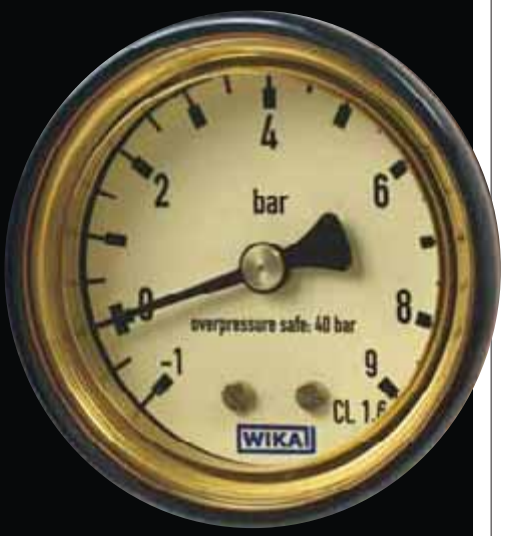
Capacitive level sensors are used for the non-contact detection of any objects. In contrast to inductive sensors, which only detect metallic objects, capacitive sensors can also detect nonmetallic materials.

The KN/KI5 level switch from ifm electronic enables safe and reliable detection of powders and liquids even through nonmetallic vessels. The sensors feature high operational reliability due to increased noise immunity caused for example by frequency converters, mobile telephones or electrostatic discharge. The user has the choice between DC and AC units. The universal switching output can be programmed to be either NC or NO.

The sensors have an adjustable sensing range up to 15 mm through a potentiometer and the resistant plastic housing makes them suitable for various applications. Different connection options include cable, connector or terminals, and types with a programmable output function are available. The sensors have a 10 to 36 V DC or 20 to 250 V AC/DC voltage range.

For more information contact Chris Cronjé, ifm electronic, +27 (0)12 450 0370, chris.cronje@ifm.com, www.ifm.com/za

PRESSURE GAUGE FOR SANITARY APPLICATIONS



The diaphragm pressure gauge from WIKA has been designed for pressure measurement in hygienic processes such as those in sanitary applications. Unlike diaphragm seals, it uses a dry measuring cell. This means, if the diaphragm should leak, no system fill fluid escapes into the process. Thus the new model PG43SA pressure gauge increases process safety.

In addition to this, its design with a welded, flush diaphragm and surface roughness of ≤ 0.4 µm ensures very good cleanability. No deposits are left on the process connection. The entire measuring instrument can be steam-sterilised (autoclaved) at 134°C and is suitable for SIP/CIP processes.

By integrating the centre-back-mount process connection into the case a compact, robust and hygienic design of the pressure gauge is enabled. The instrument, available in nominal sizes of 40 and 63, is thus ideally suited for space-saving installation, harsh operating conditions and cleaning from outside.

For more information contact WIKA Instruments, +27 (0)11 621 0000, sales@wika.co.za, www.wika.co.za

STACK GAS MONITORING WITH THERMO SCIENTIFIC

The Thermo Scientific Model 60i is a full-extractive, multigas analyser that features non-dispersive infrared (NDIR) optical filter technology to ensure accurate, precise measurement of up to six gases, including CO, CO_2, NO, NO_2, and SO_2. The Model 60i also accommodates an optional O_2 measurement capability using either an affordable electrochemical cell or, when high accuracy and reliability are required, paramagnetic technology.

In addition, the unit is designed with built-in safeguards that protect the analyser in the event of a sample conditioning system malfunction.

The high sensitivity of the Model 60i requires as little as one quarter the flow rate of similar instruments, exposing the analyser and sample conditioning system to less particulate and water for more dependable operation. It offers a wide dynamic range and configuration requirements have been minimised.

The new Thermo Scientific unit is a 'smart' analyser with on-board electronics that are capable of controlling a calibrator, probe controller and heated sample line all to reduce space requirements. With its efficient and uncomplicated gas cell and electronic system layout, plus the analyser's two-piece, slide-apart case and interior fold-down divider panel for rapid component access, the Model 60i is easy to use, maintain and service. The unit is designed to meet U.S. EPA 40CFR Part 60 requirements.

For more information contact Ero Electronic SA, +27 (0)11 869 1101, ero_sa@kingsley.co.za, www.eroelectronic.edx.co.za

LOW COST TEMPERATURE SWITCH AND TRANSMITTER

Allpronix has introduced a programmable temperature transmitter and switch from its EMA range of innovative product solutions. The TA1001 is IP67 rated and complete with local indication and two outputs. The unit can be set for a temperature range and the relays programmed to provide alarms in continuous monitoring applications.

The temperature range is -40 to 150°C with an accuracy of 0,2%. The unit is supplied standard with two outputs: output 1 is a PNP NO/NC or NPN NO/NC for alarming; the second is a 0-10 V or, 4-20 mA signal for continuous monitoring. The unit is powered from 18-36 VDC and can operate in an ambient temperature range from -25 to 70°C.

The unit is mounted using an M18 x 1.5 thread and is of 4 wire electrical design. Visual indication by means of an LED display makes set-up easy.



For more information contact Allpronix, +27 (0)11 795 9500, sales@allpronix.com, www.allpronix.com

GIGAVAC SEALED CONTACTORS



Denver Technical Products continues to expand its range of Gigavac GX Series EPIC (Extended Performance Impervious Ceramic) sealed AC and DC contactors capable of switching from 12 to 750 V.

These small RoHS compliant hermetically sealed contactors can be mounted in any position and are suitable for harsh environments including under water and at temperatures from –55 to 85°C. They are a preferred contactor for heavy trucks and equipment, boats, light rail, mine and factory automation, power management systems and solar, wind and wave power systems where long life and high reliability are required.

For more information contact Denver Technical Products, +27 (0)11 626 2023, denvertech@pixie.co.za, www.denvertech.co.za

LOCKING MALE CONNECTORS SIMPLIFY PANEL FEEDTHROUGHS

Wago has expanded its Mid Multi Connection system (MCS-MIDI) to include locking male connectors with snap-in flanges. Using these male connectors, panel feedthroughs for 5 – 2.5 mm panel wall thickness can now be quickly and easily implemented without screws. This saves the user valuable time and money as the recess in the panel is less complicated and does not require any drilled holes for anchors such as screws. As a result, the number of parts required for the feedthroughs is reduced, along with the logistics involved.

The male connectors with snap-in flanges are available with pin spacing of 5 mm, 5.08 mm, 7.5 mm and 7.62 mm and can accommodate conductor sizes of 0.08 to 2.5 mm^2 (AWG 28–12). These connectors are designed for the following rated values: 12 A and up to 630 V for pin spacings of 5 and 5.08 mm; 12 A and up to 1 000 V for pin spacings of 7.5 and 7,62 mm. This wide application range means that male connectors can be combined with all versions of the Wago female connectors available on the market. As a result the function of an existing screw locking mechanism or of locking levers can be retained.



For more information contact Christo Grobler, Shorrock Automation, +27 (0)12 345 4449, sales@shorrock.co.za, www.shorrock.co.za

FLAME SENSORS WITH FIBRE OPTIC SYSTEMS



A range of flame sensors with fibre optic systems, designed for various spectral ranges from UV to IR, is available from Durag. The type D-LE 701/703 flame sensors can be used in conjunction with Durag's D-UG 110/120/660 controllers and D-GF 150/200 automatic combustion units.

Flame sensors with fibre optics are available in two different models. The D-LE 701 series flame sensor's electronics are housed in metal, separate from the fibre optics, while the D-LE 703 series is connected directly to the fibre optics.

The flame sensors only process the signals that correspond to the flickering of the flame, ensuring the sensor is resistant to the influence of constant ambient light sources. Each is equipped with various operational controls to adjust the sensor to the particular combustion conditions of a given application.

Detectors used in the UV flame sensors have a spectral range of either 190 to 520 nm or 280 to 410 nm. All blue burning flames, such as gas flames, where the portion of visible radiant energy is low, can be monitored with these flame sensors.

Detectors in the IR flame sensors offer a spectral sensitivity of 300 to 1100 nm or 780 to 1800 nm, depending on the model. IR flame sensors with a sensitivity of up to 1800 nm have proven themselves in monitoring gas and oil flames in low NO_x combustion applications such as flue gas recirculation processes.

For more information contact Colwyn Savings, OEN Enterprises, +27 (0)11 675 4447, colwyn@oenenterprises.co.za, www.oenenterprises.co.za

PORTABLE LIGHTNING DETECTOR BY STRIKEALERT

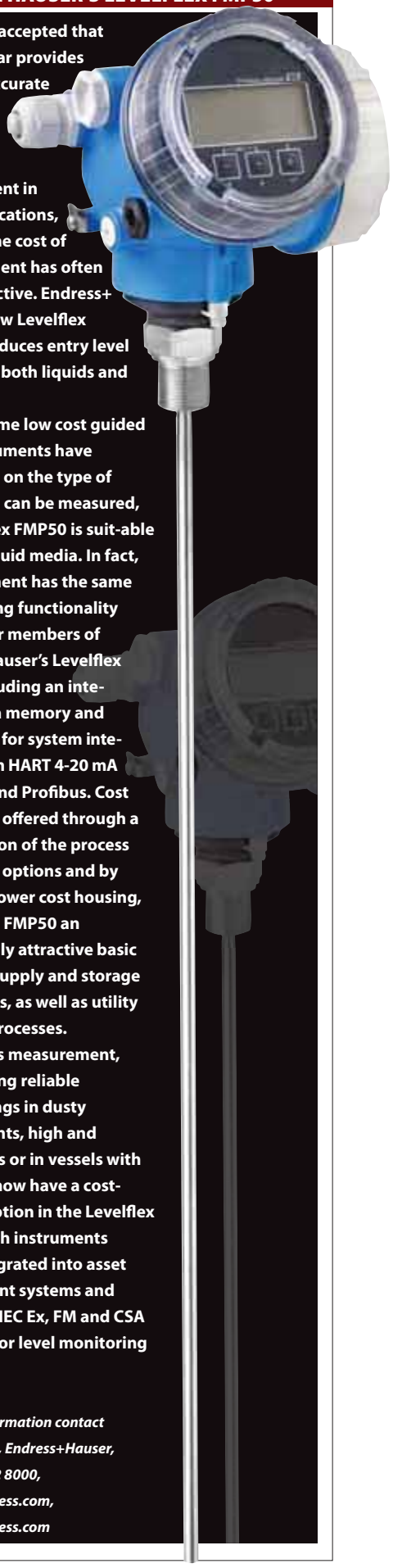
TufTek has introduced a dependable and easy-to-use portable lightning detector. StrikeAlert is a small and affordable unit that provides early warning of approaching lightning strikes from as far away as 60 kilometres. An audible alarm sounds when there is a strike and a corresponding LED illuminates at lightning distances of 32-64 km, 20-39 km, 10-20 km and within 10 km.

The unit is ideal for outdoor working activities and housed in a small pager-like casing that can be clipped on a belt. It is supplied by two AAA batteries.



For more information contact TufTek, +27 (0)11 794 2279, sales@tuftek.co.za

ENDRESS+HAUSER'S LEVELFLEX FMP50



It is widely accepted that guided radar provides the most accurate and reliable level measurement in many applications, however, the cost of the equipment has often been restrictive. Endress+Hauser's new Levelflex range introduces entry level options for both liquids and solids.

While some low cost guided radar instruments have restrictions on the type of liquids that can be measured, the Levelflex FMP50 is suit-able for most liquid media. In fact, the instrument has the same wide ranging functionality as the other members of Endress+Hauser's Levelflex family, including an integrated data memory and availability for system integration with HART 4-20 mA analogue and Profibus. Cost savings are offered through a simplification of the process connection options and by utilising a lower cost housing, making the FMP50 an economically attractive basic device for supply and storage applications, as well as utility metering processes.

For solids measurement, users seeking reliable level readings in dusty environments, high and narrow silos or in vessels with obstacles, now have a cost-effective option in the Levelflex FMP56. Both instruments can be integrated into asset management systems and hold ATEX, IEC Ex, FM and CSA approvals for level monitoring up to SIL2.

For more information contact Jan Gerritsen, Endress+Hauser, +27 (0)11 262 8000, info@za.endress.com, www.za.endress.com

TERMINAL BLOCKS WITH PUSHBUTTONS

Wago has launched the new 2060 Series of unique one-, two- and three-pole SMD terminal blocks. Equipped with integrated pushbuttons, the 2060 Series provides easy insertion and removal of all conductor types. A total height of just 4.5 mm makes it ideal for both LED applications and installation in low profile electronic components.

The terminal blocks are extremely compact and are ideal for surface-mount PCB applications. They feature CAGE CLAMPS connection technology, allowing easy and time-saving push-in termination of solid conductors.

Fine stranded conductors can be conveniently inserted via integrated pushbuttons. The design of the 2060 Series combines side entry wiring with top of unit, space saving pushbutton actuation.

Wago's 2060 Series features extremely low profile and light colour housings that minimise on-board LED shadowing. The SMD PCB terminal blocks have 4 mm pin spacing and can be assembled without losing any poles. This greatly reduces the number of versions required, while minimising supply, inventory and production costs.

The PCB terminal blocks are supplied in tape and reel packaging for fully automated assembly. They accommodate conductor sizes 0,2–0,75 mm^2 (AWG 24–18) and are rated at 9 A, up to 320 V/2,5 kV/2 (overvoltage category II).

For more information contact Christo Grobler, Shorrock Automation, +27 (0)12 345 4449, sales@shorrock.co.za, www.shorrock.co.za



EMERSON'S DIGITAL DP LEVEL ARCHITECTURE



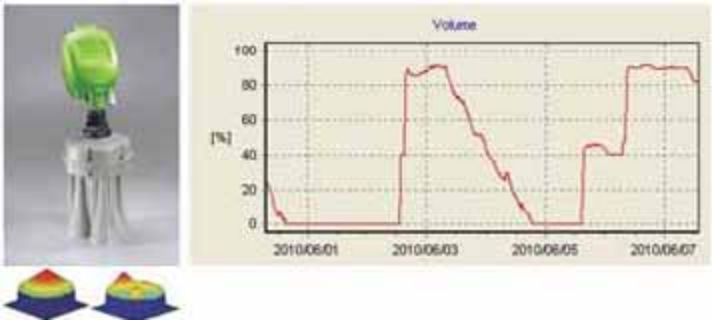
Emerson Process Management has released the industry's first digital DP Level architecture, the Rosemount 3051S Electronic Remote Sensor system. With over ten patents, the 3051S ERS system is a breakthrough technology that replaces mechanical impulse piping with two 3051S pressure sensors linked together electronically. Differential pressure is calculated in one of the two sensors and is transmitted using a standard two-wire 4-20 mA HART signal.

Ideal applications for the 3051S ERS system include tall vessels, distillation towers, and other installations that traditionally have required excessive lengths of impulse piping or capillary. When used in these types of applications, the 3051S ERS system can deliver:

- Improved performance: the 3051S ERS system replaces mechanical components with a digital architecture, resulting in faster response time and a more stable and repeatable measurement even in applications with wide varying temperatures.
- Simplified installation and maintenance: cold weather installations often require heat tracing or insulation. Impulse lines need to be checked for leaks, condensation, evaporation, and plugging. The digital architecture of the 3051S ERS system eliminates these practices, resulting in cleaner installations with less maintenance and upkeep.
- Additional process insight and diagnostics: the 3051S ERS system is a MultiVariable solution that provides additional process information for optimised control. In addition to the DP calculation, the 3051S ERS provides real-time access to the readings from each pressure sensor and a scaled output for tank level or volume measurement.

For more information contact Mark Tapson, Emerson Process Management, +27 (0)11 451 3700, mark.tapson@emerson.com, www.emersonprocess.com

VOLUME IN A FLOUR SILO

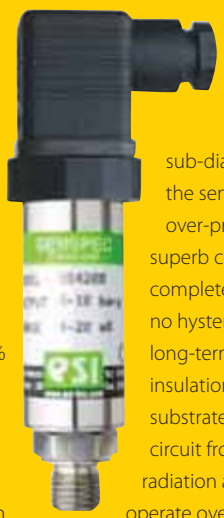


According to Allpronix, flour is a difficult product to measure using technologies such as ultrasonic, radar, laser and even guided wave radar. The ultrasonic, radar and laser all have problems due to the amount of dust present in the silo, especially during filling. The guided wave radar only measures a single point inside the silo to give an estimated level and not volume.

Allpronix's APM S-version scanner has been installed for nearly 18 months on four flour silos 14 m high and 5 m wide at a bakery in Pretoria and have been working problem free and accurately ever since. The end user can now manage inventory as well as monitor deliveries and usage of each silo. Subsequently, the entire bakery is busy replacing its old systems with the new accurate and reliable APM scanner.

For more information contact Allpronix, +27 (0)11 795 9500, sales@allpronix.com, www.allpronix.com

GENERAL-PURPOSE PRESSURE TRANSDUCERS



The Genspec GS4200 series of pressure transducers and transmitters is designed to meet the operational requirements of demanding pressure measurement applications where good quality, fast delivery and low cost are of the highest priority.

The series is available in pressure ranges from 0-500 mbar to 0-1500 bar. Accuracy is ± 0,25% with an over pressure limit of twice the rated pressure range. The standard output of 4-20 mA and easy access for recalibration contribute to the excellent design. All models are supplied with integral ¼ in (6,3 mm) BSP pressure connections. Optional connections are available. The all-titanium alloy wetted parts offer unbeatable corrosion resistance. Versions are also available offering IP66 sealing for installations requiring high levels of environmental protection.

The unique silicon-on-sapphire sensor technology provides outstanding performance and gives excellent stability over a wide temperature range. The advanced sensor design consists of a piezoresistive silicon strain gauge circuit, which is epitaxially grown onto the surface of a sapphire diaphragm to form a single crystalline structure. The sapphire sensor element is then molecularly bonded to a titanium alloy sub-diaphragm. This enables the sensor to endure higher over-pressures and provides superb corrosion resistance. The completed sensor exhibits virtually no hysteresis and has excellent long-term stability. With outstanding insulation properties, the sapphire substrate protects the strain gauge circuit from electromagnetic pulse radiation and allows the sensor to operate over a very wide temperature range without loss of performance.

Applications for the GS4200 series include the continuous monitoring of hydraulic systems with oil, gas, water and other process liquids, industrial, medical and aerospace industries. Also ideal for the measurement and control of pressure in refrigeration, pneumatic, compressor, HVAC and engine monitoring systems.

For more information contact Temperature Controls, +27 (0)11 791 6000, sales@tempcon.co.za, www.tempcon.co.za

PROCESS MOISTURE ANALYSER



The versatile Ametek Model 5000 is a highly sensitive moisture analyser for process applications where the continuous, reliable measurement of moisture in vapours and gases is required. The Model 5000 measures moisture online by monitoring the vibrational frequency change of a hygroscopically sensitised quartz crystal that is exposed alternately to wet sample and dry reference gas for 30 second periods. The reference gas is a sample gas passed through a molecular sieve dryer to remove virtually all its moisture content.

Many moisture analysers are susceptible to attack by the process stream. Unsaturated hydrocarbons, compressor oils, oxygen and carbon dioxide can permanently damage or destroy trace moisture sensors. Even worse, a fouled sensor can become completely unresponsive to moisture yet continue to give an output. This 'dying-in-place' behaviour leaves the process unmonitored.

The Model 5000 gives 'bullet-proof' performance in critical moisture applications. Its unique differential measurement principle, combined with continuous diagnostics, ensures reliable performance.

The field unit is explosion proof and is certified to comply with numerous international standards. It contains a sample cell, switching valves and associated electronic circuits mounted internally and a flow meter, flow control valves and back pressure regulator mounted externally.

The controller is also designed for use in industrial areas and is available with several standard and optional certifications. It provides all control and data processing, as well as a keypad and display interface. Outputs include RS-485 bidirectional serial data, two fully configurable 4-20 mA analog signals and three configurable alarms. It can be separated from the field unit by up to 600 metres. The external dryer supplies the analyser with reference gas by using a molecular sieve to dry the actual process.

For more information contact Colwyn Savings, OEN Enterprises, +27 (0)11 675 4447, colwyn@oenenterprises.co.za, www.oenenterprises.co.za

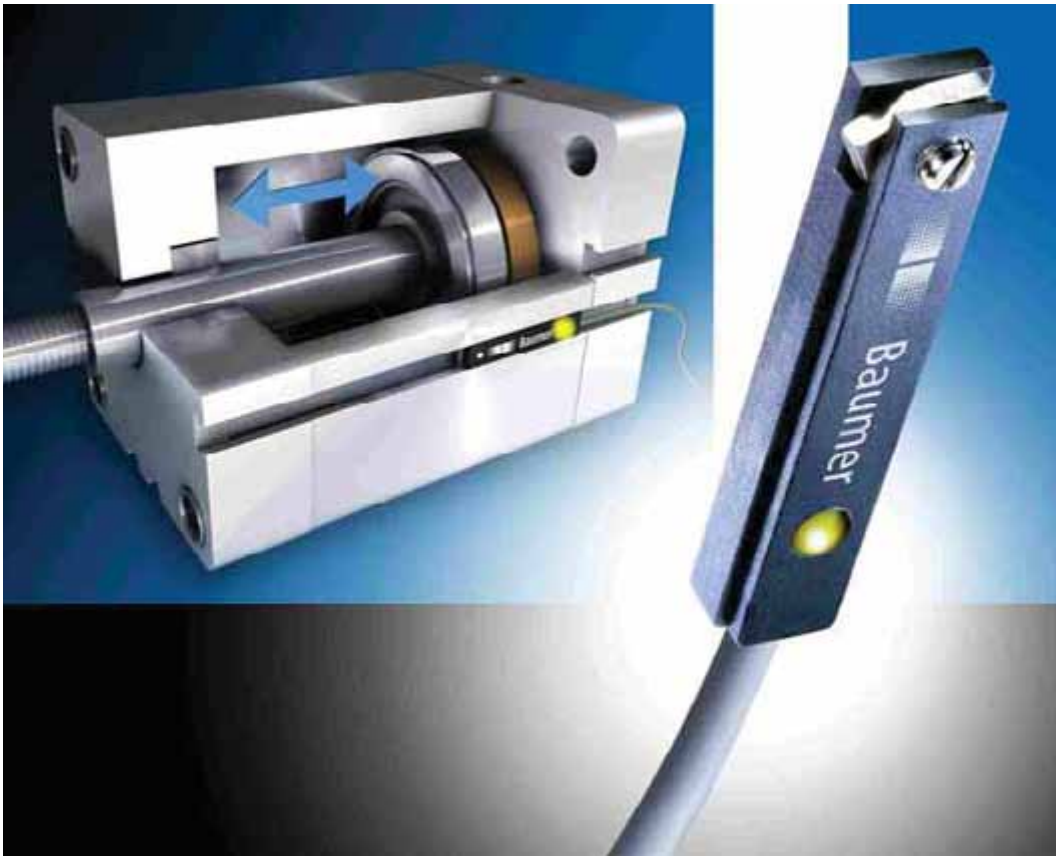
NEW ULTRASONIC FLOWMETER RANGE

Measuring and Control Instruments has announced the introduction of a new range of ultrasonic flowmeters. The product line consists of ultrasonic transmit-time flow measurement devices, which includes clamp-on, insertion, flanged, centre-insertion, handheld and ultrasonic Doppler flowmeters that can be used with liquids containing bubbles or small amounts of suspended solids.

Innovative research and development and advanced manufacturing and calibration equipment (to an accuracy of ±0,1%) makes these flowmeters ideal for applications in mines, water treatment, power plants and chemical factories. All flowmeters are calibrated in the standards flow laboratory before shipment. Data loggers and software are available for on-site recording of flow measurements.

For more information contact Bianca Joubert, Measuring and Control Instruments, +27 (0)11 615 3854, sales@mcinsstruments.co.za, www.mcinstruments.co.za

HIGH-PERFORMANCE MAGNETIC CYLINDER SENSORS



Due to ever increasing market requirements, Baumer has launched a new generation of magnetic cylinder sensors for detecting the exact position of pistons in pneumatic cylinders.

The new sensors, available for T and C slots, have been enhanced in performance and reliability. For instance, the sensitivity was optimised to produce a more precise and reproducible switching characteristic. Additionally, high-quality materials have been used to make the sensors more robust and cables now consist of PUR for trouble-free use in oily environments. Since the magnetic sensor has no moving parts product life is maximised as there is no wear.

The T slot sensors can be quickly installed from above. No cylinder disassembly is required, even if both front faces are equipped with installation panels. Thanks to the wide selection of mounting accessories, the sensor can be conveniently attached to all common cylinder types. The result is a reduction of the required sensor variants resulting in a simplification of warehousing. All sensors are available as PNP or NPN types with cable or fly-lead connectors.

For more information contact Temperature Controls, +27 (0)11 791 6000, sales@tempcon.co.za, www.tempcon.co.za

DUAL MOUNT EARTH LEAKAGE RELAY

The GA range of earth leakage relays from NewElec can now retrofit onto other popular earth leakage protection relay panel cutouts. The 94 mm x 71 mm panel cutout, as well as the standard 134 mm x 71 mm NewElec GA cutout, are both catered for with an innovative dual mount housing in the GAxxxELX model. The relay caters for factory selectable trip relay configurations to meet failsafe and non failsafe operations; specific IDMT curve selections; and indication which discriminates between earth fault and earth leakage.

The one-solution relay is designed to fit into two panel cutout types. If there is no existing panel cutout, it is easy to order the usual GA range product and use the NewElec panel cutout dimensions. If the user wishes to retrofit or has already cut the panel to suit other brand relays, then the GAxxx ELX relay is ideal. It is a simple matter to place the GAxxxELX relay behind the existing cut out, place the NewElec ELX, and secure the front control panel using the screws and neoprene seal provided.

The GAxxxELX relay offers versatility in choice and is offered in 30, 250, 375, 500 or 1000 mA trip sensing thresholds.

For more information contact Luc Dutrieux, NewElec, +27 (0)12 327 1729, sales@newelec.co.za, www.newelec.co.za

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Adroit Technologies	+27 (0)11 658 8100	info@adroit.co.za	www.adroit.co.za	5*,6
Alien Systems & Technologies	+27 (0)11 949 1157	sales@astafrica.com	www.astafrica.com	IFC*,82,83
Allpronix	+27 (0)11 795 9500	sales@allpronix.com	www.allpronix.com	49*,96,99
APC by Schneider Electric	+27 (0)11 254 6400			41
ApplyIT	+27 (0)31 514 7300	bredins@applyit.com	www.applyit.com	58
ASSTech Proc. Elec. & Instr.	+27 (0)11 708 9200	info@asstech.co.za	www.asstech.co.za	44*,70
Beckhoff Automation	+27 (0)11 795 2898	c.muller@beckhoff.com	www.beckhoff.co.za	54,92
Comtest Distribution	+27 (0)11 608 8520	info@comtest.co.za	www.comtest.co.za	71
Countapulse Controls	+27 (0)11 615 7556	clive@countapulse.co.za	www.countapulse.co.za	64,78,87*
Current Automation	+27 (0)11 462 4253	ettienne@rectifier.co.za	www.rectifier.co.za	94,stitched insert*
Denver Technical Products	+27 (0)11 626 2023	denvertech@pixie.co.za	www.denvertech.co.za	31*,96
Eaton Electric	+27 (0)11 824 7400	robhare@eaton.com	www.eaton.com	39*,45
Emerson Process Management	+27 (0)11 451 3700	mark.tapson@emerson.com	www.emersonprocess.com	10,53*,98
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Ero Electronic SA	+27 (0)11 869 1101	ero_sa@kingsley.co.za	www.eroelectronic.edx.co.za	35*,95
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Extech Safety Systems	+27 (0)11 791 6000	extech@jhbmail.co.za	www.extech.co.za	48,49,76,77*,80
Festo	08600 FESTO (33786)	sales.interaction@za.festo.com	www.festo.com	75*
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Guduza System Technologies	0861 115 441	kkruger@gst-guduza.co.za	www.gst-guduza.co.za	65*,77
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Invensys Operations Management	+27 (0)11 607 8100	tech@invensys.co.za	www.iom.invensys.co.za	10,72,73,74,loose insert*
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Mantech Electronics	+27 (0)11 493 9307	jn@mantech.co.za	www.mantech.co.za	38
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PSY Systems	+27 (0)11 782 5449	info@psysystems.com	www.psysystems.com	24
R&C Instrumentation	086 111 4217	sales@randci.co.za	www.randci.co.za	32,35*
RET Automation Controls	+27 (0)11 453 2468	rodney.topham@retautomation.com	www.retautomation.com	7*,19*,52,67*,68
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RJ Programmable Techniques	+27 (0)11 781 0777	sales@rjprogtech.co.za	www.rjprogtech.co.za	29*,30,31,52
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